



(12) **United States Patent**
Xu et al.

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(54) **POWER CONTROL PROCEDURES WITH COMMON BEAMS**

(71) Applicant: **Ofinno, LLC**, Reston, VA (US)

(72) Inventors: **Kai Xu**, Great Falls, VA (US); **Hua Zhou**, Vienna, VA (US); **Esmael Hejazi Dinan**, McLean, VA (US); **Ali Cagatay Cirik**, Chantilly, VA (US); **Hyoungsuk Jeon**, Centreville, VA (US); **Nazanin Rastegardoost**, McLean, VA (US)

(73) Assignee: **Ofinno, LLC**, Reston, VA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 433 days.

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H04W 52/14 (2009.01)
H04W 52/08 (2009.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04W 52/146** (2013.01); **H04W 52/08** (2013.01); **H04W 52/242** (2013.01); **H04W 72/1263** (2013.01); **H04W 72/23** (2023.01)

(58) **Field of Classification Search**
CPC H04W 72/232; H04W 72/1268; H04W 52/146; H04W 72/21; H04W 72/23;
(Continued)

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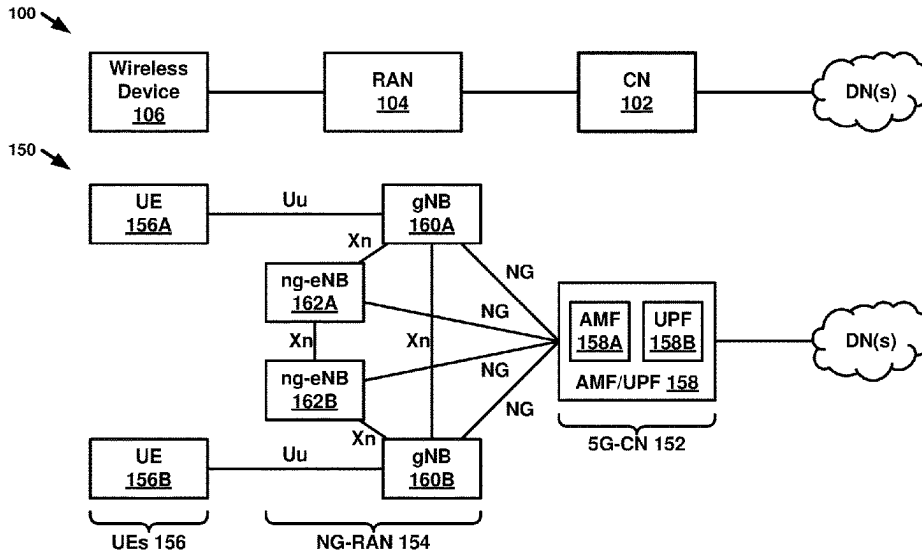
Primary Examiner — Khalid W Shaheed

(74) *Attorney, Agent, or Firm* — Xin Gao; Kavon Nasabzadeh; Jacob L. Mangan

(57) **ABSTRACT**

A wireless device may receive downlink control information (DCI) indicating a first TCI state of TCI states. The first TCI state may be associated with a first power control parameter and may be applicable after a first time instance for uplink channel transmissions. The DCI may comprise an uplink grant scheduling a transport block at a second time instance and indicating a second power control parameter. The wireless device may select, from the first power control parameter and the second power control parameter, a power control parameter based on a comparison of the first time instance with the second time instance. The wireless device may determine, based on the selecting, a transmit power. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

20 Claims, 29 Drawing Sheets



- (51) **Int. Cl.**
H04W 52/24 (2009.01)
H04W 72/1263 (2023.01)
H04W 72/23 (2023.01)
- (58) **Field of Classification Search**
 CPC H04W 72/0473; H04W 52/06; H04W
 72/231; H04L 5/0094
 See application file for complete search history.

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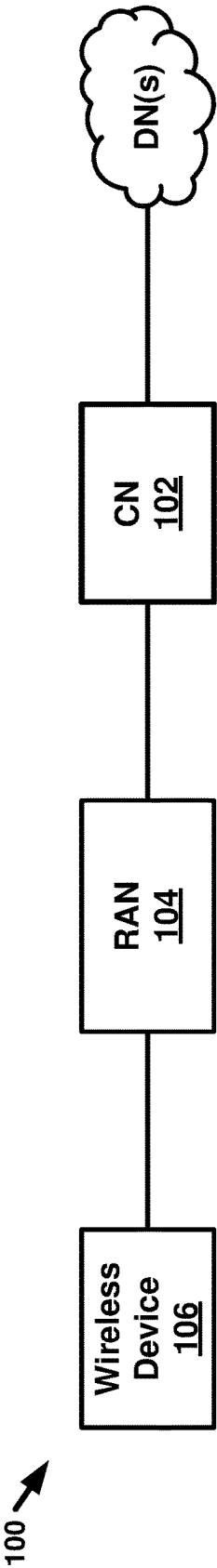


FIG. 1A

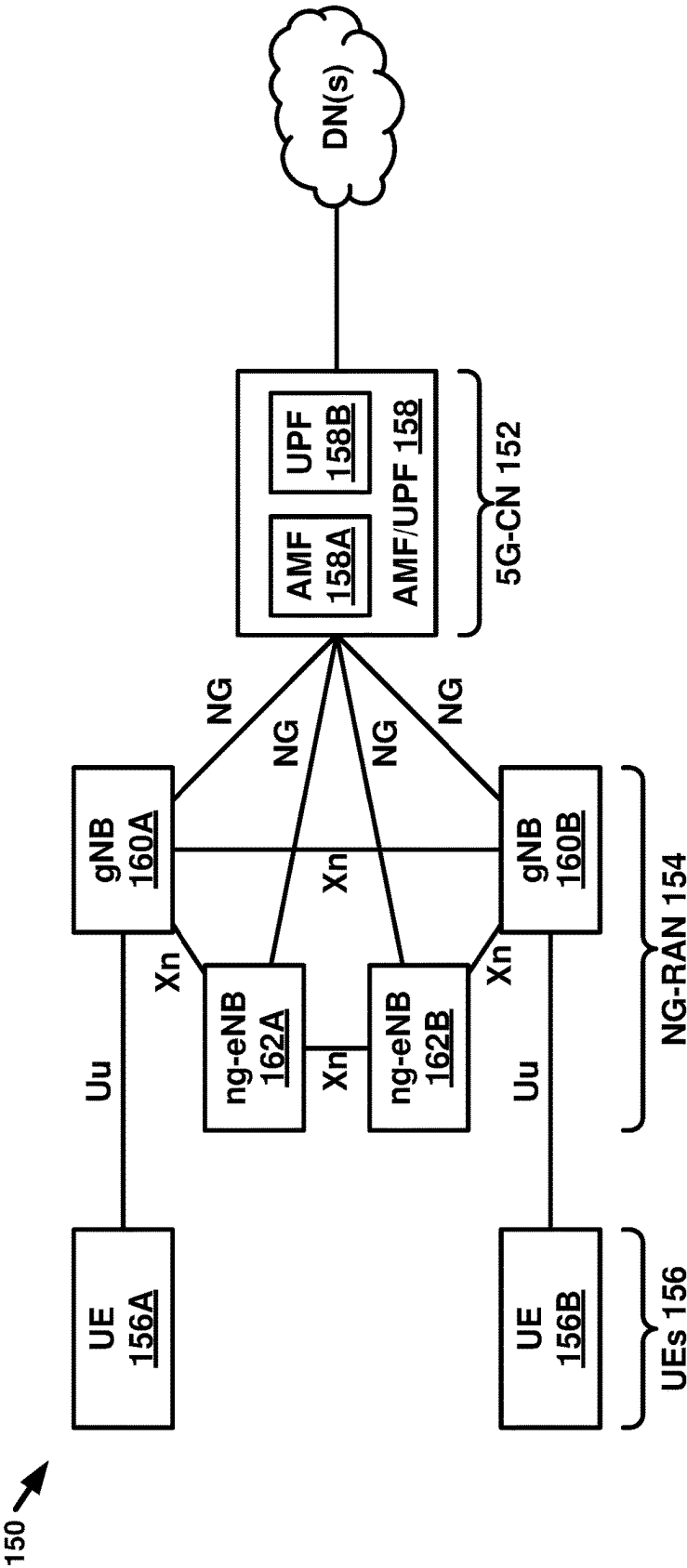


FIG. 1B

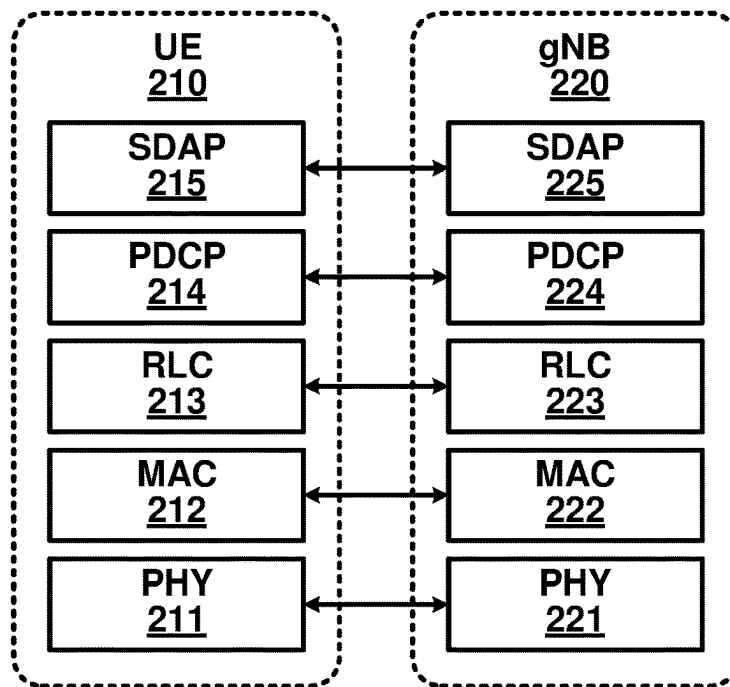


FIG. 2A

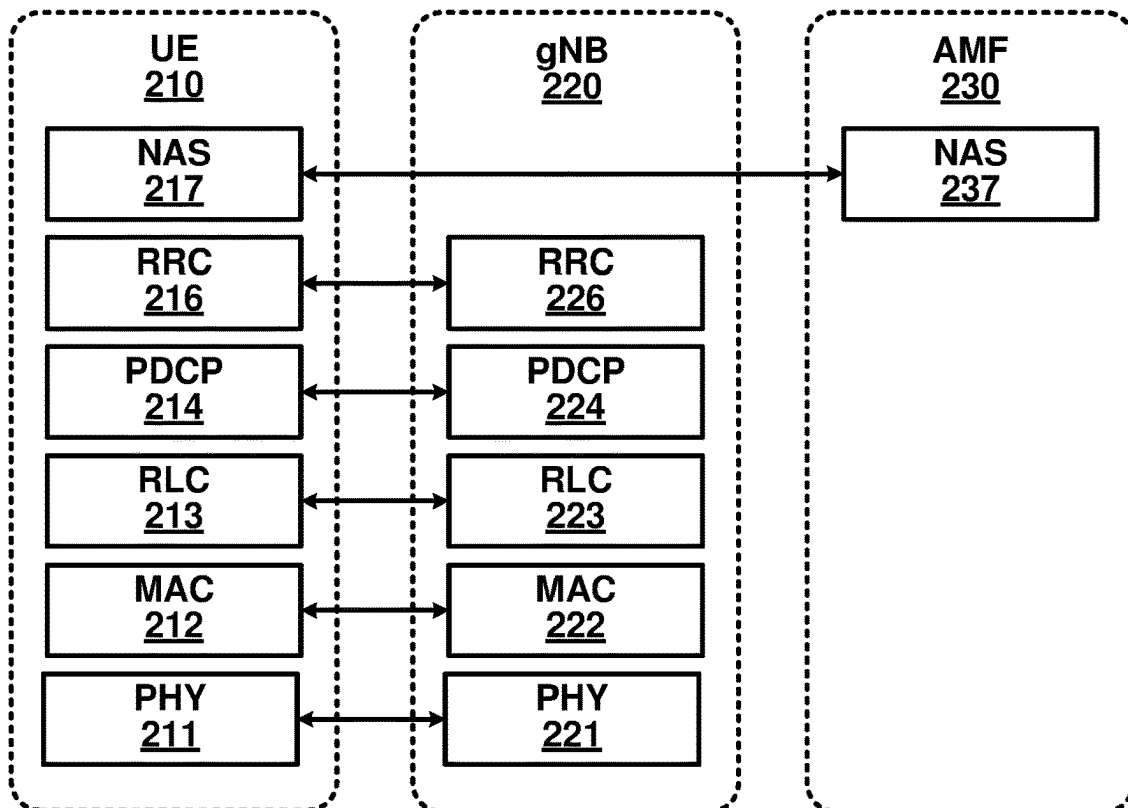


FIG. 2B

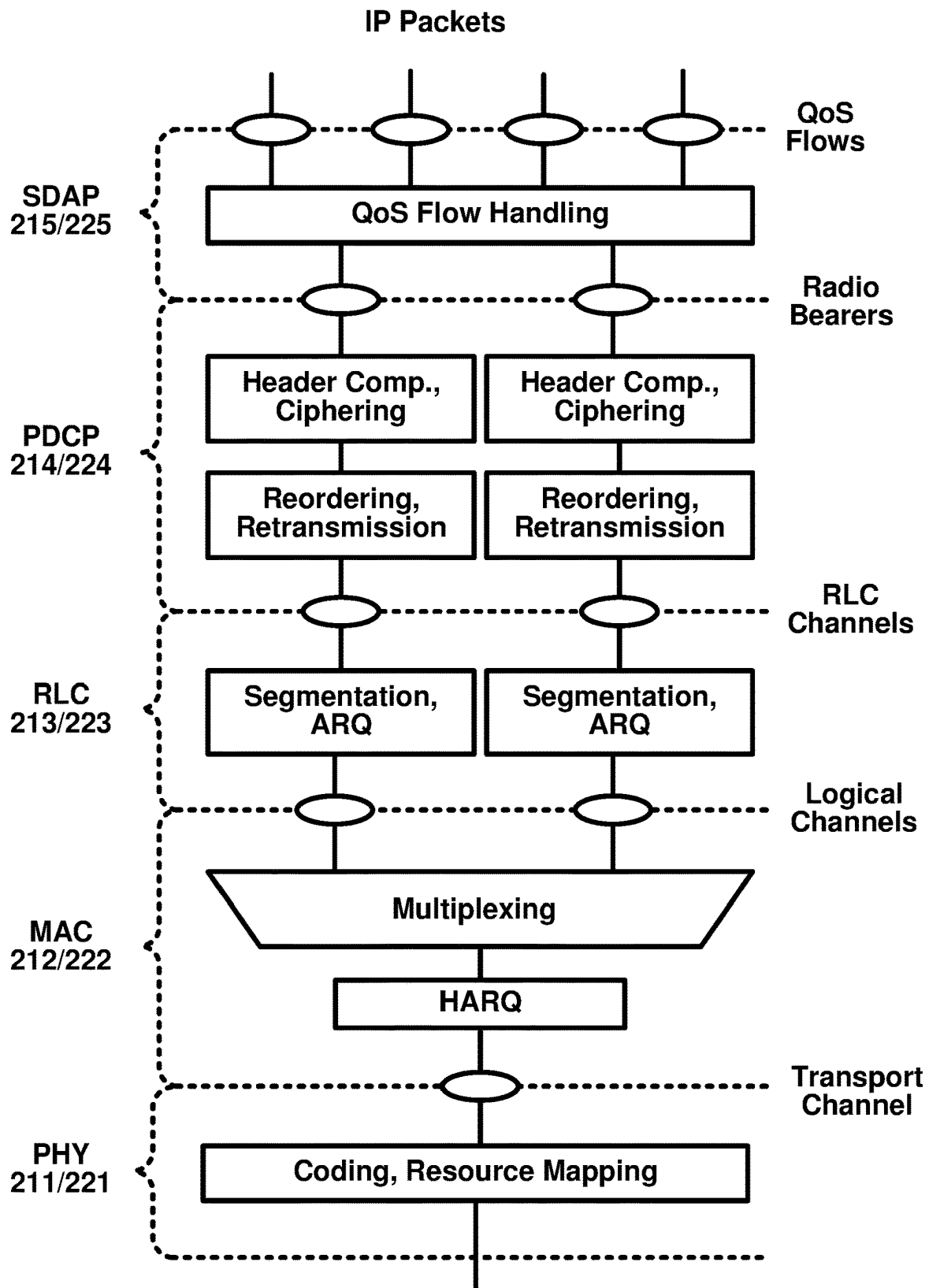


FIG. 3

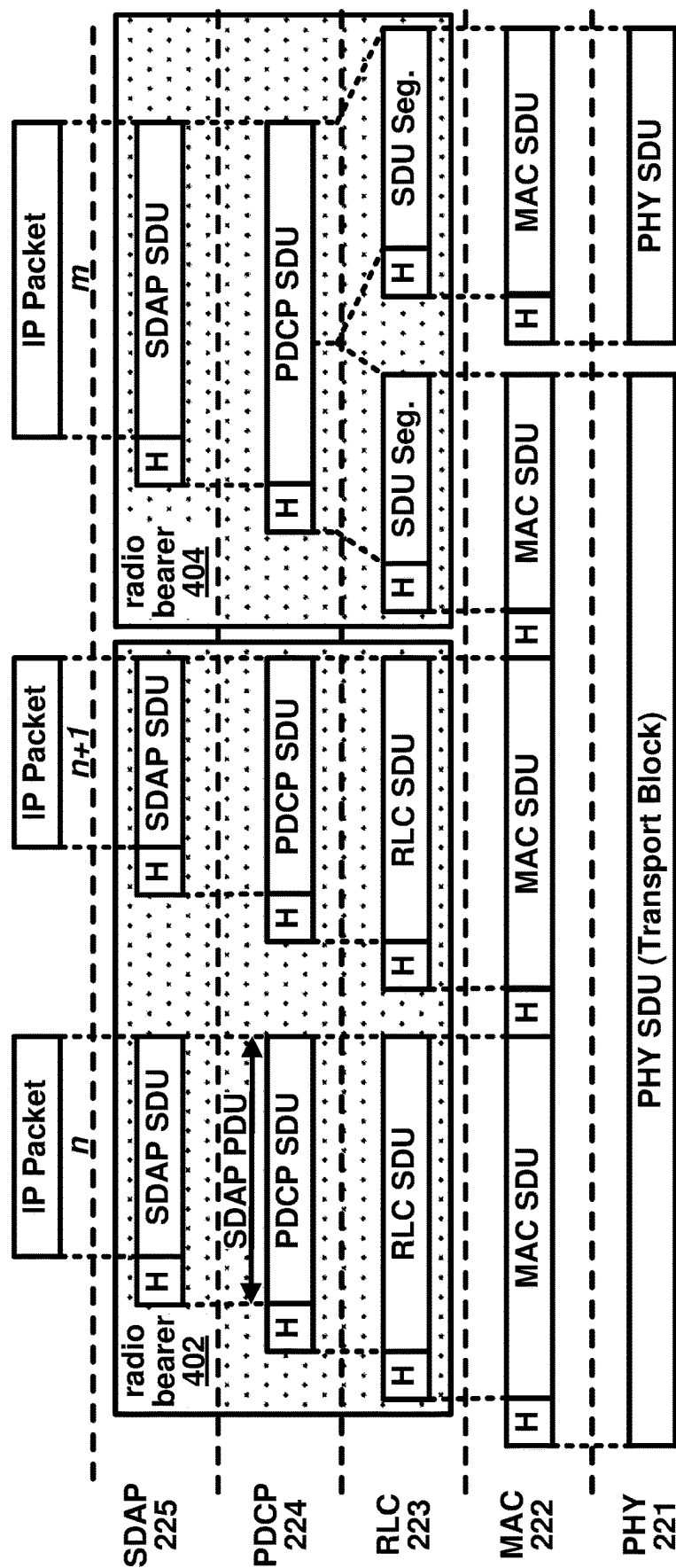


FIG. 4A

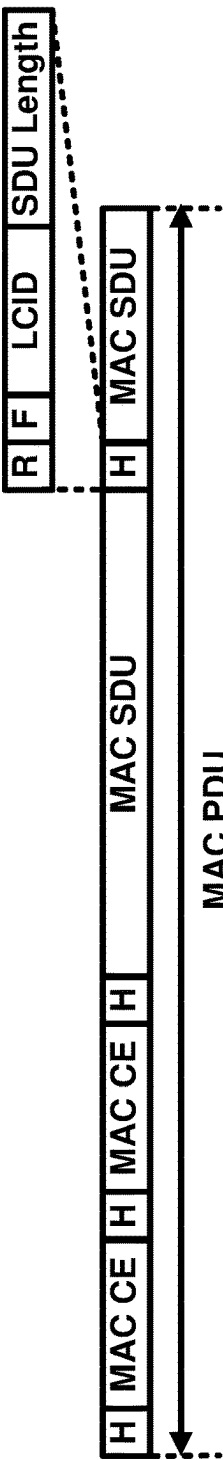
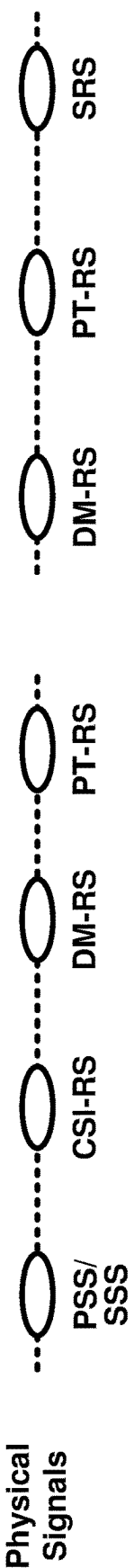
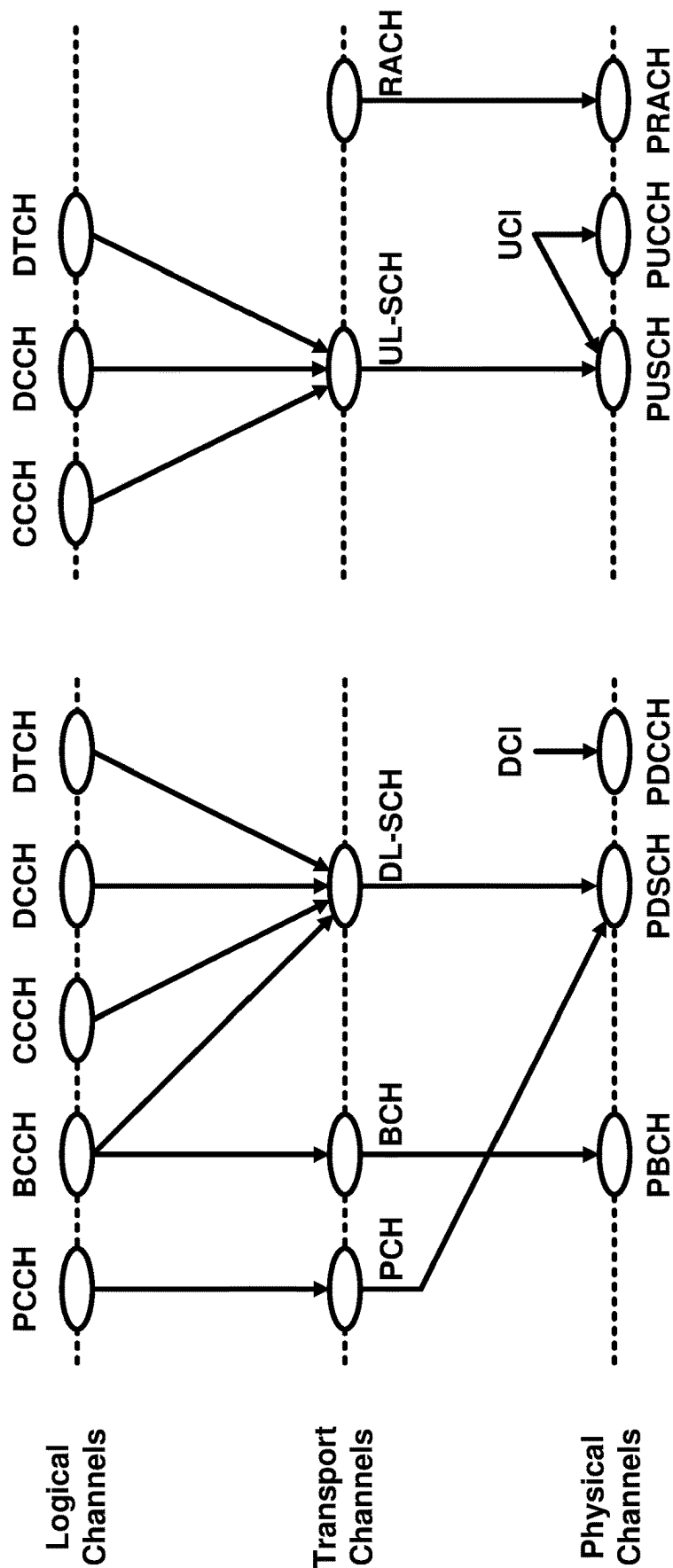


FIG. 4B



Downlink

Uplink

FIG. 5A

FIG. 5B

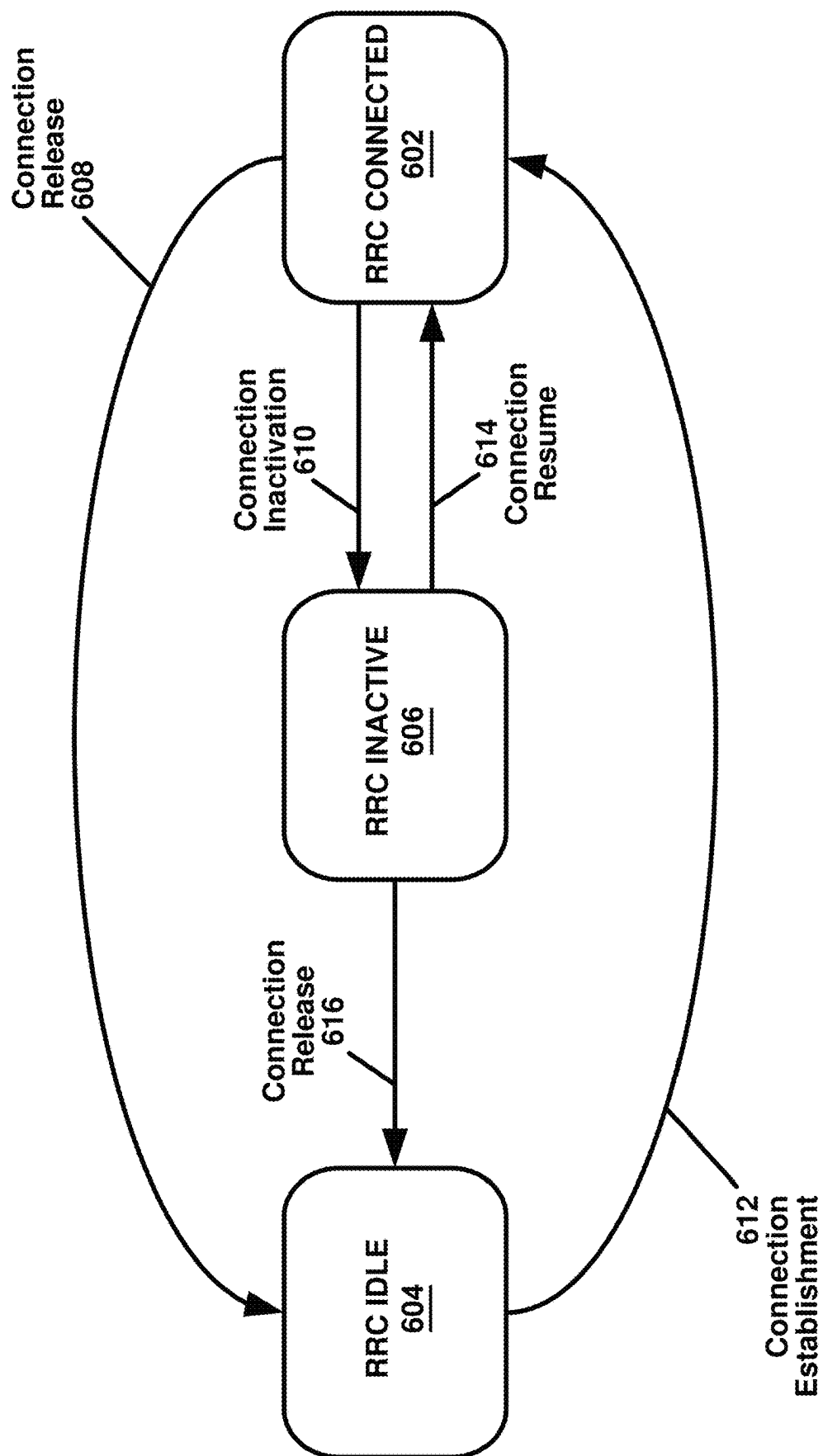


FIG. 6

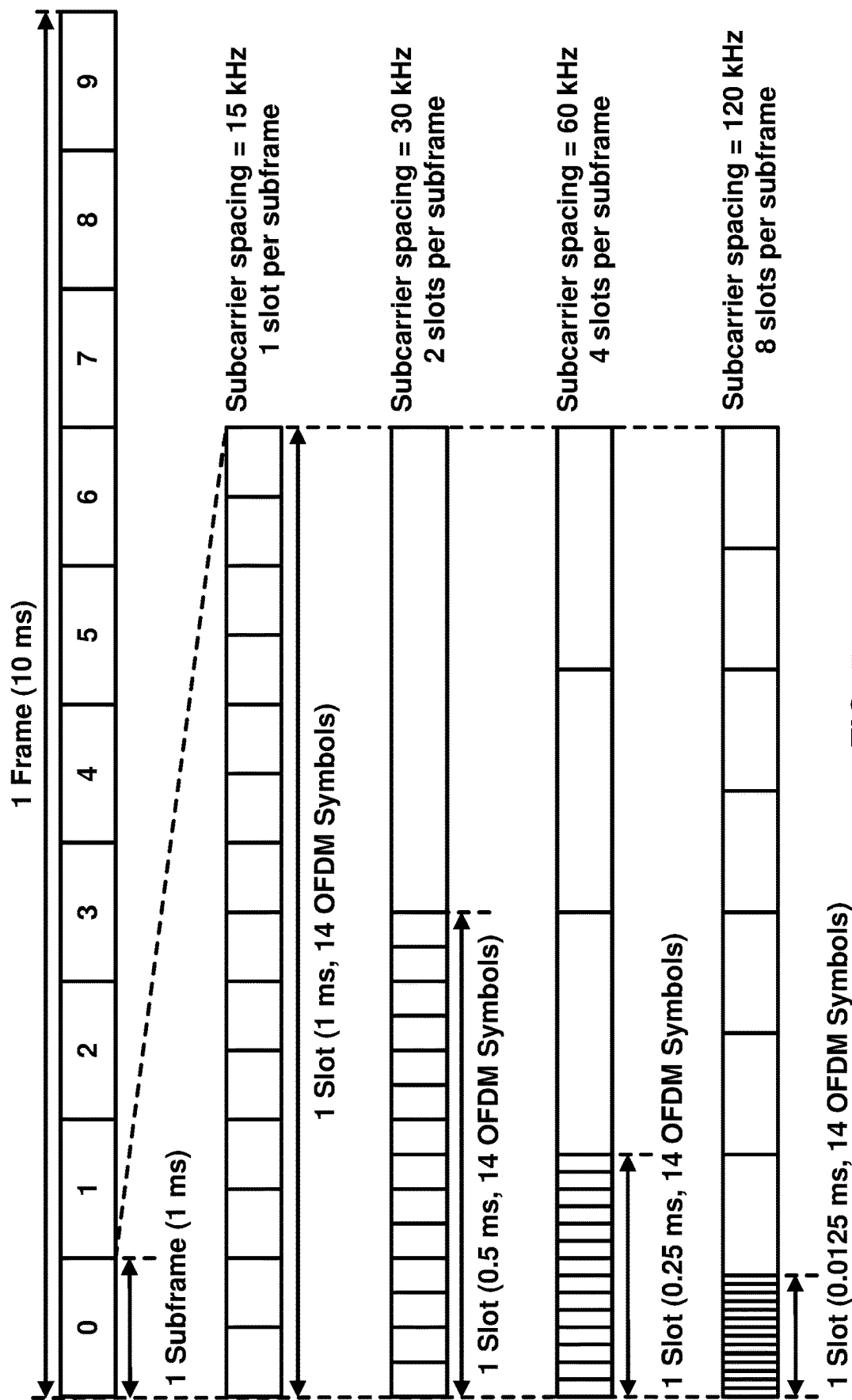


FIG. 7

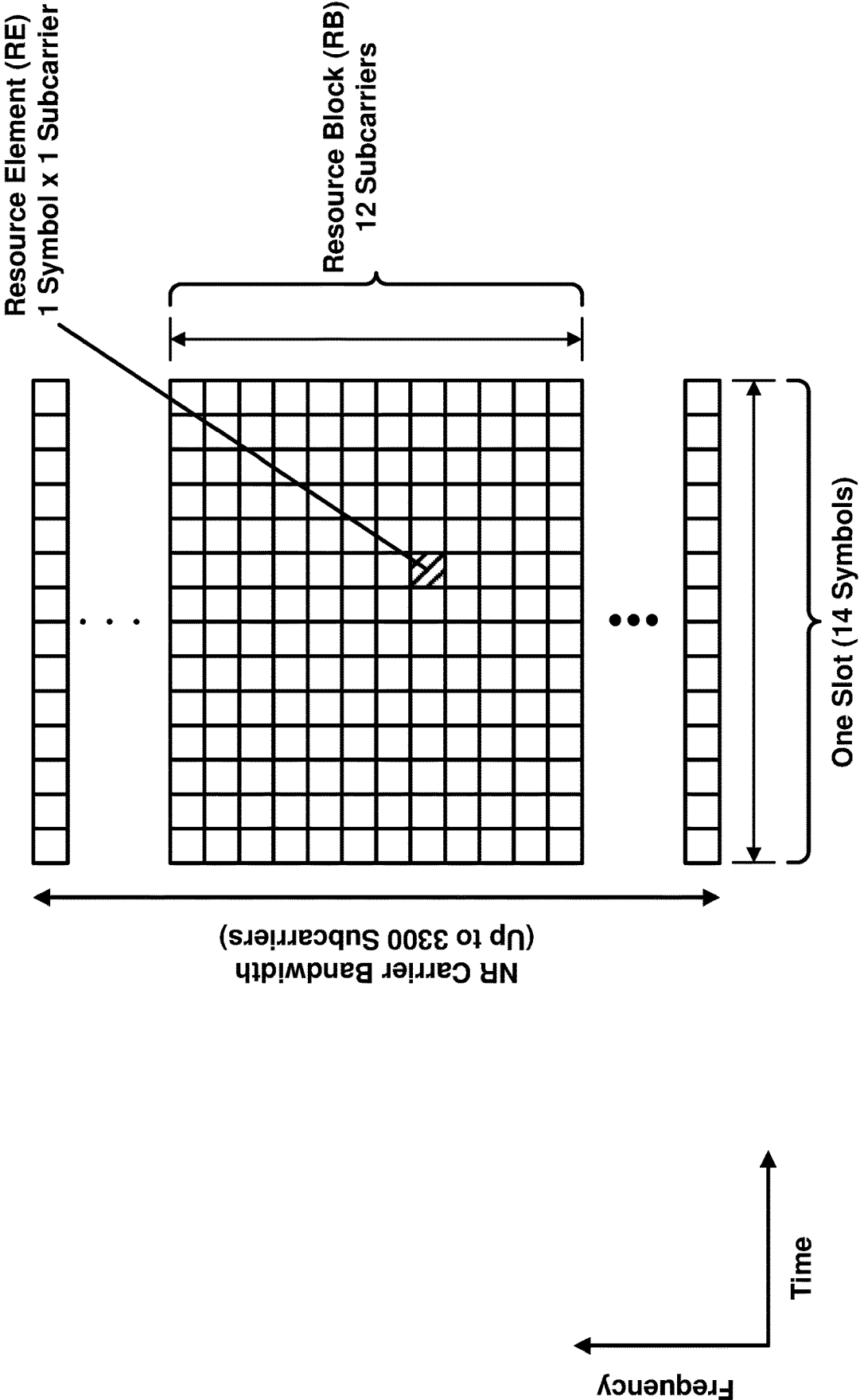


FIG. 8

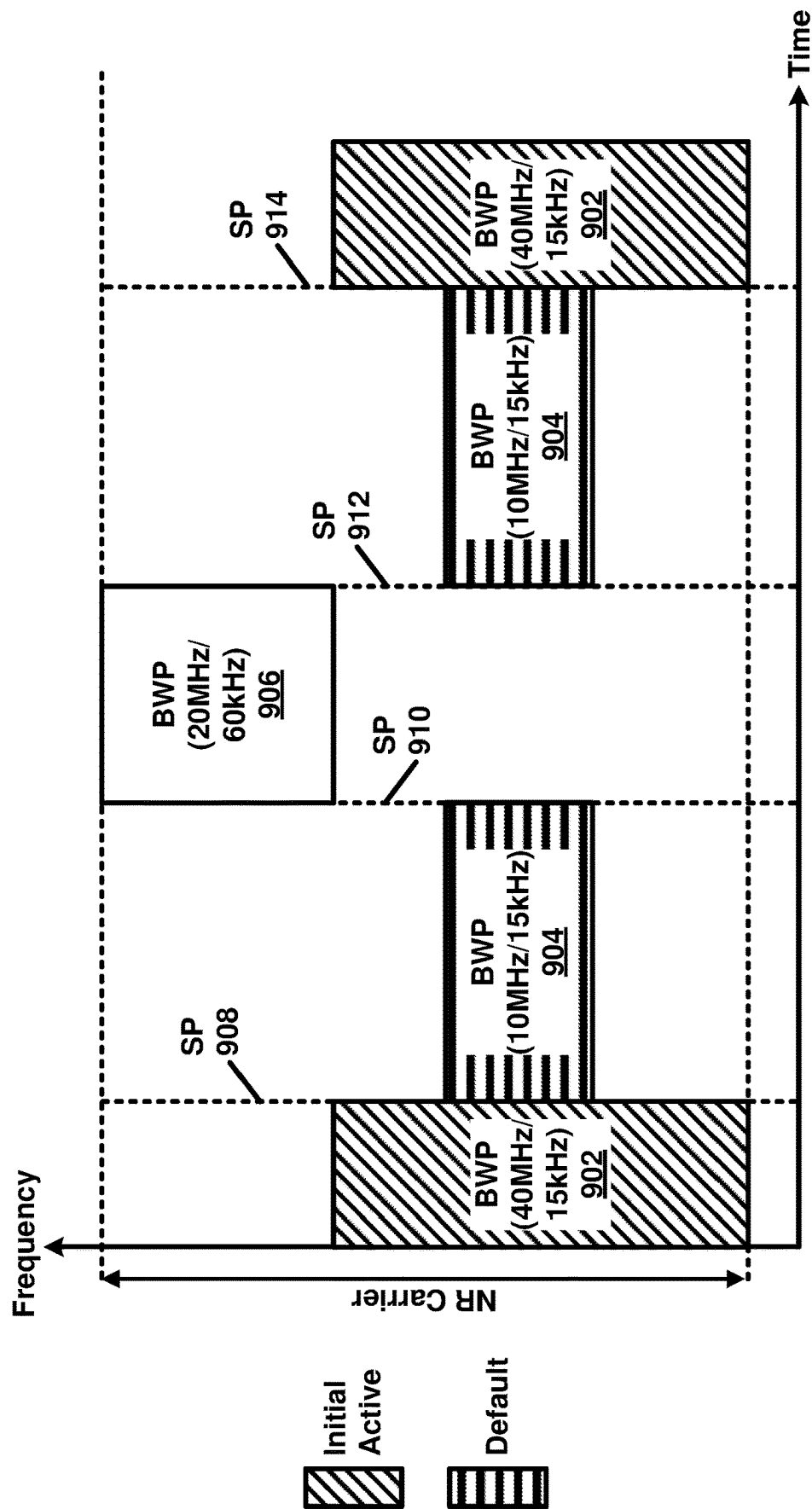


FIG. 9

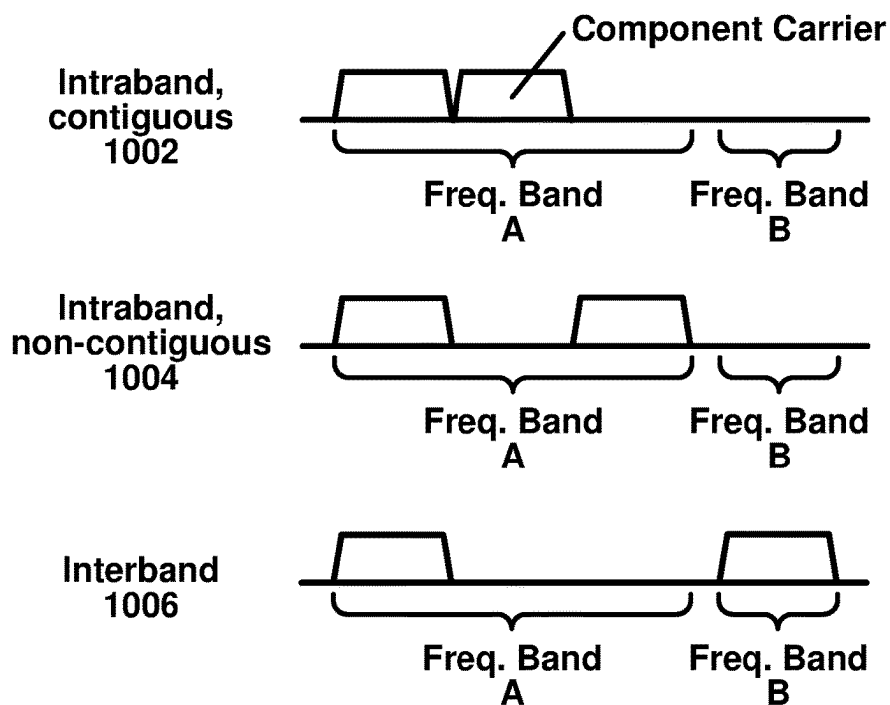


FIG. 10A

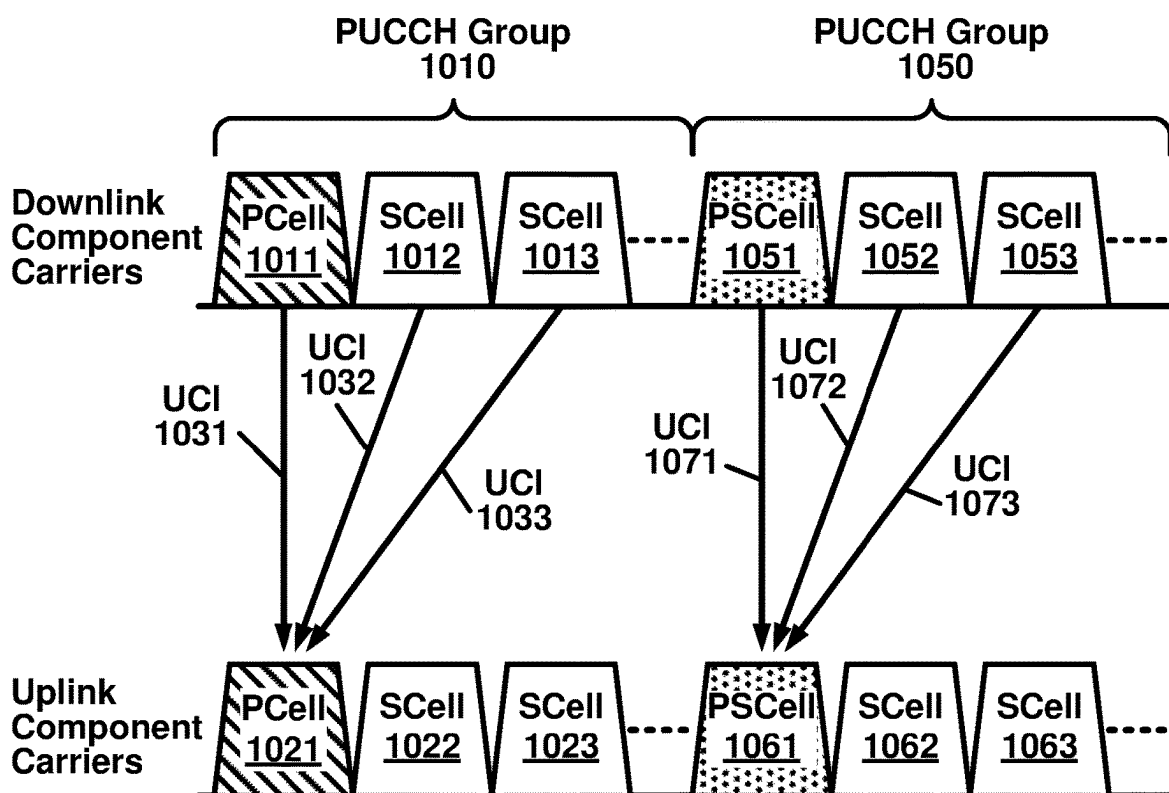


FIG. 10B

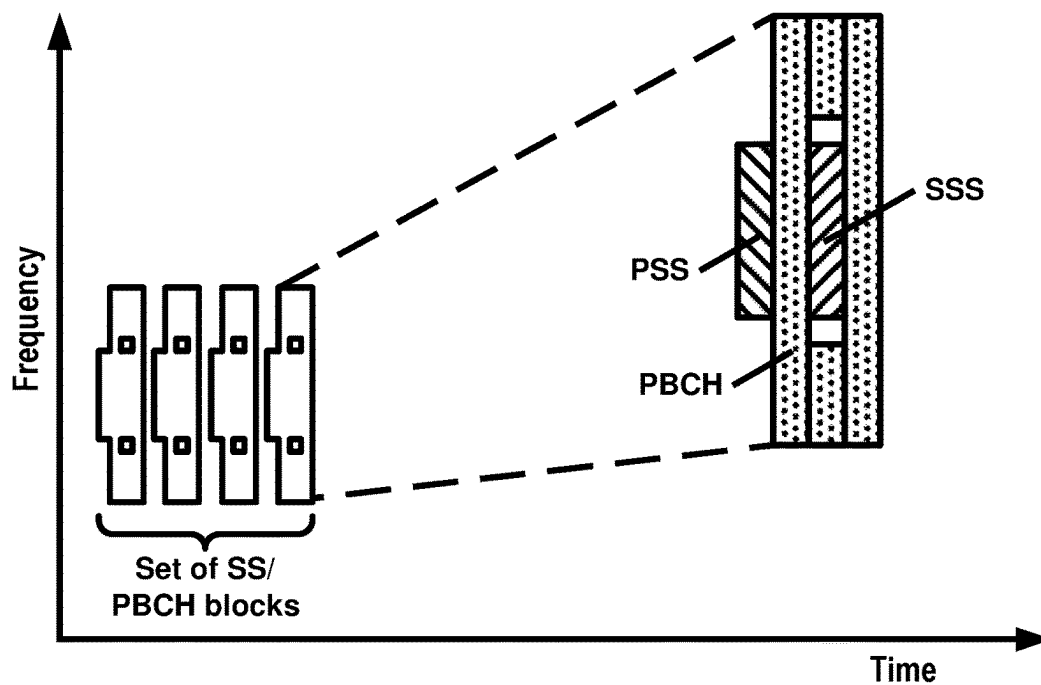


FIG. 11A

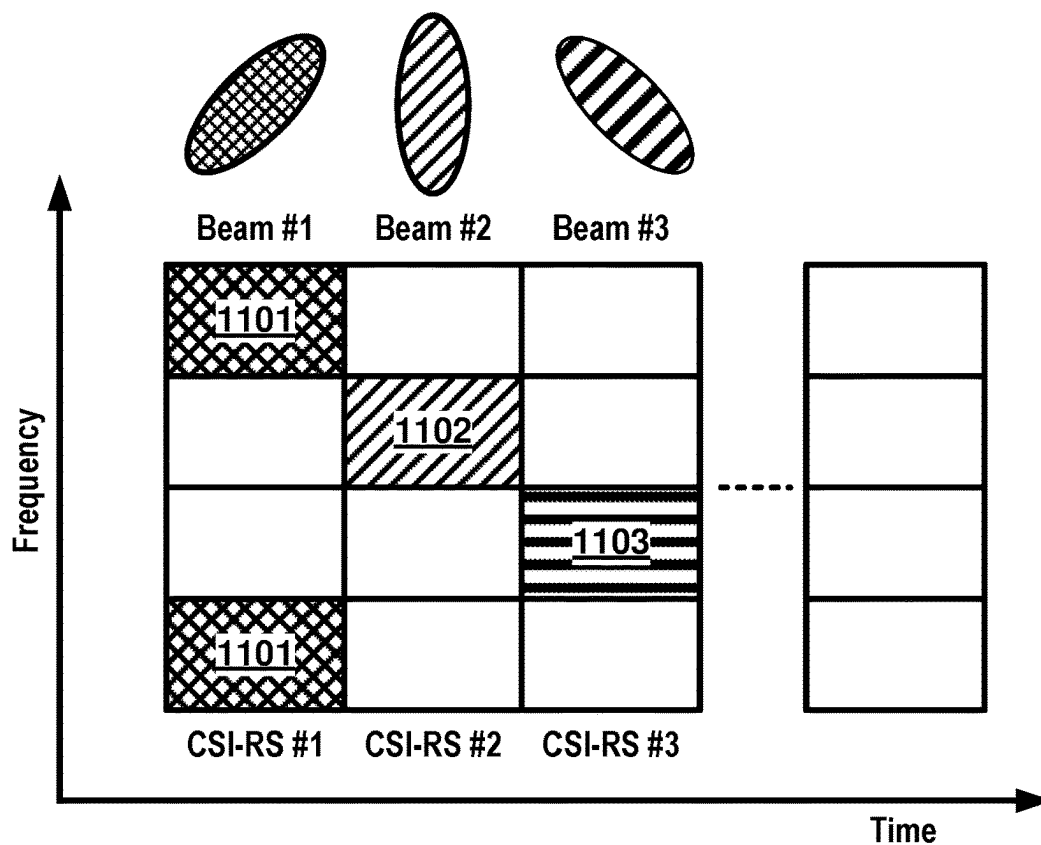


FIG. 11B

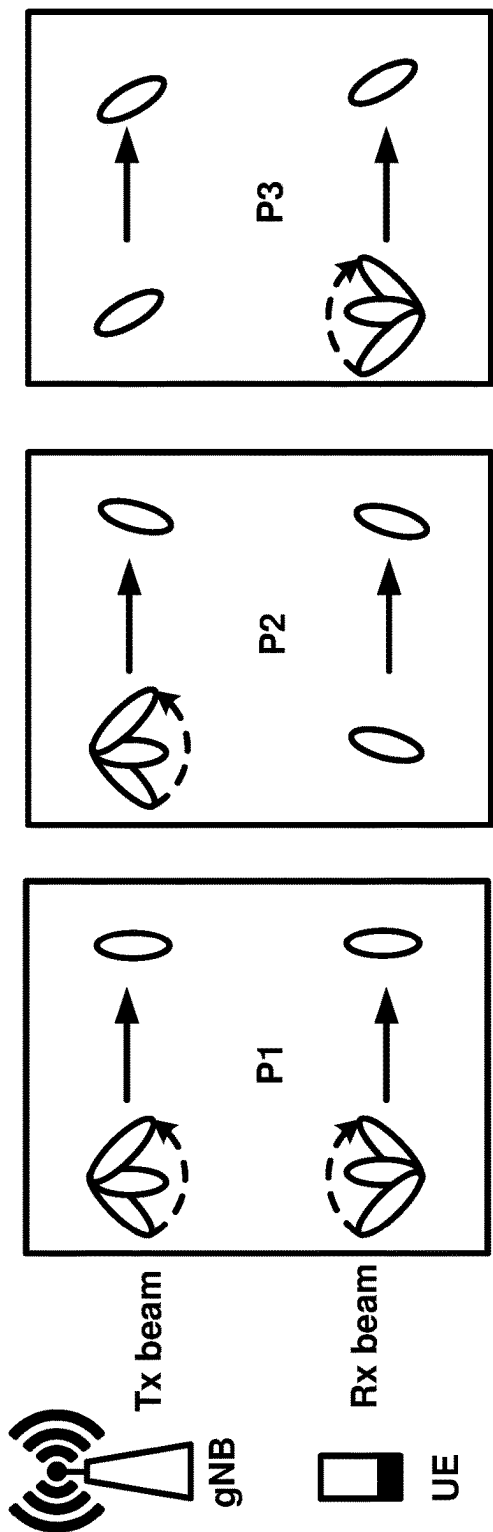


FIG. 12A

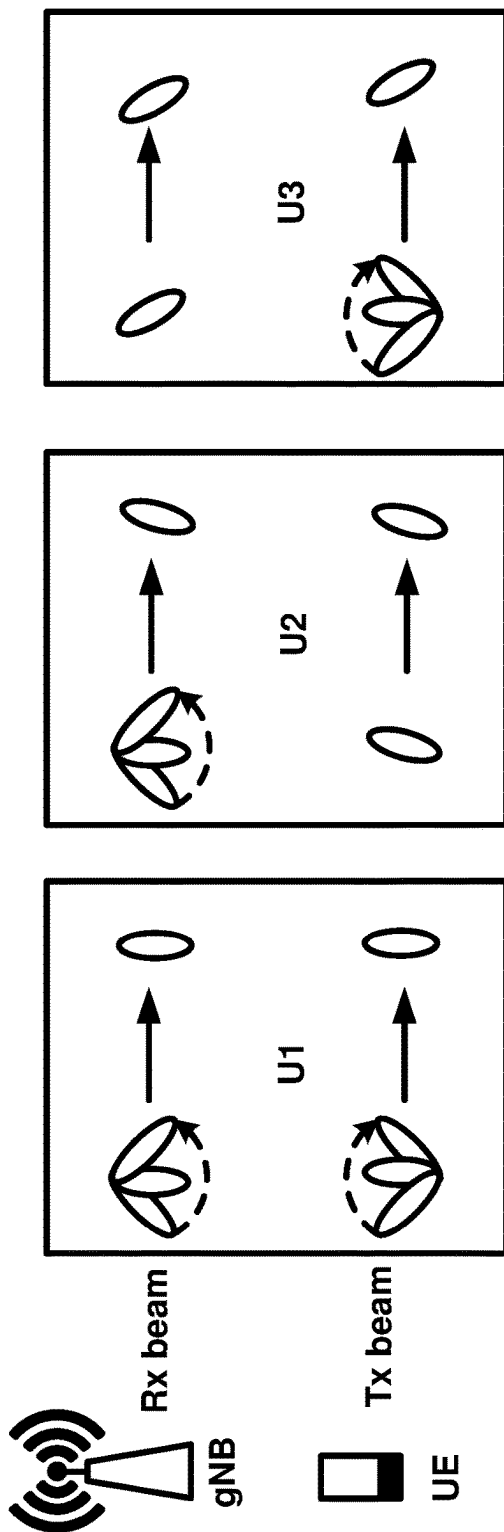


FIG. 12B

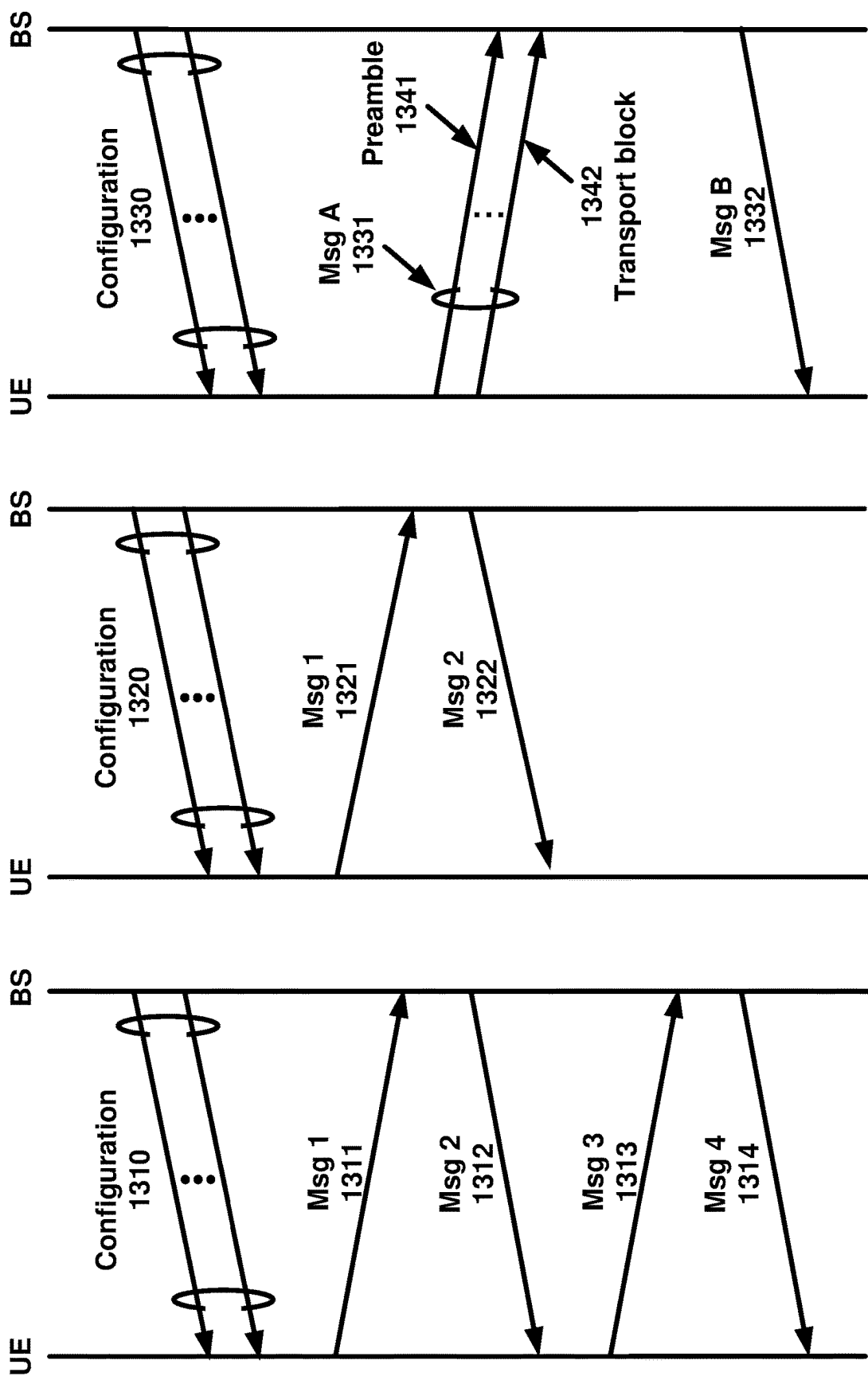


FIG. 13C

FIG. 13B

FIG. 13A

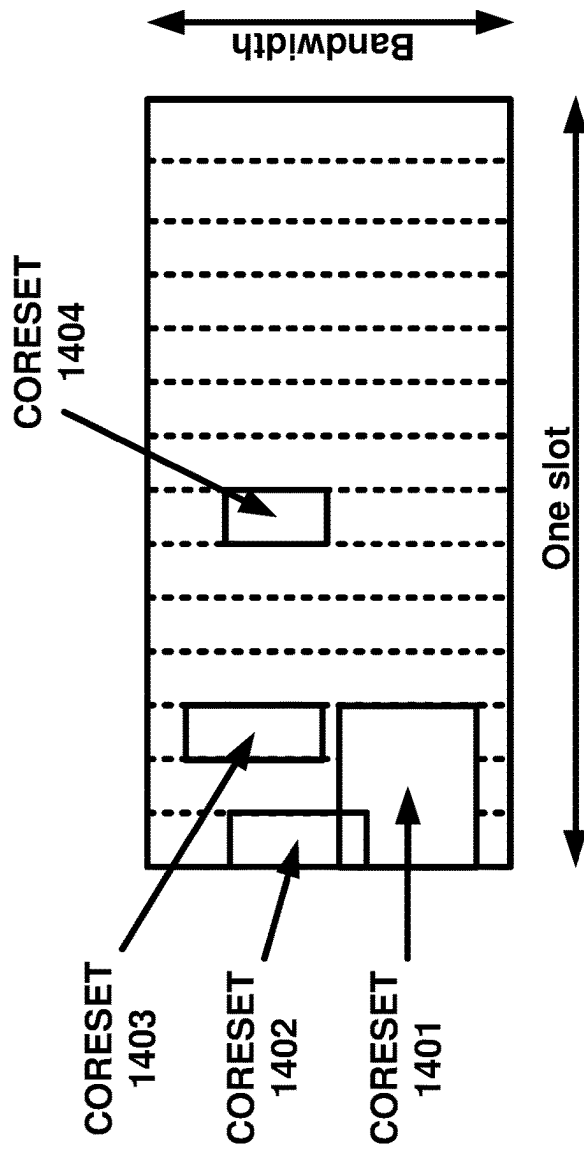


FIG. 14A

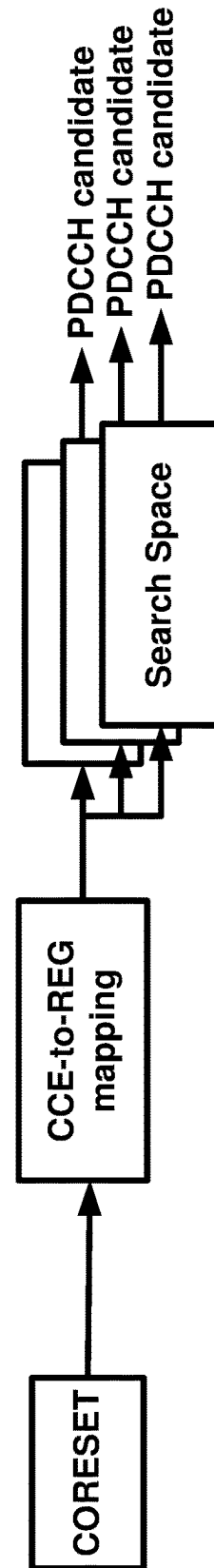


FIG. 14B

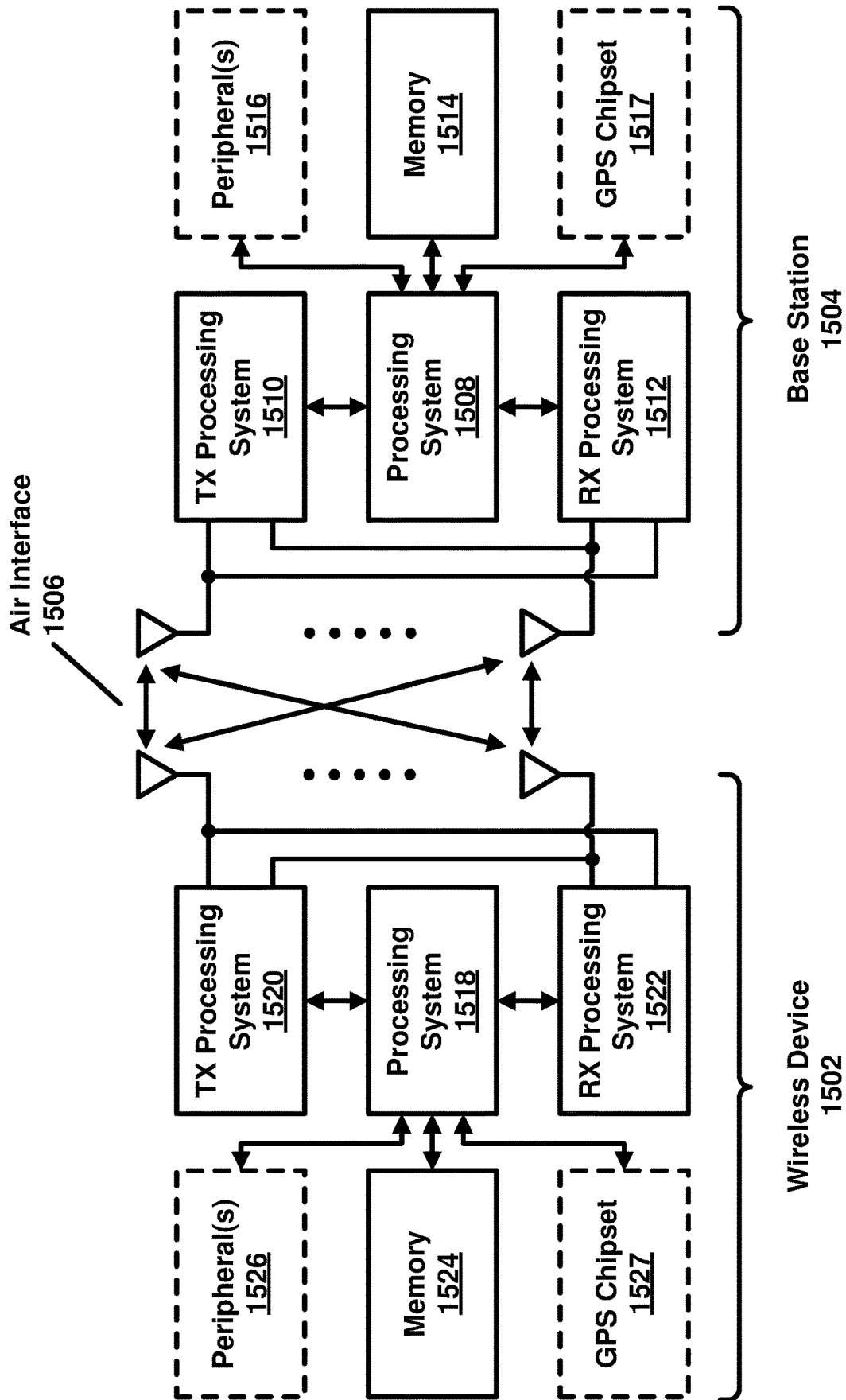


FIG. 15

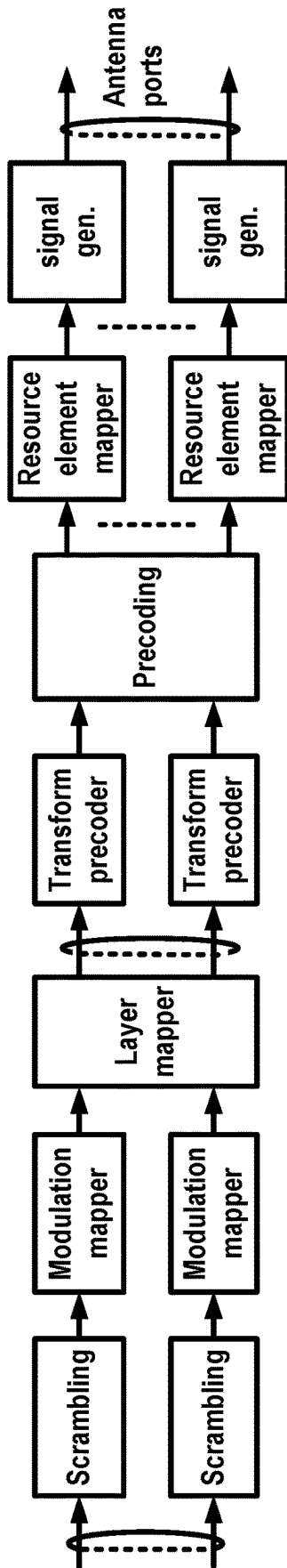


FIG. 16A

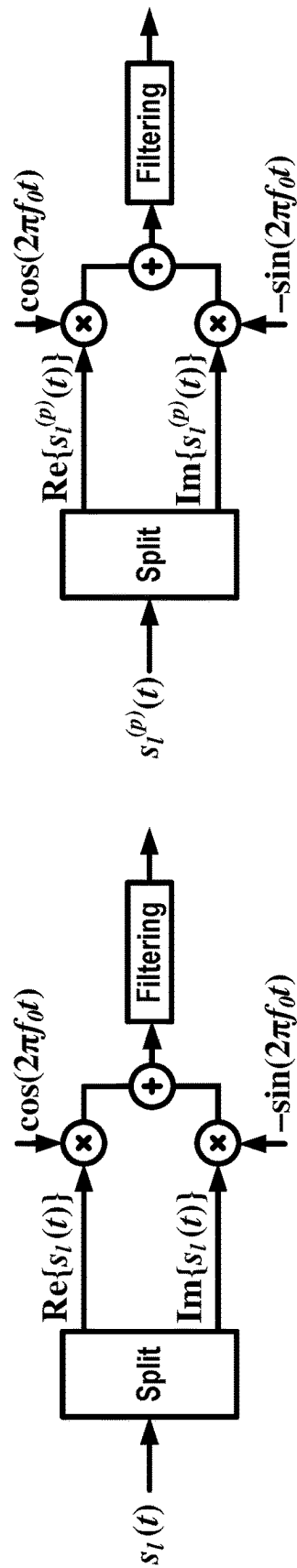


FIG. 16B

FIG. 16D

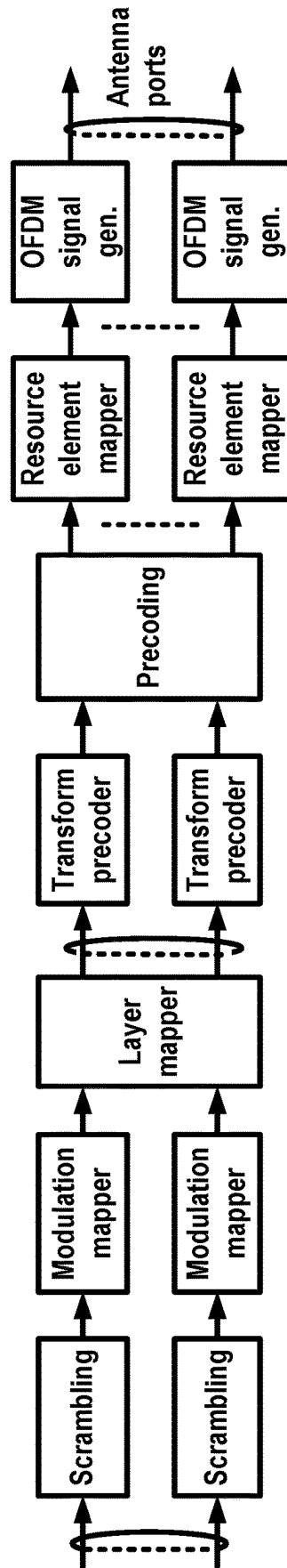


FIG. 16C

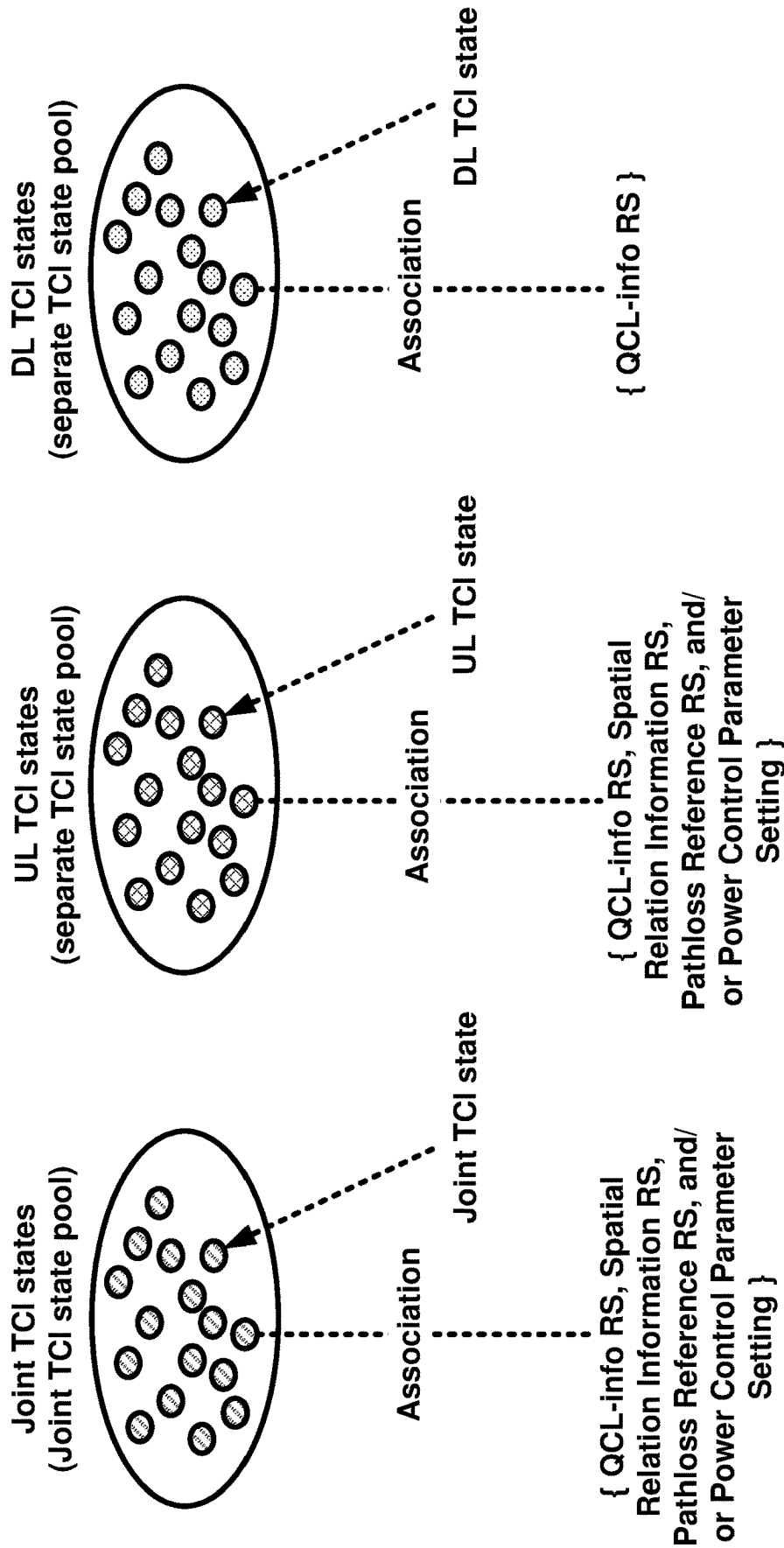


FIG. 17

UL-TCI-State information element

```

PathlossReferenceRS-Id          UL-TCI-State-Id,
PowerControlParameterSetting-Id  UL-TCI-State-Id,
OR
UL-TCI-State ::=
    ul-tci-StateId
    qcl-Type1
    qcl-Type2
OPTIONAL, -- Need R
PathlossReferenceRS-Id
PowerControlParameterSetting-Id
    ...
}

```

FIG. 18A*Joint-TCI-State* information element

```

PathlossReferenceRS-Id          Joint-TCI-State-Id,
PowerControlParameterSetting-Id Joint-TCI-State-Id,
OR
Joint-TCI-State ::=
    joint-tci-StateId
    qcl-Type1
    qcl-Type2
OPTIONAL, -- Need R
PathlossReferenceRS-Id
PowerControlParameterSetting-Id
    ...
}

```

FIG. 18B

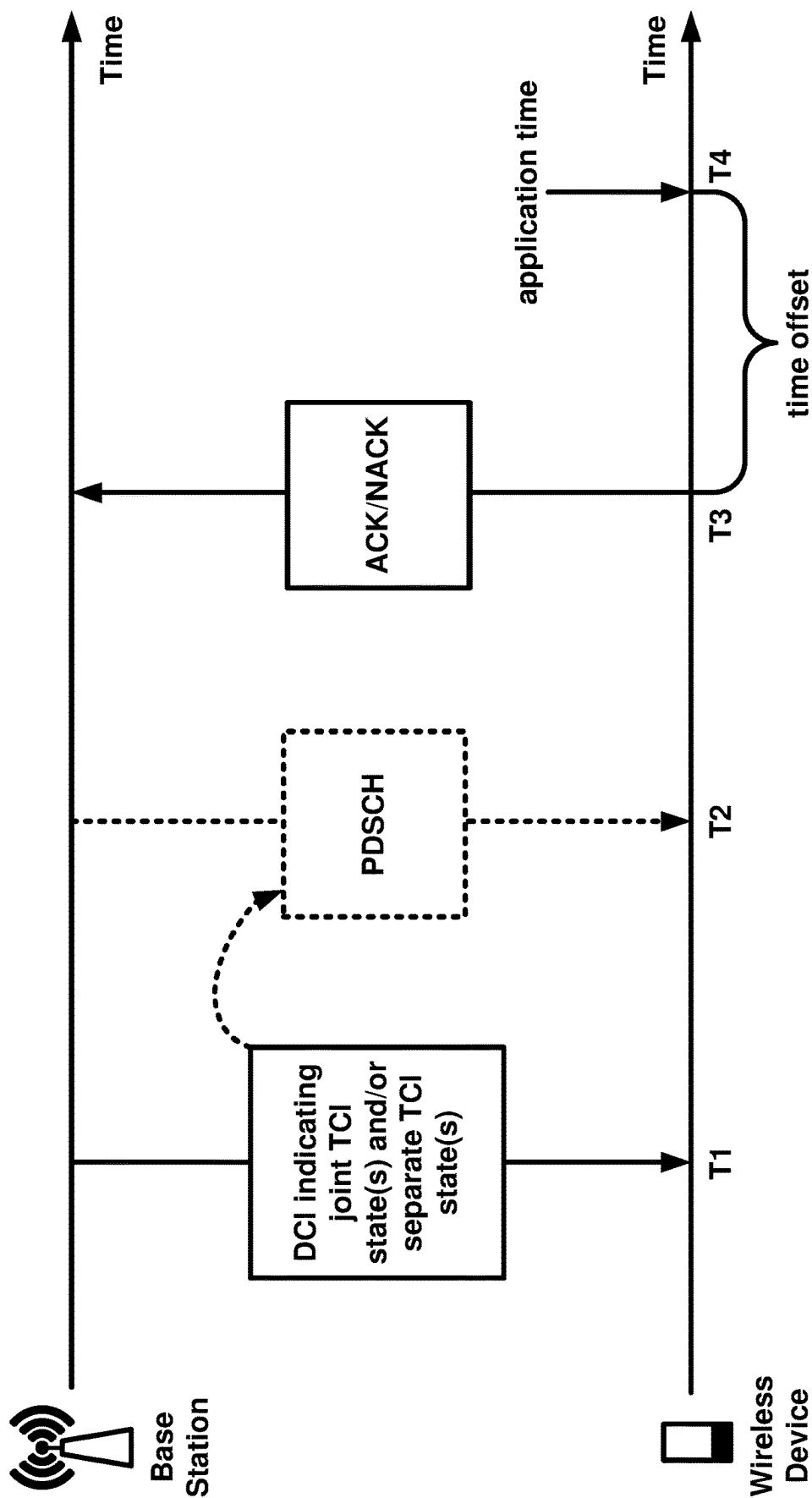


FIG. 19

```

PUSCH-PowerControl ::=
    tpc-Accumulation
    OPTIONAL, -- Need S
    ...
    sri-PUSCH-MappingToAddModList
    Mappings)) OF SRI-PUSCH-PowerControl
OPTIONAL, -- Need N
    sri-PUSCH-MappingToReleaseList
    Mappings)) OF SRI-PUSCH-PowerControlId
OPTIONAL -- Need N
}
SRI-PUSCH-PowerControl ::=
    sri-PUSCH-PowerControlId
    sri-PUSCH-PathlossReferencers-Id
    sri-P0-PUSCH-AlphaSetId
    sri-PUSCH-ClosedLoopIndex
}
SEQUENCE {
    ENUMERATED { disabled }
SEQUENCE (SIZE (1..maxNrofSRI-PUSCH-
SEQUENCE (SIZE (1..maxNrofSRI-PUSCH-
SEQUENCE {
    SRI-PUSCH-PowerControlId,
    PUSCH-PathlossReferencers-Id,
    P0-PUSCH-AlphaSetId,
    ENUMERATED { i0, i1 }
}

```

FIG. 20

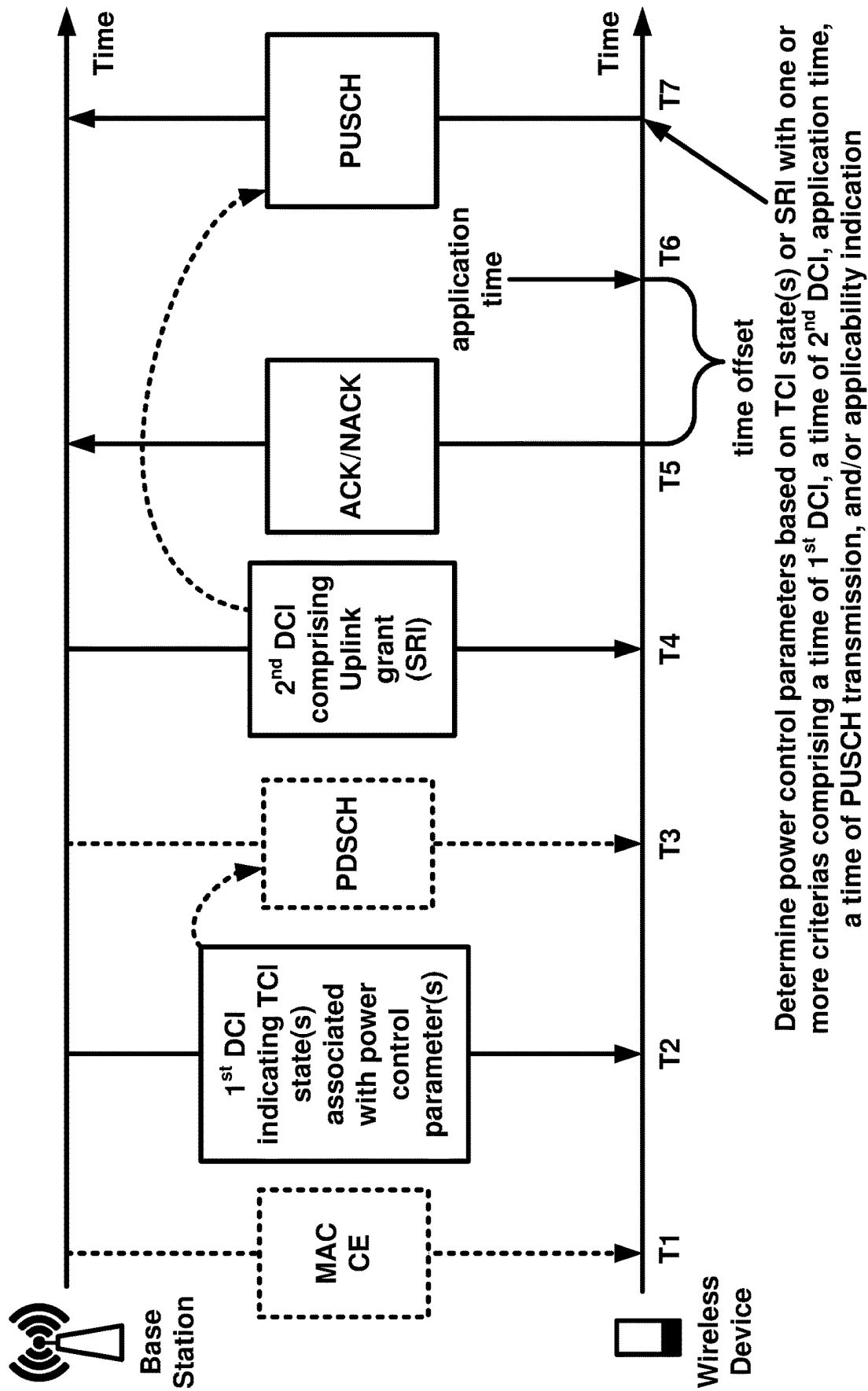


FIG. 21

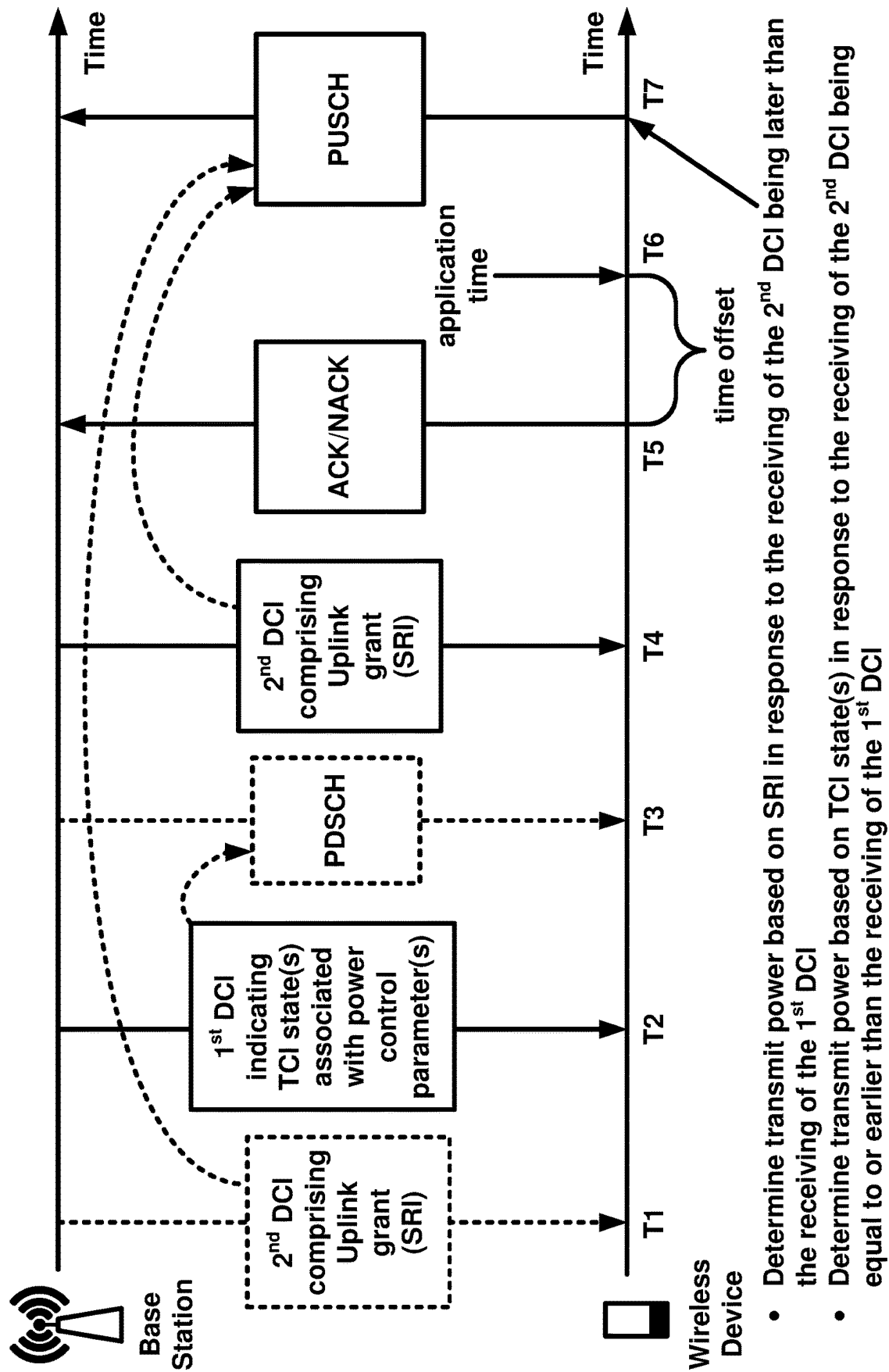


FIG. 22

- Determine transmit power based on SRI in response to the receiving of the 2nd DCI being later than the receiving of the 1st DCI
- Determine transmit power based on TCI state(s) in response to the receiving of the 2nd DCI being equal to or earlier than the receiving of the 1st DCI

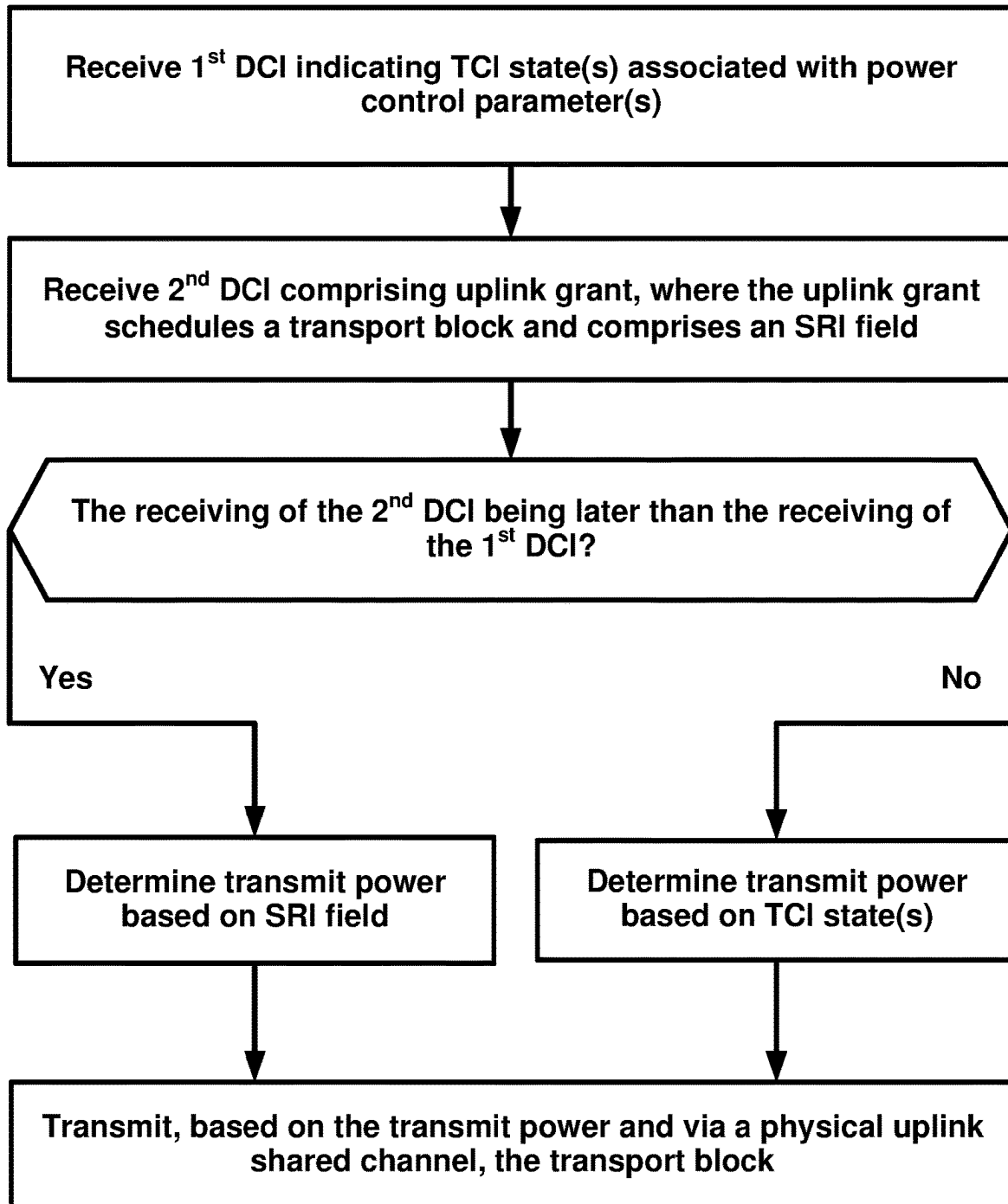


FIG. 23

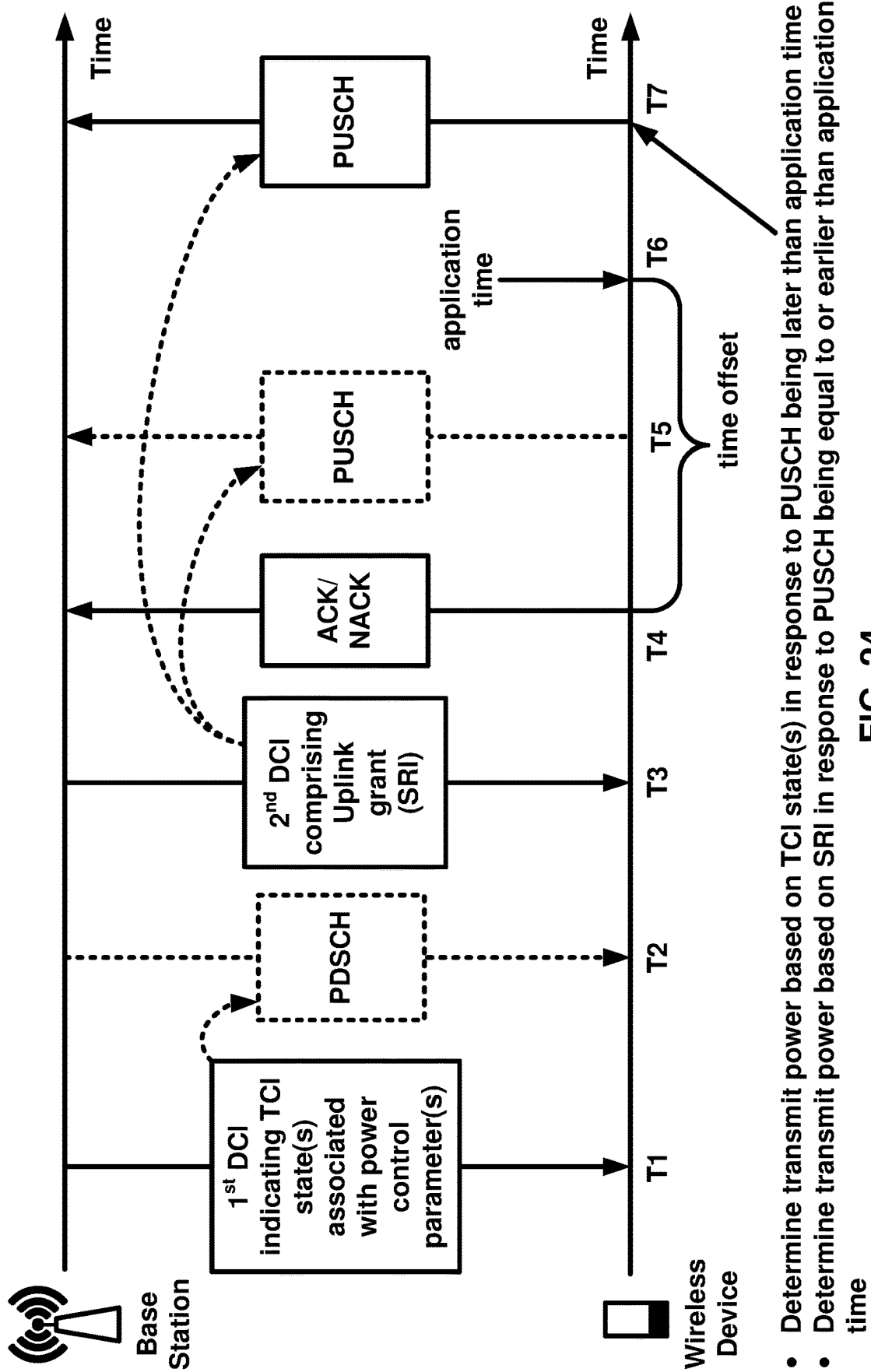


FIG. 24

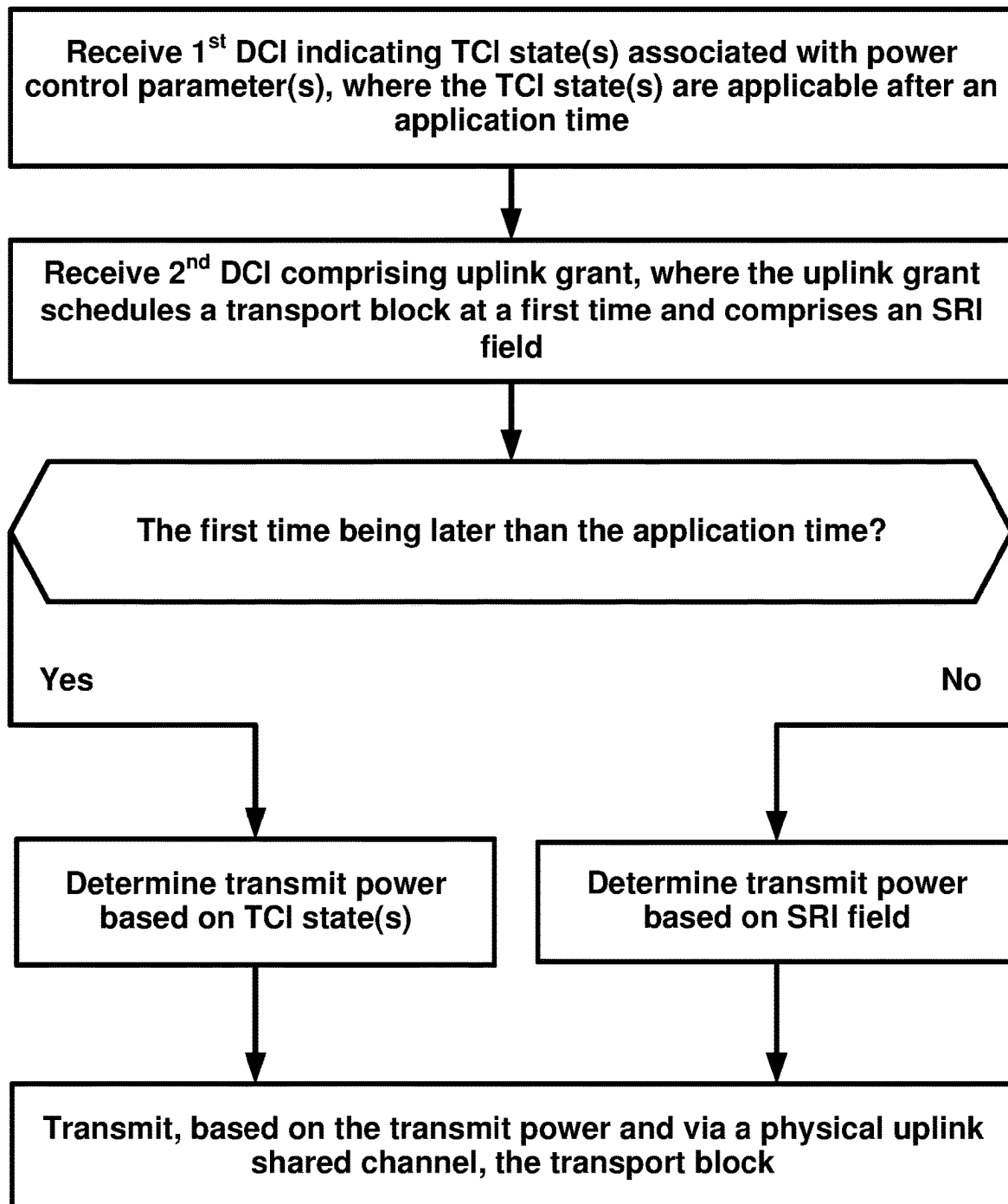


FIG. 25

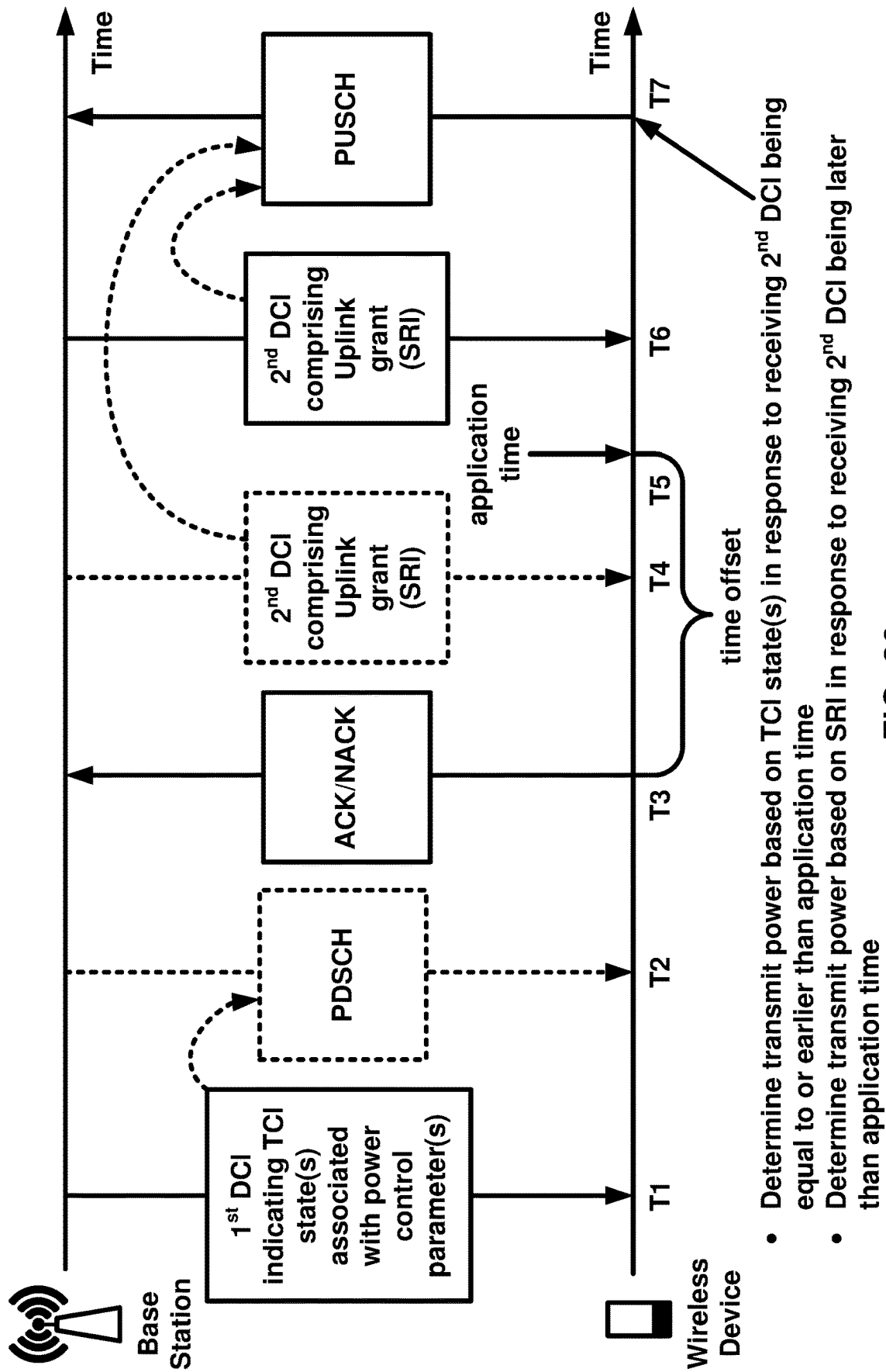


FIG. 26

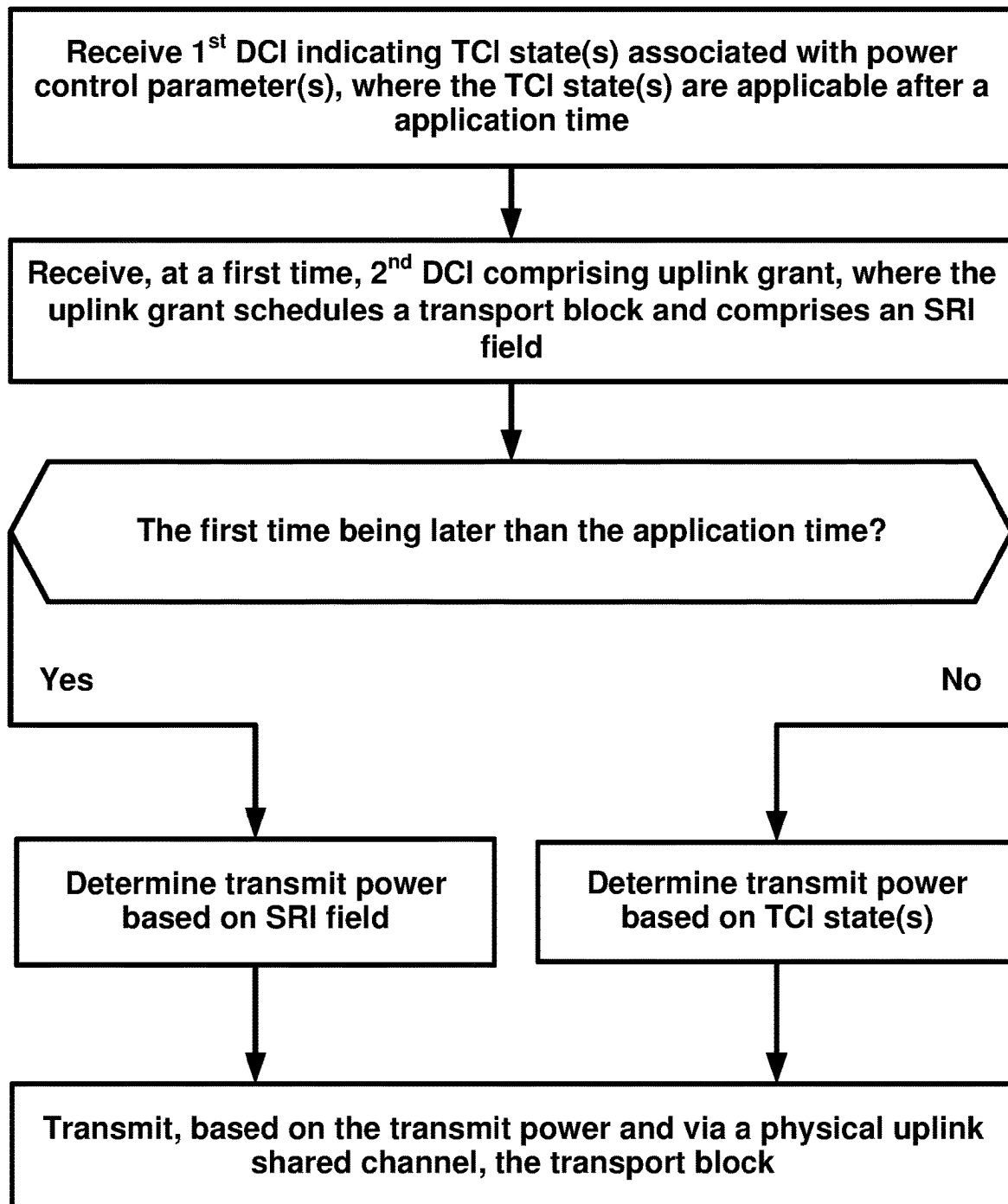


FIG. 27

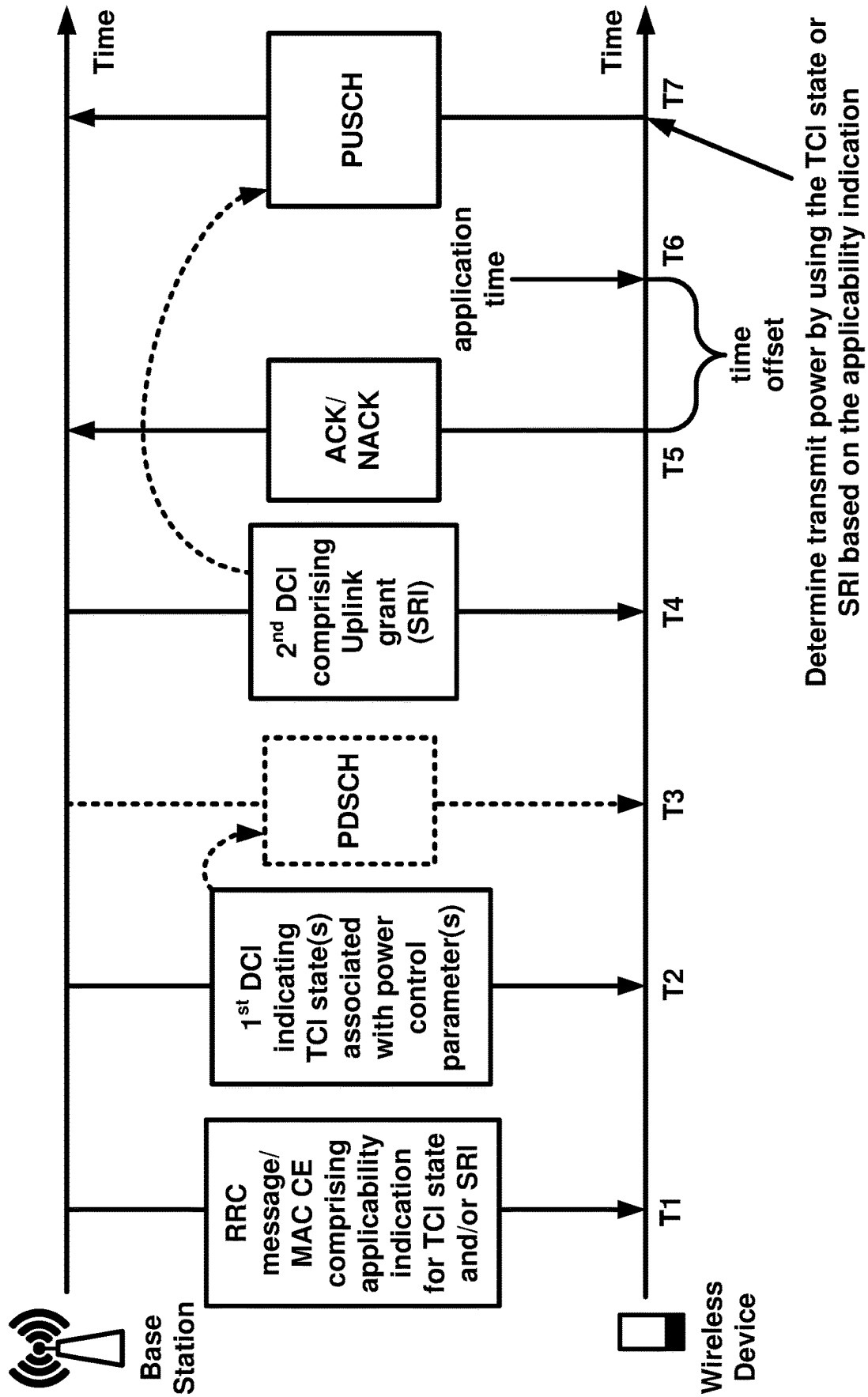


FIG. 28

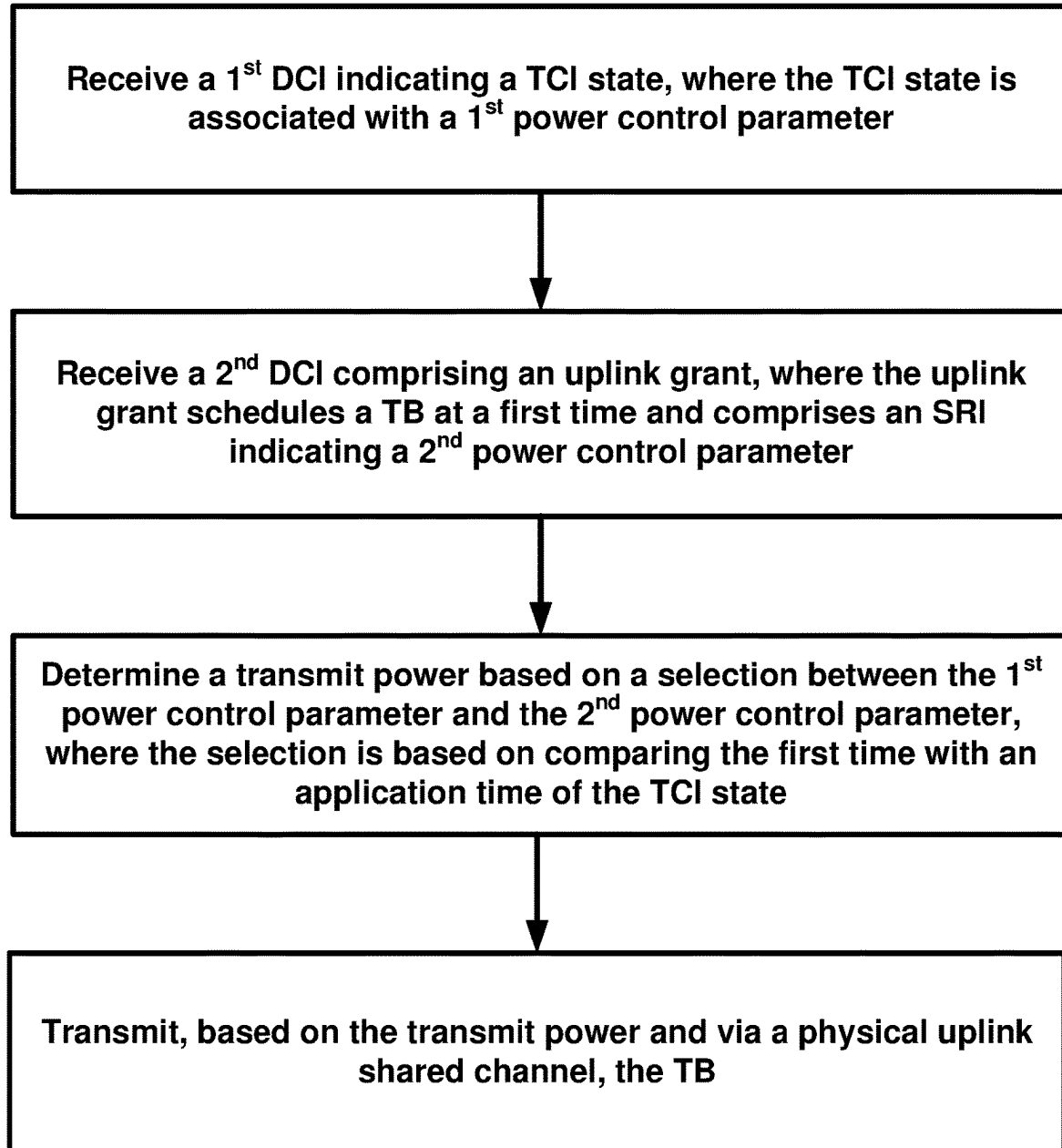


FIG. 29

POWER CONTROL PROCEDURES WITH COMMON BEAMS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/248,095, filed Sep. 24, 2021, which is hereby incorporated by reference in its entirety.

BRIEF DESCRIPTION OF THE DRAWINGS

Examples of several of the various embodiments of the present disclosure are described herein with reference to the drawings.

FIG. 1A and FIG. 1B illustrate example mobile communication networks in which embodiments of the present disclosure may be implemented.

FIG. 2A and FIG. 2B respectively illustrate a New Radio (NR) user plane and control plane protocol stack.

FIG. 3 illustrates an example of services provided between protocol layers of the NR user plane protocol stack of FIG. 2A.

FIG. 4A illustrates an example downlink data flow through the NR user plane protocol stack of FIG. 2A.

FIG. 4B illustrates an example format of a MAC sub-header in a MAC PDU.

FIG. 5A and FIG. 5B respectively illustrate a mapping between logical channels, transport channels, and physical channels for the downlink and uplink.

FIG. 6 is an example diagram showing RRC state transitions of a UE.

FIG. 7 illustrates an example configuration of an NR frame into which OFDM symbols are grouped.

FIG. 8 illustrates an example configuration of a slot in the time and frequency domain for an NR carrier.

FIG. 9 illustrates an example of bandwidth adaptation using three configured BWPs for an NR carrier.

FIG. 10A illustrates three carrier aggregation configurations with two component carriers.

FIG. 10B illustrates an example of how aggregated cells may be configured into one or more PUCCH groups.

FIG. 11A illustrates an example of an SS/PBCH block structure and location.

FIG. 11B illustrates an example of CSI-RSs that are mapped in the time and frequency domains.

FIG. 12A and FIG. 12B respectively illustrate examples of three downlink and uplink beam management procedures.

FIG. 13A, FIG. 13B, and FIG. 13C respectively illustrate a four-step contention-based random access procedure, a two-step contention-free random access procedure, and another two-step random access procedure.

FIG. 14A illustrates an example of CORESET configurations for a bandwidth part.

FIG. 14B illustrates an example of a CCE-to-REG mapping for DCI transmission on a CORESET and PDCCH processing.

FIG. 15 illustrates an example of a wireless device in communication with a base station.

FIG. 16A, FIG. 16B, FIG. 16C, and FIG. 16D illustrate example structures for uplink and downlink transmission.

FIG. 17 illustrates joint TCI states and separate TCI states as per an aspect of an example embodiment of the present disclosure.

FIG. 18A illustrates association between separate TCI states and power control parameters as per an aspect of an example embodiment of the present disclosure.

FIG. 18B illustrates association between joint TCI states and power control parameters as per an aspect of an example embodiment of the present disclosure.

FIG. 19 illustrates indications of joint TCI states and/or separate TCI states as per an aspect of an example embodiment of the present disclosure.

FIG. 20 illustrates a configuration of power control parameters with SRI as per an aspect of an example embodiment of the present disclosure.

FIG. 21 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 22 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 23 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 24 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 25 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 26 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 27 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 28 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

FIG. 29 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure.

DETAILED DESCRIPTION

In the present disclosure, various embodiments are presented as examples of how the disclosed techniques may be implemented and/or how the disclosed techniques may be practiced in environments and scenarios. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the scope. In fact, after reading the description, it will be apparent to one skilled in the relevant art how to implement alternative embodiments. The present embodiments should not be limited by any of the described exemplary embodiments. The embodiments of the present disclosure will be described with reference to the accompanying drawings. Limitations, features, and/or elements from the disclosed example embodiments may be combined to create further embodiments within the scope of the disclosure. Any figures which highlight the functionality and advantages, are presented for example purposes only. The disclosed architecture is sufficiently flexible and configurable, such that it may be utilized in ways other than that shown. For example, the actions listed in any flowchart may be re-ordered or only optionally used in some embodiments.

Embodiments may be configured to operate as needed. The disclosed mechanism may be performed when certain criteria are met, for example, in a wireless device, a base station, a radio environment, a network, a combination of the above, and/or the like. Example criteria may be based, at least in part, on for example, wireless device or network node configurations, traffic load, initial system set up, packet

sizes, traffic characteristics, a combination of the above, and/or the like. When the one or more criteria are met, various example embodiments may be applied. Therefore, it may be possible to implement example embodiments that selectively implement disclosed protocols.

A base station may communicate with a mix of wireless devices. Wireless devices and/or base stations may support multiple technologies, and/or multiple releases of the same technology. Wireless devices may have some specific capability(ies) depending on wireless device category and/or capability(ies). When this disclosure refers to a base station communicating with a plurality of wireless devices, this disclosure may refer to a subset of the total wireless devices in a coverage area. This disclosure may refer to, for example, a plurality of wireless devices of a given LTE or 5G release with a given capability and in a given sector of the base station. The plurality of wireless devices in this disclosure may refer to a selected plurality of wireless devices, and/or a subset of total wireless devices in a coverage area which perform according to disclosed methods, and/or the like. There may be a plurality of base stations or a plurality of wireless devices in a coverage area that may not comply with the disclosed methods, for example, those wireless devices or base stations may perform based on older releases of LTE or 5G technology.

In this disclosure, “a” and “an” and similar phrases are to be interpreted as “at least one” and “one or more.” Similarly, any term that ends with the suffix “(s)” is to be interpreted as “at least one” and “one or more.” In this disclosure, the term “may” is to be interpreted as “may, for example.” In other words, the term “may” is indicative that the phrase following the term “may” is an example of one of a multitude of suitable possibilities that may, or may not, be employed by one or more of the various embodiments. The terms “comprises” and “consists of”, as used herein, enumerate one or more components of the element being described. The term “comprises” is interchangeable with “includes” and does not exclude unenumerated components from being included in the element being described. By contrast, “consists of” provides a complete enumeration of the one or more components of the element being described. The term “based on”, as used herein, should be interpreted as “based at least in part on” rather than, for example, “based solely on”. The term “and/or” as used herein represents any possible combination of enumerated elements. For example, “A, B, and/or C” may represent A; B; C; A and B; A and C; B and C; or A, B, and C.

If A and B are sets and every element of A is an element of B, A is called a subset of B. In this specification, only non-empty sets and subsets are considered. For example, possible subsets of B={cell1, cell2} are: {cell1}, {cell2}, and {cell1, cell2}. The phrase “based on” (or equally “based at least on”) is indicative that the phrase following the term “based on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “in response to” (or equally “in response at least to”) is indicative that the phrase following the phrase “in response to” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “depending on” (or equally “depending at least to”) is indicative that the phrase following the phrase “depending on” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments. The phrase “employing/using” (or equally “employing/using at least”) is indicative that the phrase following the phrase

“employing/using” is an example of one of a multitude of suitable possibilities that may, or may not, be employed to one or more of the various embodiments.

The term configured may relate to the capacity of a device whether the device is in an operational or non-operational state. Configured may refer to specific settings in a device that effect the operational characteristics of the device whether the device is in an operational or non-operational state. In other words, the hardware, software, firmware, registers, memory values, and/or the like may be “configured” within a device, whether the device is in an operational or nonoperational state, to provide the device with specific characteristics. Terms such as “a control message to cause in a device” may mean that a control message has parameters that may be used to configure specific characteristics or may be used to implement certain actions in the device, whether the device is in an operational or non-operational state.

In this disclosure, parameters (or equally called, fields, or Information elements: IEs) may comprise one or more information objects, and an information object may comprise one or more other objects. For example, if parameter (IE) N comprises parameter (IE) M, and parameter (IE) M comprises parameter (IE) K, and parameter (IE) K comprises parameter (information element) J. Then, for example, N comprises K, and N comprises J. In an example embodiment, when one or more messages comprise a plurality of parameters, it implies that a parameter in the plurality of parameters is in at least one of the one or more messages, but does not have to be in each of the one or more messages.

Many features presented are described as being optional through the use of “may” or the use of parentheses. For the sake of brevity and legibility, the present disclosure does not explicitly recite each and every permutation that may be obtained by choosing from the set of optional features. The present disclosure is to be interpreted as explicitly disclosing all such permutations. For example, a system described as having three optional features may be embodied in seven ways, namely with just one of the three possible features, with any two of the three possible features or with three of the three possible features.

Many of the elements described in the disclosed embodiments may be implemented as modules. A module is defined here as an element that performs a defined function and has a defined interface to other elements. The modules described in this disclosure may be implemented in hardware, software in combination with hardware, firmware, wetware (e.g. hardware with a biological element) or a combination thereof, which may be behaviorally equivalent. For example, modules may be implemented as a software routine written in a computer language configured to be executed by a hardware machine (such as C, C++, Fortran, Java, Basic, Matlab or the like) or a modeling/simulation program such as Simulink, Stateflow, GNU Octave, or Lab VIEWMathScript. It may be possible to implement modules using physical hardware that incorporates discrete or programmable analog, digital and/or quantum hardware. Examples of programmable hardware comprise: computers, microcontrollers, microprocessors, application-specific integrated circuits (ASICs); field programmable gate arrays (FPGAs); and complex programmable logic devices (CPLDs). Computers, microcontrollers and microprocessors are programmed using languages such as assembly, C, C++ or the like. FPGAs, ASICs and CPLDs are often programmed using hardware description languages (HDL) such as VHSIC hardware description language (VHDL) or Ver-

ilog that configure connections between internal hardware modules with lesser functionality on a programmable device. The mentioned technologies are often used in combination to achieve the result of a functional module.

FIG. 1A illustrates an example of a mobile communication network **100** in which embodiments of the present disclosure may be implemented. The mobile communication network **100** may be, for example, a public land mobile network (PLMN) run by a network operator. As illustrated in FIG. 1A, the mobile communication network **100** includes a core network (CN) **102**, a radio access network (RAN) **104**, and a wireless device **106**.

The CN **102** may provide the wireless device **106** with an interface to one or more data networks (DNs), such as public DNs (e.g., the Internet), private DNs, and/or intra-operator DNs. As part of the interface functionality, the CN **102** may set up end-to-end connections between the wireless device **106** and the one or more DNs, authenticate the wireless device **106**, and provide charging functionality.

The RAN **104** may connect the CN **102** to the wireless device **106** through radio communications over an air interface. As part of the radio communications, the RAN **104** may provide scheduling, radio resource management, and retransmission protocols. The communication direction from the RAN **104** to the wireless device **106** over the air interface is known as the downlink and the communication direction from the wireless device **106** to the RAN **104** over the air interface is known as the uplink. Downlink transmissions may be separated from uplink transmissions using frequency division duplexing (FDD), time-division duplexing (TDD), and/or some combination of the two duplexing techniques.

The term wireless device may be used throughout this disclosure to refer to and encompass any mobile device or fixed (non-mobile) device for which wireless communication is needed or usable. For example, a wireless device may be a telephone, smart phone, tablet, computer, laptop, sensor, meter, wearable device, Internet of Things (IoT) device, vehicle road side unit (RSU), relay node, automobile, and/or any combination thereof. The term wireless device encompasses other terminology, including user equipment (UE), user terminal (UT), access terminal (AT), mobile station, handset, wireless transmit and receive unit (WTRU), and/or wireless communication device.

The RAN **104** may include one or more base stations (not shown). The term base station may be used throughout this disclosure to refer to and encompass a Node B (associated with UMTS and/or 3G standards), an Evolved Node B (eNB, associated with E-UTRA and/or 4G standards), a remote radio head (RRH), a baseband processing unit coupled to one or more RRHs, a repeater node or relay node used to extend the coverage area of a donor node, a Next Generation Evolved Node B (ng-eNB), a Generation Node B (gNB, associated with NR and/or 5G standards), an access point (AP, associated with, for example, WiFi or any other suitable wireless communication standard), and/or any combination thereof. A base station may comprise at least one gNB Central Unit (gNB-CU) and at least one gNB Distributed Unit (gNB-DU).

A base station included in the RAN **104** may include one or more sets of antennas for communicating with the wireless device **106** over the air interface. For example, one or more of the base stations may include three sets of antennas to respectively control three cells (or sectors). The size of a cell may be determined by a range at which a receiver (e.g., a base station receiver) can successfully receive the transmissions from a transmitter (e.g., a wireless device trans-

mitter) operating in the cell. Together, the cells of the base stations may provide radio coverage to the wireless device **106** over a wide geographic area to support wireless device mobility.

In addition to three-sector sites, other implementations of base stations are possible. For example, one or more of the base stations in the RAN **104** may be implemented as a sectorized site with more or less than three sectors. One or more of the base stations in the RAN **104** may be implemented as an access point, as a baseband processing unit coupled to several remote radio heads (RRHs), and/or as a repeater or relay node used to extend the coverage area of a donor node. A baseband processing unit coupled to RRHs may be part of a centralized or cloud RAN architecture, where the baseband processing unit may be either centralized in a pool of baseband processing units or virtualized. A repeater node may amplify and rebroadcast a radio signal received from a donor node. A relay node may perform the same/similar functions as a repeater node but may decode the radio signal received from the donor node to remove noise before amplifying and rebroadcasting the radio signal.

The RAN **104** may be deployed as a homogenous network of macrocell base stations that have similar antenna patterns and similar high-level transmit powers. The RAN **104** may be deployed as a heterogeneous network. In heterogeneous networks, small cell base stations may be used to provide small coverage areas, for example, coverage areas that overlap with the comparatively larger coverage areas provided by macrocell base stations. The small coverage areas may be provided in areas with high data traffic (or so-called "hotspots") or in areas with weak macrocell coverage. Examples of small cell base stations include, in order of decreasing coverage area, microcell base stations, picocell base stations, and femtocell base stations or home base stations.

The Third-Generation Partnership Project (3GPP) was formed in 1998 to provide global standardization of specifications for mobile communication networks similar to the mobile communication network **100** in FIG. 1A. To date, 3GPP has produced specifications for three generations of mobile networks: a third generation (3G) network known as Universal Mobile Telecommunications System (UMTS), a fourth generation (4G) network known as Long-Term Evolution (LTE), and a fifth generation (5G) network known as 5G System (5GS). Embodiments of the present disclosure are described with reference to the RAN of a 3GPP 5G network, referred to as next-generation RAN (NG-RAN). Embodiments may be applicable to RANs of other mobile communication networks, such as the RAN **104** in FIG. 1A, the RANs of earlier 3G and 4G networks, and those of future networks yet to be specified (e.g., a 3GPP 6G network). NG-RAN implements 5G radio access technology known as New Radio (NR) and may be provisioned to implement 4G radio access technology or other radio access technologies, including non-3GPP radio access technologies.

FIG. 1B illustrates another example mobile communication network **150** in which embodiments of the present disclosure may be implemented. Mobile communication network **150** may be, for example, a PLMN run by a network operator. As illustrated in FIG. 1B, mobile communication network **150** includes a 5G core network (5G-CN) **152**, an NG-RAN **154**, and UEs **156A** and **156B** (collectively UEs **156**). These components may be implemented and operate in the same or similar manner as corresponding components described with respect to FIG. 1A.

The 5G-CN **152** provides the UEs **156** with an interface to one or more DNs, such as public DNs (e.g., the Internet),

private DNs, and/or intra-operator DNs. As part of the interface functionality, the 5G-CN **152** may set up end-to-end connections between the UEs **156** and the one or more DNs, authenticate the UEs **156**, and provide charging functionality. Compared to the CN of a 3GPP 4G network, the basis of the 5G-CN **152** may be a service-based architecture. This means that the architecture of the nodes making up the 5G-CN **152** may be defined as network functions that offer services via interfaces to other network functions. The network functions of the 5G-CN **152** may be implemented in several ways, including as network elements on dedicated or shared hardware, as software instances running on dedicated or shared hardware, or as virtualized functions instantiated on a platform (e.g., a cloud-based platform).

As illustrated in FIG. 1B, the 5G-CN **152** includes an Access and Mobility Management Function (AMF) **158A** and a User Plane Function (UPF) **158B**, which are shown as one component AMF/UPF **158** in FIG. 1B for ease of illustration. The UPF **158B** may serve as a gateway between the NG-RAN **154** and the one or more DNs. The UPF **158B** may perform functions such as packet routing and forwarding, packet inspection and user plane policy rule enforcement, traffic usage reporting, uplink classification to support routing of traffic flows to the one or more DNs, quality of service (QoS) handling for the user plane (e.g., packet filtering, gating, uplink/downlink rate enforcement, and uplink traffic verification), downlink packet buffering, and downlink data notification triggering. The UPF **158B** may serve as an anchor point for intra-/inter-Radio Access Technology (RAT) mobility, an external protocol (or packet) data unit (PDU) session point of interconnect to the one or more DNs, and/or a branching point to support a multi-homed PDU session. The UEs **156** may be configured to receive services through a PDU session, which is a logical connection between a UE and a DN.

The AMF **158A** may perform functions such as Non-Access Stratum (NAS) signaling termination, NAS signaling security, Access Stratum (AS) security control, inter-CN node signaling for mobility between 3GPP access networks, idle mode UE reachability (e.g., control and execution of paging retransmission), registration area management, intra-system and inter-system mobility support, access authentication, access authorization including checking of roaming rights, mobility management control (subscription and policies), network slicing support, and/or session management function (SMF) selection. NAS may refer to the functionality operating between a CN and a UE, and AS may refer to the functionality operating between the UE and a RAN.

The 5G-CN **152** may include one or more additional network functions that are not shown in FIG. 1B for the sake of clarity. For example, the 5G-CN **152** may include one or more of a Session Management Function (SMF), an NR Repository Function (NRF), a Policy Control Function (PCF), a Network Exposure Function (NEF), a Unified Data Management (UDM), an Application Function (AF), and/or an Authentication Server Function (AUSF).

The NG-RAN **154** may connect the 5G-CN **152** to the UEs **156** through radio communications over the air interface. The NG-RAN **154** may include one or more gNBs, illustrated as gNB **160A** and gNB **160B** (collectively gNBs **160**) and/or one or more ng-eNBs, illustrated as ng-eNB **162A** and ng-eNB **162B** (collectively ng-eNBs **162**). The gNBs **160** and ng-eNBs **162** may be more generically referred to as base stations. The gNBs **160** and ng-eNBs **162** may include one or more sets of antennas for communicating with the UEs **156** over an air interface. For example, one or more of the gNBs **160** and/or one or more of the ng-eNBs

162 may include three sets of antennas to respectively control three cells (or sectors). Together, the cells of the gNBs **160** and the ng-eNBs **162** may provide radio coverage to the UEs **156** over a wide geographic area to support UE mobility.

As shown in FIG. 1B, the gNBs **160** and/or the ng-eNBs **162** may be connected to the 5G-CN **152** by means of an NG interface and to other base stations by an Xn interface. The NG and Xn interfaces may be established using direct physical connections and/or indirect connections over an underlying transport network, such as an internet protocol (IP) transport network. The gNBs **160** and/or the ng-eNBs **162** may be connected to the UEs **156** by means of a Uu interface. For example, as illustrated in FIG. 1B, gNB **160A** may be connected to the UE **156A** by means of a Uu interface. The NG, Xn, and Uu interfaces are associated with a protocol stack. The protocol stacks associated with the interfaces may be used by the network elements in FIG. 1B to exchange data and signaling messages and may include two planes: a user plane and a control plane. The user plane may handle data of interest to a user. The control plane may handle signaling messages of interest to the network elements.

The gNBs **160** and/or the ng-eNBs **162** may be connected to one or more AMF/UPF functions of the 5G-CN **152**, such as the AMF/UPF **158**, by means of one or more NG interfaces. For example, the gNB **160A** may be connected to the UPF **158B** of the AMF/UPF **158** by means of an NG-User plane (NG-U) interface. The NG-U interface may provide delivery (e.g., non-guaranteed delivery) of user plane PDUs between the gNB **160A** and the UPF **158B**. The gNB **160A** may be connected to the AMF **158A** by means of an NG-Control plane (NG-C) interface. The NG-C interface may provide, for example, NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, and configuration transfer and/or warning message transmission.

The gNBs **160** may provide NR user plane and control plane protocol terminations towards the UEs **156** over the Uu interface. For example, the gNB **160A** may provide NR user plane and control plane protocol terminations toward the UE **156A** over a Uu interface associated with a first protocol stack. The ng-eNBs **162** may provide Evolved UMTS Terrestrial Radio Access (E-UTRA) user plane and control plane protocol terminations towards the UEs **156** over a Uu interface, where E-UTRA refers to the 3GPP 4G radio-access technology. For example, the ng-eNB **162B** may provide E-UTRA user plane and control plane protocol terminations towards the UE **156B** over a Uu interface associated with a second protocol stack.

The 5G-CN **152** was described as being configured to handle NR and 4G radio accesses. It will be appreciated by one of ordinary skill in the art that it may be possible for NR to connect to a 4G core network in a mode known as “non-standalone operation.” In non-standalone operation, a 4G core network is used to provide (or at least support) control-plane functionality (e.g., initial access, mobility, and paging). Although only one AMF/UPF **158** is shown in FIG. 1B, one gNB or ng-eNB may be connected to multiple AMF/UPF nodes to provide redundancy and/or to load share across the multiple AMF/UPF nodes.

As discussed, an interface (e.g., Uu, Xn, and NG interfaces) between the network elements in FIG. 1B may be associated with a protocol stack that the network elements use to exchange data and signaling messages. A protocol stack may include two planes: a user plane and a control plane. The user plane may handle data of interest to a user,

and the control plane may handle signaling messages of interest to the network elements.

FIG. 2A and FIG. 2B respectively illustrate examples of NR user plane and NR control plane protocol stacks for the Uu interface that lies between a UE 210 and a gNB 220. The protocol stacks illustrated in FIG. 2A and FIG. 2B may be the same or similar to those used for the Uu interface between, for example, the UE 156A and the gNB 160A shown in FIG. 1B.

FIG. 2A illustrates a NR user plane protocol stack comprising five layers implemented in the UE 210 and the gNB 220. At the bottom of the protocol stack, physical layers (PHYs) 211 and 221 may provide transport services to the higher layers of the protocol stack and may correspond to layer 1 of the Open Systems Interconnection (OSI) model. The next four protocols above PHYs 211 and 221 comprise media access control layers (MACs) 212 and 222, radio link control layers (RLCs) 213 and 223, packet data convergence protocol layers (PDCPs) 214 and 224, and service data application protocol layers (SDAPs) 215 and 225. Together, these four protocols may make up layer 2, or the data link layer, of the OSI model.

FIG. 3 illustrates an example of services provided between protocol layers of the NR user plane protocol stack. Starting from the top of FIG. 2A and FIG. 3, the SDAPs 215 and 225 may perform QoS flow handling. The UE 210 may receive services through a PDU session, which may be a logical connection between the UE 210 and a DN. The PDU session may have one or more QoS flows. A UPF of a CN (e.g., the UPF 158B) may map IP packets to the one or more QoS flows of the PDU session based on QoS requirements (e.g., in terms of delay, data rate, and/or error rate). The SDAPs 215 and 225 may perform mapping/de-mapping between the one or more QoS flows and one or more data radio bearers. The mapping/de-mapping between the QoS flows and the data radio bearers may be determined by the SDAP 225 at the gNB 220. The SDAP 215 at the UE 210 may be informed of the mapping between the QoS flows and the data radio bearers through reflective mapping or control signaling received from the gNB 220. For reflective mapping, the SDAP 225 at the gNB 220 may mark the downlink packets with a QoS flow indicator (QFI), which may be observed by the SDAP 215 at the UE 210 to determine the mapping/de-mapping between the QoS flows and the data radio bearers.

The PDCPs 214 and 224 may perform header compression/decompression to reduce the amount of data that needs to be transmitted over the air interface, ciphering/deciphering to prevent unauthorized decoding of data transmitted over the air interface, and integrity protection (to ensure control messages originate from intended sources). The PDCPs 214 and 224 may perform retransmissions of undelivered packets, in-sequence delivery and reordering of packets, and removal of packets received in duplicate due to, for example, an intra-gNB handover. The PDCPs 214 and 224 may perform packet duplication to improve the likelihood of the packet being received and, at the receiver, remove any duplicate packets. Packet duplication may be useful for services that require high reliability.

Although not shown in FIG. 3, PDCPs 214 and 224 may perform mapping/de-mapping between a split radio bearer and RLC channels in a dual connectivity scenario. Dual connectivity is a technique that allows a UE to connect to two cells or, more generally, two cell groups: a master cell group (MCG) and a secondary cell group (SCG). A split bearer is when a single radio bearer, such as one of the radio bearers provided by the PDCPs 214 and 224 as a service to

the SDAPs 215 and 225, is handled by cell groups in dual connectivity. The PDCPs 214 and 224 may map/de-map the split radio bearer between RLC channels belonging to cell groups.

The RLCs 213 and 223 may perform segmentation, retransmission through Automatic Repeat Request (ARQ), and removal of duplicate data units received from MACs 212 and 222, respectively. The RLCs 213 and 223 may support three transmission modes: transparent mode (TM); unacknowledged mode (UM); and acknowledged mode (AM). Based on the transmission mode an RLC is operating, the RLC may perform one or more of the noted functions. The RLC configuration may be per logical channel with no dependency on numerologies and/or Transmission Time Interval (TTI) durations. As shown in FIG. 3, the RLCs 213 and 223 may provide RLC channels as a service to PDCPs 214 and 224, respectively.

The MACs 212 and 222 may perform multiplexing/demultiplexing of logical channels and/or mapping between logical channels and transport channels. The multiplexing/demultiplexing may include multiplexing/demultiplexing of data units, belonging to the one or more logical channels, into/from Transport Blocks (TBs) delivered to/from the PHYs 211 and 221. The MAC 222 may be configured to perform scheduling, scheduling information reporting, and priority handling between UEs by means of dynamic scheduling. Scheduling may be performed in the gNB 220 (at the MAC 222) for downlink and uplink. The MACs 212 and 222 may be configured to perform error correction through Hybrid Automatic Repeat Request (HARQ) (e.g., one HARQ entity per carrier in case of Carrier Aggregation (CA)), priority handling between logical channels of the UE 210 by means of logical channel prioritization, and/or padding. The MACs 212 and 222 may support one or more numerologies and/or transmission timings. In an example, mapping restrictions in a logical channel prioritization may control which numerology and/or transmission timing a logical channel may use. As shown in FIG. 3, the MACs 212 and 222 may provide logical channels as a service to the RLCs 213 and 223.

The PHYs 211 and 221 may perform mapping of transport channels to physical channels and digital and analog signal processing functions for sending and receiving information over the air interface. These digital and analog signal processing functions may include, for example, coding/decoding and modulation/demodulation. The PHYs 211 and 221 may perform multi-antenna mapping. As shown in FIG. 3, the PHYs 211 and 221 may provide one or more transport channels as a service to the MACs 212 and 222.

FIG. 4A illustrates an example downlink data flow through the NR user plane protocol stack. FIG. 4A illustrates a downlink data flow of three IP packets (n, n+1, and m) through the NR user plane protocol stack to generate two TBs at the gNB 220. An uplink data flow through the NR user plane protocol stack may be similar to the downlink data flow depicted in FIG. 4A.

The downlink data flow of FIG. 4A begins when SDAP 225 receives the three IP packets from one or more QoS flows and maps the three packets to radio bearers. In FIG. 4A, the SDAP 225 maps IP packets n and n+1 to a first radio bearer 402 and maps IP packet m to a second radio bearer 404. An SDAP header (labeled with an "H" in FIG. 4A) is added to an IP packet. The data unit from/to a higher protocol layer is referred to as a service data unit (SDU) of the lower protocol layer and the data unit to/from a lower protocol layer is referred to as a protocol data unit (PDU) of the higher protocol layer. As shown in FIG. 4A, the data unit

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from the SDAP 225 is an SDU of lower protocol layer PDCP 224 and is a PDU of the SDAP 225.

The remaining protocol layers in FIG. 4A may perform their associated functionality (e.g., with respect to FIG. 3), add corresponding headers, and forward their respective outputs to the next lower layer. For example, the PDCP 224 may perform IP-header compression and ciphering and forward its output to the RLC 223. The RLC 223 may optionally perform segmentation (e.g., as shown for IP packet m in FIG. 4A) and forward its output to the MAC 222. The MAC 222 may multiplex a number of RLC PDUs and may attach a MAC subheader to an RLC PDU to form a transport block. In NR, the MAC subheaders may be distributed across the MAC PDU, as illustrated in FIG. 4A. In LTE, the MAC subheaders may be entirely located at the beginning of the MAC PDU. The NR MAC PDU structure may reduce processing time and associated latency because the MAC PDU subheaders may be computed before the full MAC PDU is assembled.

FIG. 4B illustrates an example format of a MAC subheader in a MAC PDU. The MAC subheader includes: an SDU length field for indicating the length (e.g., in bytes) of the MAC SDU to which the MAC subheader corresponds; a logical channel identifier (LCID) field for identifying the logical channel from which the MAC SDU originated to aid in the demultiplexing process; a flag (F) for indicating the size of the SDU length field; and a reserved bit (R) field for future use.

FIG. 4B further illustrates MAC control elements (CEs) inserted into the MAC PDU by a MAC, such as MAC 223 or MAC 222. For example, FIG. 4B illustrates two MAC CEs inserted into the MAC PDU. MAC CEs may be inserted at the beginning of a MAC PDU for downlink transmissions (as shown in FIG. 4B) and at the end of a MAC PDU for uplink transmissions. MAC CEs may be used for in-band control signaling. Example MAC CEs include: scheduling-related MAC CEs, such as buffer status reports and power headroom reports; activation/deactivation MAC CEs, such as those for activation/deactivation of PDCP duplication detection, channel state information (CSI) reporting, sounding reference signal (SRS) transmission, and prior configured components; discontinuous reception (DRX) related MAC CEs; timing advance MAC CEs; and random access related MAC CEs. A MAC CE may be preceded by a MAC subheader with a similar format as described for MAC SDUs and may be identified with a reserved value in the LCID field that indicates the type of control information included in the MAC CE.

Before describing the NR control plane protocol stack, logical channels, transport channels, and physical channels are first described as well as a mapping between the channel types. One or more of the channels may be used to carry out functions associated with the NR control plane protocol stack described later below.

FIG. 5A and FIG. 5B illustrate, for downlink and uplink respectively, a mapping between logical channels, transport channels, and physical channels. Information is passed through channels between the RLC, the MAC, and the PHY of the NR protocol stack. A logical channel may be used between the RLC and the MAC and may be classified as a control channel that carries control and configuration information in the NR control plane or as a traffic channel that carries data in the NR user plane. A logical channel may be classified as a dedicated logical channel that is dedicated to a specific UE or as a common logical channel that may be used by more than one UE. A logical channel may also be

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defined by the type of information it carries. The set of logical channels defined by NR include, for example:

- a paging control channel (PCCH) for carrying paging messages used to page a UE whose location is not known to the network on a cell level;
- a broadcast control channel (BCCH) for carrying system information messages in the form of a master information block (MIB) and several system information blocks (SIBs), wherein the system information messages may be used by the UEs to obtain information about how a cell is configured and how to operate within the cell;
- a common control channel (CCCH) for carrying control messages together with random access;
- a dedicated control channel (DCCH) for carrying control messages to/from a specific the UE to configure the UE; and
- a dedicated traffic channel (DTCH) for carrying user data to/from a specific the UE.

Transport channels are used between the MAC and PHY layers and may be defined by how the information they carry is transmitted over the air interface. The set of transport channels defined by NR include, for example:

- a paging channel (PCH) for carrying paging messages that originated from the PCCH;
- a broadcast channel (BCH) for carrying the MIB from the BCCH;
- a downlink shared channel (DL-SCH) for carrying downlink data and signaling messages, including the SIBs from the BCCH;
- an uplink shared channel (UL-SCH) for carrying uplink data and signaling messages; and
- a random access channel (RACH) for allowing a UE to contact the network without any prior scheduling.

The PHY may use physical channels to pass information between processing levels of the PHY. A physical channel may have an associated set of time-frequency resources for carrying the information of one or more transport channels. The PHY may generate control information to support the low-level operation of the PHY and provide the control information to the lower levels of the PHY via physical control channels, known as L1/L2 control channels. The set of physical channels and physical control channels defined by NR include, for example:

- a physical broadcast channel (PBCH) for carrying the MIB from the BCH;
- a physical downlink shared channel (PDSCH) for carrying downlink data and signaling messages from the DL-SCH, as well as paging messages from the PCH;
- a physical downlink control channel (PDCCH) for carrying downlink control information (DCI), which may include downlink scheduling commands, uplink scheduling grants, and uplink power control commands;
- a physical uplink shared channel (PUSCH) for carrying uplink data and signaling messages from the UL-SCH and in some instances uplink control information (UCI) as described below;
- a physical uplink control channel (PUCCH) for carrying UCI, which may include HARQ acknowledgments, channel quality indicators (CQI), pre-coding matrix indicators (PMI), rank indicators (RI), and scheduling requests (SR); and
- a physical random access channel (PRACH) for random access.

Similar to the physical control channels, the physical layer generates physical signals to support the low-level operation of the physical layer. As shown in FIG. 5A and

FIG. 5B, the physical layer signals defined by NR include: primary synchronization signals (PSS), secondary synchronization signals (SSS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), sounding reference signals (SRS), and phase-tracking reference signals (PT-RS). These physical layer signals will be described in greater detail below.

FIG. 2B illustrates an example NR control plane protocol stack. As shown in FIG. 2B, the NR control plane protocol stack may use the same/similar first four protocol layers as the example NR user plane protocol stack. These four protocol layers include the PHYs **211** and **221**, the MACs **212** and **222**, the RLCs **213** and **223**, and the PDCPs **214** and **224**. Instead of having the SDAPs **215** and **225** at the top of the stack as in the NR user plane protocol stack, the NR control plane stack has radio resource controls (RRCs) **216** and **226** and NAS protocols **217** and **237** at the top of the NR control plane protocol stack.

The NAS protocols **217** and **237** may provide control plane functionality between the UE **210** and the AMF **230** (e.g., the AMF **158A**) or, more generally, between the UE **210** and the CN. The NAS protocols **217** and **237** may provide control plane functionality between the UE **210** and the AMF **230** via signaling messages, referred to as NAS messages. There is no direct path between the UE **210** and the AMF **230** through which the NAS messages can be transported. The NAS messages may be transported using the AS of the Uu and NG interfaces. NAS protocols **217** and **237** may provide control plane functionality such as authentication, security, connection setup, mobility management, and session management.

The RRCs **216** and **226** may provide control plane functionality between the UE **210** and the gNB **220** or, more generally, between the UE **210** and the RAN. The RRCs **216** and **226** may provide control plane functionality between the UE **210** and the gNB **220** via signaling messages, referred to as RRC messages. RRC messages may be transmitted between the UE **210** and the RAN using signaling radio bearers and the same/similar PDCP, RLC, MAC, and PHY protocol layers. The MAC may multiplex control-plane and user-plane data into the same transport block (TB). The RRCs **216** and **226** may provide control plane functionality such as: broadcast of system information related to AS and NAS; paging initiated by the CN or the RAN; establishment, maintenance and release of an RRC connection between the UE **210** and the RAN; security functions including key management; establishment, configuration, maintenance and release of signaling radio bearers and data radio bearers; mobility functions; QoS management functions; the UE measurement reporting and control of the reporting; detection of and recovery from radio link failure (RLF); and/or NAS message transfer. As part of establishing an RRC connection, RRCs **216** and **226** may establish an RRC context, which may involve configuring parameters for communication between the UE **210** and the RAN.

FIG. 6 is an example diagram showing RRC state transitions of a UE. The UE may be the same or similar to the wireless device **106** depicted in FIG. 1A, the UE **210** depicted in FIG. 2A and FIG. 2B, or any other wireless device described in the present disclosure. As illustrated in FIG. 6, a UE may be in at least one of three RRC states: RRC connected **602** (e.g., RRC_CONNECTED), RRC idle **604** (e.g., RRC_IDLE), and RRC inactive **606** (e.g., RRC_INACTIVE).

In RRC connected **602**, the UE has an established RRC context and may have at least one RRC connection with a base station. The base station may be similar to one of the

one or more base stations included in the RAN **104** depicted in FIG. 1A, one of the gNBs **160** or ng-eNBs **162** depicted in FIG. 1B, the gNB **220** depicted in FIG. 2A and FIG. 2B, or any other base station described in the present disclosure. The base station with which the UE is connected may have the RRC context for the UE. The RRC context, referred to as the UE context, may comprise parameters for communication between the UE and the base station. These parameters may include, for example: one or more AS contexts; one or more radio link configuration parameters; bearer configuration information (e.g., relating to a data radio bearer, signaling radio bearer, logical channel, QoS flow, and/or PDU session); security information; and/or PHY, MAC, RLC, PDCP, and/or SDAP layer configuration information. While in RRC connected **602**, mobility of the UE may be managed by the RAN (e.g., the RAN **104** or the NG-RAN **154**). The UE may measure the signal levels (e.g., reference signal levels) from a serving cell and neighboring cells and report these measurements to the base station currently serving the UE. The UE's serving base station may request a handover to a cell of one of the neighboring base stations based on the reported measurements. The RRC state may transition from RRC connected **602** to RRC idle **604** through a connection release procedure **608** or to RRC inactive **606** through a connection inactivation procedure **610**.

In RRC idle **604**, an RRC context may not be established for the UE. In RRC idle **604**, the UE may not have an RRC connection with the base station. While in RRC idle **604**, the UE may be in a sleep state for the majority of the time (e.g., to conserve battery power). The UE may wake up periodically (e.g., once in every discontinuous reception cycle) to monitor for paging messages from the RAN. Mobility of the UE may be managed by the UE through a procedure known as cell reselection. The RRC state may transition from RRC idle **604** to RRC connected **602** through a connection establishment procedure **612**, which may involve a random access procedure as discussed in greater detail below.

In RRC inactive **606**, the RRC context previously established is maintained in the UE and the base station. This allows for a fast transition to RRC connected **602** with reduced signaling overhead as compared to the transition from RRC idle **604** to RRC connected **602**. While in RRC inactive **606**, the UE may be in a sleep state and mobility of the UE may be managed by the UE through cell reselection. The RRC state may transition from RRC inactive **606** to RRC connected **602** through a connection resume procedure **614** or to RRC idle **604** through a connection release procedure **616** that may be the same as or similar to connection release procedure **608**.

An RRC state may be associated with a mobility management mechanism. In RRC idle **604** and RRC inactive **606**, mobility is managed by the UE through cell reselection. The purpose of mobility management in RRC idle **604** and RRC inactive **606** is to allow the network to be able to notify the UE of an event via a paging message without having to broadcast the paging message over the entire mobile communications network. The mobility management mechanism used in RRC idle **604** and RRC inactive **606** may allow the network to track the UE on a cell-group level so that the paging message may be broadcast over the cells of the cell group that the UE currently resides within instead of the entire mobile communication network. The mobility management mechanisms for RRC idle **604** and RRC inactive **606** track the UE on a cell-group level. They may do so using different granularities of grouping. For example, there may be three levels of cell-grouping granularity: individual

cells; cells within a RAN area identified by a RAN area identifier (RAI); and cells within a group of RAN areas, referred to as a tracking area and identified by a tracking area identifier (TAI).

Tracking areas may be used to track the UE at the CN level. The CN (e.g., the CN 102 or the 5G-CN 152) may provide the UE with a list of TAIs associated with a UE registration area. If the UE moves, through cell reselection, to a cell associated with a TAI not included in the list of TAIs associated with the UE registration area, the UE may perform a registration update with the CN to allow the CN to update the UE's location and provide the UE with a new the UE registration area.

RAN areas may be used to track the UE at the RAN level. For a UE in RRC inactive 606 state, the UE may be assigned a RAN notification area. A RAN notification area may comprise one or more cell identities, a list of RAIs, or a list of TAIs. In an example, a base station may belong to one or more RAN notification areas. In an example, a cell may belong to one or more RAN notification areas. If the UE moves, through cell reselection, to a cell not included in the RAN notification area assigned to the UE, the UE may perform a notification area update with the RAN to update the UE's RAN notification area.

A base station storing an RRC context for a UE or a last serving base station of the UE may be referred to as an anchor base station. An anchor base station may maintain an RRC context for the UE at least during a period of time that the UE stays in a RAN notification area of the anchor base station and/or during a period of time that the UE stays in RRC inactive 606.

A gNB, such as gNBs 160 in FIG. 1B, may be split in two parts: a central unit (gNB-CU), and one or more distributed units (gNB-DU). A gNB-CU may be coupled to one or more gNB-DUs using an F1 interface. The gNB-CU may comprise the RRC, the PDCP, and the SDAP. A gNB-DU may comprise the RLC, the MAC, and the PHY.

In NR, the physical signals and physical channels (discussed with respect to FIG. 5A and FIG. 5B) may be mapped onto orthogonal frequency divisional multiplexing (OFDM) symbols. OFDM is a multicarrier communication scheme that transmits data over F orthogonal subcarriers (or tones). Before transmission, the data may be mapped to a series of complex symbols (e.g., M-quadrature amplitude modulation (M-QAM) or M-phase shift keying (M-PSK) symbols), referred to as source symbols, and divided into F parallel symbol streams. The F parallel symbol streams may be treated as though they are in the frequency domain and used as inputs to an Inverse Fast Fourier Transform (IFFT) block that transforms them into the time domain. The IFFT block may take in F source symbols at a time, one from each of the F parallel symbol streams, and use each source symbol to modulate the amplitude and phase of one of F sinusoidal basis functions that correspond to the F orthogonal subcarriers. The output of the IFFT block may be F time-domain samples that represent the summation of the F orthogonal subcarriers. The F time-domain samples may form a single OFDM symbol. After some processing (e.g., addition of a cyclic prefix) and up-conversion, an OFDM symbol provided by the IFFT block may be transmitted over the air interface on a carrier frequency. The F parallel symbol streams may be mixed using an FFT block before being processed by the IFFT block. This operation produces Discrete Fourier Transform (DFT)-precoded OFDM symbols and may be used by UEs in the uplink to reduce the peak to average power ratio (PAPR). Inverse processing may

be performed on the OFDM symbol at a receiver using an FFT block to recover the data mapped to the source symbols.

FIG. 7 illustrates an example configuration of an NR frame into which OFDM symbols are grouped. An NR frame may be identified by a system frame number (SFN). The SFN may repeat with a period of 1024 frames. As illustrated, one NR frame may be 10 milliseconds (ms) in duration and may include 10 subframes that are 1 ms in duration. A subframe may be divided into slots that include, for example, 14 OFDM symbols per slot.

The duration of a slot may depend on the numerology used for the OFDM symbols of the slot. In NR, a flexible numerology is supported to accommodate different cell deployments (e.g., cells with carrier frequencies below 1 GHz up to cells with carrier frequencies in the mm-wave range). A numerology may be defined in terms of subcarrier spacing and cyclic prefix duration. For a numerology in NR, subcarrier spacings may be scaled up by powers of two from a baseline subcarrier spacing of 15 kHz, and cyclic prefix durations may be scaled down by powers of two from a baseline cyclic prefix duration of 4.7 μ s. For example, NR defines numerologies with the following subcarrier spacing/cyclic prefix duration combinations: 15 kHz/4.7 μ s; 30 kHz/2.3 μ s; 60 kHz/1.2 μ s; 120 kHz/0.59 μ s; and 240 kHz/0.29 μ s.

A slot may have a fixed number of OFDM symbols (e.g., 14 OFDM symbols). A numerology with a higher subcarrier spacing has a shorter slot duration and, correspondingly, more slots per subframe. FIG. 7 illustrates this numerology-dependent slot duration and slots-per-subframe transmission structure (the numerology with a subcarrier spacing of 240 kHz is not shown in FIG. 7 for ease of illustration). A subframe in NR may be used as a numerology-independent time reference, while a slot may be used as the unit upon which uplink and downlink transmissions are scheduled. To support low latency, scheduling in NR may be decoupled from the slot duration and start at any OFDM symbol and last for as many symbols as needed for a transmission. These partial slot transmissions may be referred to as mini-slot or subslot transmissions.

FIG. 8 illustrates an example configuration of a slot in the time and frequency domain for an NR carrier. The slot includes resource elements (REs) and resource blocks (RBs). An RE is the smallest physical resource in NR. An RE spans one OFDM symbol in the time domain by one subcarrier in the frequency domain as shown in FIG. 8. An RB spans twelve consecutive REs in the frequency domain as shown in FIG. 8. An NR carrier may be limited to a width of 275 RBs or $275 \times 12 = 3300$ subcarriers. Such a limitation, if used, may limit the NR carrier to 50, 100, 200, and 400 MHz for subcarrier spacings of 15, 30, 60, and 120 kHz, respectively, where the 400 MHz bandwidth may be set based on a 400 MHz per carrier bandwidth limit.

FIG. 8 illustrates a single numerology being used across the entire bandwidth of the NR carrier. In other example configurations, multiple numerologies may be supported on the same carrier.

NR may support wide carrier bandwidths (e.g., up to 400 MHz for a subcarrier spacing of 120 kHz). Not all UEs may be able to receive the full carrier bandwidth (e.g., due to hardware limitations). Also, receiving the full carrier bandwidth may be prohibitive in terms of UE power consumption. In an example, to reduce power consumption and/or for other purposes, a UE may adapt the size of the UE's receive bandwidth based on the amount of traffic the UE is scheduled to receive. This is referred to as bandwidth adaptation.

NR defines bandwidth parts (BWPs) to support UEs not capable of receiving the full carrier bandwidth and to support bandwidth adaptation. In an example, a BWP may be defined by a subset of contiguous RBs on a carrier. A UE may be configured (e.g., via RRC layer) with one or more downlink BWPs and one or more uplink BWPs per serving cell (e.g., up to four downlink BWPs and up to four uplink BWPs per serving cell). At a given time, one or more of the configured BWPs for a serving cell may be active. These one or more BWPs may be referred to as active BWPs of the serving cell. When a serving cell is configured with a secondary uplink carrier, the serving cell may have one or more first active BWPs in the uplink carrier and one or more second active BWPs in the secondary uplink carrier.

For unpaired spectra, a downlink BWP from a set of configured downlink BWPs may be linked with an uplink BWP from a set of configured uplink BWPs if a downlink BWP index of the downlink BWP and an uplink BWP index of the uplink BWP are the same. For unpaired spectra, a UE may expect that a center frequency for a downlink BWP is the same as a center frequency for an uplink BWP.

For a downlink BWP in a set of configured downlink BWPs on a primary cell (PCell), a base station may configure a UE with one or more control resource sets (CORESETs) for at least one search space. A search space is a set of locations in the time and frequency domains where the UE may find control information. The search space may be a UE-specific search space or a common search space (potentially usable by a plurality of UEs). For example, a base station may configure a UE with a common search space, on a PCell or on a primary secondary cell (PSCell), in an active downlink BWP.

For an uplink BWP in a set of configured uplink BWPs, a BS may configure a UE with one or more resource sets for one or more PUCCH transmissions. A UE may receive downlink receptions (e.g., PDCCH or PDSCH) in a downlink BWP according to a configured numerology (e.g., subcarrier spacing and cyclic prefix duration) for the downlink BWP. The UE may transmit uplink transmissions (e.g., PUCCH or PUSCH) in an uplink BWP according to a configured numerology (e.g., subcarrier spacing and cyclic prefix length for the uplink BWP).

One or more BWP indicator fields may be provided in Downlink Control Information (DCI). A value of a BWP indicator field may indicate which BWP in a set of configured BWPs is an active downlink BWP for one or more downlink receptions. The value of the one or more BWP indicator fields may indicate an active uplink BWP for one or more uplink transmissions.

A base station may semi-statically configure a UE with a default downlink BWP within a set of configured downlink BWPs associated with a PCell. If the base station does not provide the default downlink BWP to the UE, the default downlink BWP may be an initial active downlink BWP. The UE may determine which BWP is the initial active downlink BWP based on a CORESET configuration obtained using the PBCH.

A base station may configure a UE with a BWP inactivity timer value for a PCell. The UE may start or restart a BWP inactivity timer at any appropriate time. For example, the UE may start or restart the BWP inactivity timer (a) when the UE detects a DCI indicating an active downlink BWP other than a default downlink BWP for a paired spectra operation; or (b) when a UE detects a DCI indicating an active downlink BWP or active uplink BWP other than a default downlink BWP or uplink BWP for an unpaired spectra operation. If the UE does not detect DCI during an

interval of time (e.g., 1 ms or 0.5 ms), the UE may run the BWP inactivity timer toward expiration (for example, increment from zero to the BWP inactivity timer value, or decrement from the BWP inactivity timer value to zero). When the BWP inactivity timer expires, the UE may switch from the active downlink BWP to the default downlink BWP.

In an example, a base station may semi-statically configure a UE with one or more BWPs. A UE may switch an active BWP from a first BWP to a second BWP in response to receiving a DCI indicating the second BWP as an active BWP and/or in response to an expiry of the BWP inactivity timer (e.g., if the second BWP is the default BWP).

Downlink and uplink BWP switching (where BWP switching refers to switching from a currently active BWP to a not currently active BWP) may be performed independently in paired spectra. In unpaired spectra, downlink and uplink BWP switching may be performed simultaneously. Switching between configured BWPs may occur based on RRC signaling, DCI, expiration of a BWP inactivity timer, and/or an initiation of random access.

FIG. 9 illustrates an example of bandwidth adaptation using three configured BWPs for an NR carrier. A UE configured with the three BWPs may switch from one BWP to another BWP at a switching point. In the example illustrated in FIG. 9, the BWPs include: a BWP 902 with a bandwidth of 40 MHz and a subcarrier spacing of 15 kHz; a BWP 904 with a bandwidth of 10 MHz and a subcarrier spacing of 15 kHz; and a BWP 906 with a bandwidth of 20 MHz and a subcarrier spacing of 60 kHz. The BWP 902 may be an initial active BWP, and the BWP 904 may be a default BWP. The UE may switch between BWPs at switching points. In the example of FIG. 9, the UE may switch from the BWP 902 to the BWP 904 at a switching point 908. The switching at the switching point 908 may occur for any suitable reason, for example, in response to an expiry of a BWP inactivity timer (indicating switching to the default BWP) and/or in response to receiving a DCI indicating BWP 904 as the active BWP. The UE may switch at a switching point 910 from active BWP 904 to BWP 906 in response receiving a DCI indicating BWP 906 as the active BWP. The UE may switch at a switching point 912 from active BWP 906 to BWP 904 in response to an expiry of a BWP inactivity timer and/or in response receiving a DCI indicating BWP 904 as the active BWP. The UE may switch at a switching point 914 from active BWP 904 to BWP 902 in response receiving a DCI indicating BWP 902 as the active BWP.

If a UE is configured for a secondary cell with a default downlink BWP in a set of configured downlink BWPs and a timer value, UE procedures for switching BWPs on a secondary cell may be the same/similar as those on a primary cell. For example, the UE may use the timer value and the default downlink BWP for the secondary cell in the same/similar manner as the UE would use these values for a primary cell.

To provide for greater data rates, two or more carriers can be aggregated and simultaneously transmitted to/from the same UE using carrier aggregation (CA). The aggregated carriers in CA may be referred to as component carriers (CCs). When CA is used, there are a number of serving cells for the UE, one for a CC. The CCs may have three configurations in the frequency domain.

FIG. 10A illustrates the three CA configurations with two CCs. In the intraband, contiguous configuration 1002, the two CCs are aggregated in the same frequency band (frequency band A) and are located directly adjacent to each

other within the frequency band. In the intraband, non-contiguous configuration **1004**, the two CCs are aggregated in the same frequency band (frequency band A) and are separated in the frequency band by a gap. In the interband configuration **1006**, the two CCs are located in frequency bands (frequency band A and frequency band B).

In an example, up to 32 CCs may be aggregated. The aggregated CCs may have the same or different bandwidths, subcarrier spacing, and/or duplexing schemes (TDD or FDD). A serving cell for a UE using CA may have a downlink CC. For FDD, one or more uplink CCs may be optionally configured for a serving cell. The ability to aggregate more downlink carriers than uplink carriers may be useful, for example, when the UE has more data traffic in the downlink than in the uplink.

When CA is used, one of the aggregated cells for a UE may be referred to as a primary cell (PCell). The PCell may be the serving cell that the UE initially connects to at RRC connection establishment, reestablishment, and/or handover. The PCell may provide the UE with NAS mobility information and the security input. UEs may have different PCells. In the downlink, the carrier corresponding to the PCell may be referred to as the downlink primary CC (DL PCC). In the uplink, the carrier corresponding to the PCell may be referred to as the uplink primary CC (UL PCC). The other aggregated cells for the UE may be referred to as secondary cells (SCells). In an example, the SCells may be configured after the PCell is configured for the UE. For example, an SCell may be configured through an RRC Connection Reconfiguration procedure. In the downlink, the carrier corresponding to an SCell may be referred to as a downlink secondary CC (DL SCC). In the uplink, the carrier corresponding to the SCell may be referred to as the uplink secondary CC (UL SCC).

Configured SCells for a UE may be activated and deactivated based on, for example, traffic and channel conditions. Deactivation of an SCell may mean that PDCCH and PDSCH reception on the SCell is stopped and PUSCH, SRS, and CQI transmissions on the SCell are stopped. Configured SCells may be activated and deactivated using a MAC CE with respect to FIG. 4B. For example, a MAC CE may use a bitmap (e.g., one bit per SCell) to indicate which SCells (e.g., in a subset of configured SCells) for the UE are activated or deactivated. Configured SCells may be deactivated in response to an expiration of an SCell deactivation timer (e.g., one SCell deactivation timer per SCell).

Downlink control information, such as scheduling assignments and scheduling grants, for a cell may be transmitted on the cell corresponding to the assignments and grants, which is known as self-scheduling. The DCI for the cell may be transmitted on another cell, which is known as cross-carrier scheduling. Uplink control information (e.g., HARQ acknowledgments and channel state feedback, such as CQI, PMI, and/or RI) for aggregated cells may be transmitted on the PUCCH of the PCell. For a larger number of aggregated downlink CCs, the PUCCH of the PCell may become overloaded. Cells may be divided into multiple PUCCH groups.

FIG. 10B illustrates an example of how aggregated cells may be configured into one or more PUCCH groups. A PUCCH group **1010** and a PUCCH group **1050** may include one or more downlink CCs, respectively. In the example of FIG. 10B, the PUCCH group **1010** includes three downlink CCs: a PCell **1011**, an SCell **1012**, and an SCell **1013**. The PUCCH group **1050** includes three downlink CCs in the present example: a PCell **1051**, an SCell **1052**, and an SCell **1053**. One or more uplink CCs may be configured as a PCell

1021, an SCell **1022**, and an SCell **1023**. One or more other uplink CCs may be configured as a primary SCell (PSCell) **1061**, an SCell **1062**, and an SCell **1063**. Uplink control information (UCI) related to the downlink CCs of the PUCCH group **1010**, shown as UCI **1031**, UCI **1032**, and UCI **1033**, may be transmitted in the uplink of the PCell **1021**. Uplink control information (UCI) related to the downlink CCs of the PUCCH group **1050**, shown as UCI **1071**, UCI **1072**, and UCI **1073**, may be transmitted in the uplink of the PSCell **1061**. In an example, if the aggregated cells depicted in FIG. 10B were not divided into the PUCCH group **1010** and the PUCCH group **1050**, a single uplink PCell to transmit UCI relating to the downlink CCs, and the PCell may become overloaded. By dividing transmissions of UCI between the PCell **1021** and the PSCell **1061**, overloading may be prevented.

A cell, comprising a downlink carrier and optionally an uplink carrier, may be assigned with a physical cell ID and a cell index. The physical cell ID or the cell index may identify a downlink carrier and/or an uplink carrier of the cell, for example, depending on the context in which the physical cell ID is used. A physical cell ID may be determined using a synchronization signal transmitted on a downlink component carrier. A cell index may be determined using RRC messages. In the disclosure, a physical cell ID may be referred to as a carrier ID, and a cell index may be referred to as a carrier index. For example, when the disclosure refers to a first physical cell ID for a first downlink carrier, the disclosure may mean the first physical cell ID is for a cell comprising the first downlink carrier. The same/similar concept may apply to, for example, a carrier activation. When the disclosure indicates that a first carrier is activated, the specification may mean that a cell comprising the first carrier is activated.

In CA, a multi-carrier nature of a PHY may be exposed to a MAC. In an example, a HARQ entity may operate on a serving cell. A transport block may be generated per assignment/grant per serving cell. A transport block and potential HARQ retransmissions of the transport block may be mapped to a serving cell.

In the downlink, a base station may transmit (e.g., unicast, multicast, and/or broadcast) one or more Reference Signals (RSs) to a UE (e.g., PSS, SSS, CSI-RS, DMRS, and/or PT-RS, as shown in FIG. 5A). In the uplink, the UE may transmit one or more RSs to the base station (e.g., DMRS, PT-RS, and/or SRS, as shown in FIG. 5B). The PSS and the SSS may be transmitted by the base station and used by the UE to synchronize the UE to the base station. The PSS and the SSS may be provided in a synchronization signal (SS)/physical broadcast channel (PBCH) block that includes the PSS, the SSS, and the PBCH. The base station may periodically transmit a burst of SS/PBCH blocks.

FIG. 11A illustrates an example of an SS/PBCH block's structure and location. A burst of SS/PBCH blocks may include one or more SS/PBCH blocks (e.g., 4 SS/PBCH blocks, as shown in FIG. 11A). Bursts may be transmitted periodically (e.g., every 2 frames or 20 ms). A burst may be restricted to a half-frame (e.g., a first half-frame having a duration of 5 ms). It will be understood that FIG. 11A is an example, and that these parameters (number of SS/PBCH blocks per burst, periodicity of bursts, position of burst within the frame) may be configured based on, for example: a carrier frequency of a cell in which the SS/PBCH block is transmitted; a numerology or subcarrier spacing of the cell; a configuration by the network (e.g., using RRC signaling); or any other suitable factor. In an example, the UE may assume a subcarrier spacing for the SS/PBCH block based

on the carrier frequency being monitored, unless the radio network configured the UE to assume a different subcarrier spacing.

The SS/PBCH block may span one or more OFDM symbols in the time domain (e.g., 4 OFDM symbols, as shown in the example of FIG. 11A) and may span one or more subcarriers in the frequency domain (e.g., 240 contiguous subcarriers). The PSS, the SSS, and the PBCH may have a common center frequency. The PSS may be transmitted first and may span, for example, 1 OFDM symbol and 127 subcarriers. The SSS may be transmitted after the PSS (e.g., two symbols later) and may span 1 OFDM symbol and 127 subcarriers. The PBCH may be transmitted after the PSS (e.g., across the next 3 OFDM symbols) and may span 240 subcarriers.

The location of the SS/PBCH block in the time and frequency domains may not be known to the UE (e.g., if the UE is searching for the cell). To find and select the cell, the UE may monitor a carrier for the PSS. For example, the UE may monitor a frequency location within the carrier. If the PSS is not found after a certain duration (e.g., 20 ms), the UE may search for the PSS at a different frequency location within the carrier, as indicated by a synchronization raster. If the PSS is found at a location in the time and frequency domains, the UE may determine, based on a known structure of the SS/PBCH block, the locations of the SSS and the PBCH, respectively. The SS/PBCH block may be a cell-defining SS block (CD-SSB). In an example, a primary cell may be associated with a CD-SSB. The CD-SSB may be located on a synchronization raster. In an example, a cell selection/search and/or reselection may be based on the CD-SSB.

The SS/PBCH block may be used by the UE to determine one or more parameters of the cell. For example, the UE may determine a physical cell identifier (PCI) of the cell based on the sequences of the PSS and the SSS, respectively. The UE may determine a location of a frame boundary of the cell based on the location of the SS/PBCH block. For example, the SS/PBCH block may indicate that it has been transmitted in accordance with a transmission pattern, wherein a SS/PBCH block in the transmission pattern is a known distance from the frame boundary.

The PBCH may use a QPSK modulation and may use forward error correction (FEC). The FEC may use polar coding. One or more symbols spanned by the PBCH may carry one or more DMRSs for demodulation of the PBCH. The PBCH may include an indication of a current system frame number (SFN) of the cell and/or a SS/PBCH block timing index. These parameters may facilitate time synchronization of the UE to the base station. The PBCH may include a master information block (MIB) used to provide the UE with one or more parameters.

The MIB may be used by the UE to locate remaining minimum system information (RMSI) associated with the cell. The RMSI may include a System Information Block Type 1 (SIB1). The SIB1 may contain information needed by the UE to access the cell. The UE may use one or more parameters of the MIB to monitor PDCCH, which may be used to schedule PDSCH. The PDSCH may include the SIB1. The SIB1 may be decoded using parameters provided in the MIB. The PBCH may indicate an absence of SIB1. Based on the PBCH indicating the absence of SIB1, the UE may be pointed to a frequency. The UE may search for an SS/PBCH block at the frequency to which the UE is pointed.

The UE may assume that one or more SS/PBCH blocks transmitted with a same SS/PBCH block index are quasi-co-located (QCLed) (e.g., having the same/similar Doppler

spread, Doppler shift, average gain, average delay, and/or spatial Rx parameters). The UE may not assume QCL for SS/PBCH block transmissions having different SS/PBCH block indices.

SS/PBCH blocks (e.g., those within a half-frame) may be transmitted in spatial directions (e.g., using different beams that span a coverage area of the cell). In an example, a first SS/PBCH block may be transmitted in a first spatial direction using a first beam, and a second SS/PBCH block may be transmitted in a second spatial direction using a second beam.

In an example, within a frequency span of a carrier, a base station may transmit a plurality of SS/PBCH blocks. In an example, a first PCI of a first SS/PBCH block of the plurality of SS/PBCH blocks may be different from a second PCI of a second SS/PBCH block of the plurality of SS/PBCH blocks. The PCIs of SS/PBCH blocks transmitted in different frequency locations may be different or the same.

The CSI-RS may be transmitted by the base station and used by the UE to acquire channel state information (CSI). The base station may configure the UE with one or more CSI-RSs for channel estimation or any other suitable purpose. The base station may configure a UE with one or more of the same/similar CSI-RSs. The UE may measure the one or more CSI-RSs. The UE may estimate a downlink channel state and/or generate a CSI report based on the measuring of the one or more downlink CSI-RSs. The UE may provide the CSI report to the base station. The base station may use feedback provided by the UE (e.g., the estimated downlink channel state) to perform link adaptation.

The base station may semi-statically configure the UE with one or more CSI-RS resource sets. A CSI-RS resource may be associated with a location in the time and frequency domains and a periodicity. The base station may selectively activate and/or deactivate a CSI-RS resource. The base station may indicate to the UE that a CSI-RS resource in the CSI-RS resource set is activated and/or deactivated.

The base station may configure the UE to report CSI measurements. The base station may configure the UE to provide CSI reports periodically, aperiodically, or semi-persistently. For periodic CSI reporting, the UE may be configured with a timing and/or periodicity of a plurality of CSI reports. For aperiodic CSI reporting, the base station may request a CSI report. For example, the base station may command the UE to measure a configured CSI-RS resource and provide a CSI report relating to the measurements. For semi-persistent CSI reporting, the base station may configure the UE to transmit periodically, and selectively activate or deactivate the periodic reporting. The base station may configure the UE with a CSI-RS resource set and CSI reports using RRC signaling.

The CSI-RS configuration may comprise one or more parameters indicating, for example, up to 32 antenna ports. The UE may be configured to employ the same OFDM symbols for a downlink CSI-RS and a control resource set (CORESET) when the downlink CSI-RS and CORESET are spatially QCLed and resource elements associated with the downlink CSI-RS are outside of the physical resource blocks (PRBs) configured for the CORESET. The UE may be configured to employ the same OFDM symbols for downlink CSI-RS and SS/PBCH blocks when the downlink CSI-RS and SS/PBCH blocks are spatially QCLed and resource elements associated with the downlink CSI-RS are outside of PRBs configured for the SS/PBCH blocks.

Downlink DMRSs may be transmitted by a base station and used by a UE for channel estimation. For example, the downlink DMRS may be used for coherent demodulation of

one or more downlink physical channels (e.g., PDSCH). An NR network may support one or more variable and/or configurable DMRS patterns for data demodulation. At least one downlink DMRS configuration may support a front-loaded DMRS pattern. A front-loaded DMRS may be mapped over one or more OFDM symbols (e.g., one or two adjacent OFDM symbols). A base station may semi-statically configure the UE with a number (e.g., a maximum number) of front-loaded DMRS symbols for PDSCH. A DMRS configuration may support one or more DMRS ports. For example, for single user-MIMO, a DMRS configuration may support up to eight orthogonal downlink DMRS ports per UE. For multiuser-MIMO, a DMRS configuration may support up to 4 orthogonal downlink DMRS ports per UE. A radio network may support (e.g., at least for CP-OFDM) a common DMRS structure for downlink and uplink, wherein a DMRS location, a DMRS pattern, and/or a scrambling sequence may be the same or different. The base station may transmit a downlink DMRS and a corresponding PDSCH using the same precoding matrix. The UE may use the one or more downlink DMRSs for coherent demodulation/channel estimation of the PDSCH.

In an example, a transmitter (e.g., a base station) may use a precoder matrices for a part of a transmission bandwidth. For example, the transmitter may use a first precoder matrix for a first bandwidth and a second precoder matrix for a second bandwidth. The first precoder matrix and the second precoder matrix may be different based on the first bandwidth being different from the second bandwidth. The UE may assume that a same precoding matrix is used across a set of PRBs. The set of PRBs may be denoted as a precoding resource block group (PRG).

A PDSCH may comprise one or more layers. The UE may assume that at least one symbol with DMRS is present on a layer of the one or more layers of the PDSCH. A higher layer may configure up to 3 DMRSs for the PDSCH.

Downlink PT-RS may be transmitted by a base station and used by a UE for phase-noise compensation. Whether a downlink PT-RS is present or not may depend on an RRC configuration. The presence and/or pattern of the downlink PT-RS may be configured on a UE-specific basis using a combination of RRC signaling and/or an association with one or more parameters employed for other purposes (e.g., modulation and coding scheme (MCS)), which may be indicated by DCI. When configured, a dynamic presence of a downlink PT-RS may be associated with one or more DCI parameters comprising at least MCS. An NR network may support a plurality of PT-RS densities defined in the time and/or frequency domains. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. The UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DMRS ports in a scheduled resource. Downlink PT-RS may be confined in the scheduled time/frequency duration for the UE. Downlink PT-RS may be transmitted on symbols to facilitate phase tracking at the receiver.

The UE may transmit an uplink DMRS to a base station for channel estimation. For example, the base station may use the uplink DMRS for coherent demodulation of one or more uplink physical channels. For example, the UE may transmit an uplink DMRS with a PUSCH and/or a PUCCH. The uplink DM-RS may span a range of frequencies that is similar to a range of frequencies associated with the corresponding physical channel. The base station may configure the UE with one or more uplink DMRS configurations. At least one DMRS configuration may support a front-loaded

DMRS pattern. The front-loaded DMRS may be mapped over one or more OFDM symbols (e.g., one or two adjacent OFDM symbols). One or more uplink DMRSs may be configured to transmit at one or more symbols of a PUSCH and/or a PUCCH. The base station may semi-statically configure the UE with a number (e.g., maximum number) of front-loaded DMRS symbols for the PUSCH and/or the PUCCH, which the UE may use to schedule a single-symbol DMRS and/or a double-symbol DMRS. An NR network may support (e.g., for cyclic prefix orthogonal frequency division multiplexing (CP-OFDM)) a common DMRS structure for downlink and uplink, wherein a DMRS location, a DMRS pattern, and/or a scrambling sequence for the DMRS may be the same or different.

A PUSCH may comprise one or more layers, and the UE may transmit at least one symbol with DMRS present on a layer of the one or more layers of the PUSCH. In an example, a higher layer may configure up to three DMRSs for the PUSCH.

Uplink PT-RS (which may be used by a base station for phase tracking and/or phase-noise compensation) may or may not be present depending on an RRC configuration of the UE. The presence and/or pattern of uplink PT-RS may be configured on a UE-specific basis by a combination of RRC signaling and/or one or more parameters employed for other purposes (e.g., Modulation and Coding Scheme (MCS)), which may be indicated by DCI. When configured, a dynamic presence of uplink PT-RS may be associated with one or more DCI parameters comprising at least MCS. A radio network may support a plurality of uplink PT-RS densities defined in time/frequency domain. When present, a frequency domain density may be associated with at least one configuration of a scheduled bandwidth. The UE may assume a same precoding for a DMRS port and a PT-RS port. A number of PT-RS ports may be fewer than a number of DMRS ports in a scheduled resource. For example, uplink PT-RS may be confined in the scheduled time/frequency duration for the UE.

SRS may be transmitted by a UE to a base station for channel state estimation to support uplink channel dependent scheduling and/or link adaptation. SRS transmitted by the UE may allow a base station to estimate an uplink channel state at one or more frequencies. A scheduler at the base station may employ the estimated uplink channel state to assign one or more resource blocks for an uplink PUSCH transmission from the UE. The base station may semi-statically configure the UE with one or more SRS resource sets. For an SRS resource set, the base station may configure the UE with one or more SRS resources. An SRS resource set applicability may be configured by a higher layer (e.g., RRC) parameter. For example, when a higher layer parameter indicates beam management, an SRS resource in a SRS resource set of the one or more SRS resource sets (e.g., with the same/similar time domain behavior, periodic, aperiodic, and/or the like) may be transmitted at a time instant (e.g., simultaneously). The UE may transmit one or more SRS resources in SRS resource sets. An NR network may support aperiodic, periodic and/or semi-persistent SRS transmissions. The UE may transmit SRS resources based on one or more trigger types, wherein the one or more trigger types may comprise higher layer signaling (e.g., RRC) and/or one or more DCI formats. In an example, at least one DCI format may be employed for the UE to select at least one of one or more configured SRS resource sets. An SRS trigger type 0 may refer to an SRS triggered based on a higher layer signaling. An SRS trigger type 1 may refer to an SRS triggered based on one or more DCI formats. In an example,

when PUSCH and SRS are transmitted in a same slot, the UE may be configured to transmit SRS after a transmission of a PUSCH and a corresponding uplink DMRS.

The base station may semi-statically configure the UE with one or more SRS configuration parameters indicating at least one of following: a SRS resource configuration identifier; a number of SRS ports; time domain behavior of an SRS resource configuration (e.g., an indication of periodic, semi-persistent, or aperiodic SRS); slot, mini-slot, and/or subframe level periodicity; offset for a periodic and/or an aperiodic SRS resource; a number of OFDM symbols in an SRS resource; a starting OFDM symbol of an SRS resource; an SRS bandwidth; a frequency hopping bandwidth; a cyclic shift; and/or an SRS sequence ID.

An antenna port is defined such that the channel over which a symbol on the antenna port is conveyed can be inferred from the channel over which another symbol on the same antenna port is conveyed. If a first symbol and a second symbol are transmitted on the same antenna port, the receiver may infer the channel (e.g., fading gain, multipath delay, and/or the like) for conveying the second symbol on the antenna port, from the channel for conveying the first symbol on the antenna port. A first antenna port and a second antenna port may be referred to as quasi co-located (QCLed) if one or more large-scale properties of the channel over which a first symbol on the first antenna port is conveyed may be inferred from the channel over which a second symbol on a second antenna port is conveyed. The one or more large-scale properties may comprise at least one of: a delay spread; a Doppler spread; a Doppler shift; an average gain; an average delay; and/or spatial Receiving (Rx) parameters.

Channels that use beamforming require beam management. Beam management may comprise beam measurement, beam selection, and beam indication. A beam may be associated with one or more reference signals. For example, a beam may be identified by one or more beamformed reference signals. The UE may perform downlink beam measurement based on downlink reference signals (e.g., a channel state information reference signal (CSI-RS)) and generate a beam measurement report. The UE may perform the downlink beam measurement procedure after an RRC connection is set up with a base station.

FIG. 11B illustrates an example of channel state information reference signals (CSI-RSs) that are mapped in the time and frequency domains. A square shown in FIG. 11B may span a resource block (RB) within a bandwidth of a cell. A base station may transmit one or more RRC messages comprising CSI-RS resource configuration parameters indicating one or more CSI-RSs. One or more of the following parameters may be configured by higher layer signaling (e.g., RRC and/or MAC signaling) for a CSI-RS resource configuration: a CSI-RS resource configuration identity, a number of CSI-RS ports, a CSI-RS configuration (e.g., symbol and resource element (RE) locations in a subframe), a CSI-RS subframe configuration (e.g., subframe location, offset, and periodicity in a radio frame), a CSI-RS power parameter, a CSI-RS sequence parameter, a code division multiplexing (CDM) type parameter, a frequency density, a transmission comb, quasi co-location (QCL) parameters (e.g., QCL-scrambling identity, crs-portscount, mbsfn-subframeconfiglist, csi-rs-configZPId, qcl-csi-rs-configNZPId), and/or other radio resource parameters.

The three beams illustrated in FIG. 11B may be configured for a UE in a UE-specific configuration. Three beams are illustrated in FIG. 11B (beam #1, beam #2, and beam #3), more or fewer beams may be configured. Beam #1 may

be allocated with CSI-RS 1101 that may be transmitted in one or more subcarriers in an RB of a first symbol. Beam #2 may be allocated with CSI-RS 1102 that may be transmitted in one or more subcarriers in an RB of a second symbol. Beam #3 may be allocated with CSI-RS 1103 that may be transmitted in one or more subcarriers in an RB of a third symbol. By using frequency division multiplexing (FDM), a base station may use other subcarriers in a same RB (for example, those that are not used to transmit CSI-RS 1101) to transmit another CSI-RS associated with a beam for another UE. By using time domain multiplexing (TDM), beams used for the UE may be configured such that beams for the UE use symbols from beams of other UEs.

CSI-RSs such as those illustrated in FIG. 11B (e.g., CSI-RS 1101, 1102, 1103) may be transmitted by the base station and used by the UE for one or more measurements. For example, the UE may measure a reference signal received power (RSRP) of configured CSI-RS resources. The base station may configure the UE with a reporting configuration and the UE may report the RSRP measurements to a network (for example, via one or more base stations) based on the reporting configuration. In an example, the base station may determine, based on the reported measurement results, one or more transmission configuration indication (TCI) states comprising a number of reference signals. In an example, the base station may indicate one or more TCI states to the UE (e.g., via RRC signaling, a MAC CE, and/or a DCI). The UE may receive a downlink transmission with a receive (Rx) beam determined based on the one or more TCI states. In an example, the UE may or may not have a capability of beam correspondence. If the UE has the capability of beam correspondence, the UE may determine a spatial domain filter of a transmit (Tx) beam based on a spatial domain filter of the corresponding Rx beam. If the UE does not have the capability of beam correspondence, the UE may perform an uplink beam selection procedure to determine the spatial domain filter of the Tx beam. The UE may perform the uplink beam selection procedure based on one or more sounding reference signal (SRS) resources configured to the UE by the base station. The base station may select and indicate uplink beams for the UE based on measurements of the one or more SRS resources transmitted by the UE.

In a beam management procedure, a UE may assess (e.g., measure) a channel quality of one or more beam pair links, a beam pair link comprising a transmitting beam transmitted by a base station and a receiving beam received by the UE. Based on the assessment, the UE may transmit a beam measurement report indicating one or more beam pair quality parameters comprising, e.g., one or more beam identifications (e.g., a beam index, a reference signal index, or the like), RSRP, a precoding matrix indicator (PMI), a channel quality indicator (CQI), and/or a rank indicator (RI).

FIG. 12A illustrates examples of three downlink beam management procedures: P1, P2, and P3. Procedure P1 may enable a UE measurement on transmit (Tx) beams of a transmission reception point (TRP) (or multiple TRPs), e.g., to support a selection of one or more base station Tx beams and/or UE Rx beams (shown as ovals in the top row and bottom row, respectively, of P1). Beamforming at a TRP may comprise a Tx beam sweep for a set of beams (shown, in the top rows of P1 and P2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). Beamforming at a UE may comprise an Rx beam sweep for a set of beams (shown, in the bottom rows of P1 and P3, as ovals rotated in a clockwise direction indicated by the dashed arrow). Procedure P2 may be used to enable a UE measure-

ment on Tx beams of a TRP (shown, in the top row of P2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). The UE and/or the base station may perform procedure P2 using a smaller set of beams than is used in procedure P1, or using narrower beams than the beams used in procedure P1. This may be referred to as beam refinement. The UE may perform procedure P3 for Rx beam determination by using the same Tx beam at the base station and sweeping an Rx beam at the UE.

FIG. 12B illustrates examples of three uplink beam management procedures: U1, U2, and U3. Procedure U1 may be used to enable a base station to perform a measurement on Tx beams of a UE, e.g., to support a selection of one or more UE Tx beams and/or base station Rx beams (shown as ovals in the top row and bottom row, respectively, of U1). Beamforming at the UE may include, e.g., a Tx beam sweep from a set of beams (shown in the bottom rows of U1 and U3 as ovals rotated in a clockwise direction indicated by the dashed arrow). Beamforming at the base station may include, e.g., an Rx beam sweep from a set of beams (shown, in the top rows of U1 and U2, as ovals rotated in a counter-clockwise direction indicated by the dashed arrow). Procedure U2 may be used to enable the base station to adjust its Rx beam when the UE uses a fixed Tx beam. The UE and/or the base station may perform procedure U2 using a smaller set of beams than is used in procedure P1, or using narrower beams than the beams used in procedure P1. This may be referred to as beam refinement. The UE may perform procedure U3 to adjust its Tx beam when the base station uses a fixed Rx beam.

A UE may initiate a beam failure recovery (BFR) procedure based on detecting a beam failure. The UE may transmit a BFR request (e.g., a preamble, a UCI, an SR, a MAC CE, and/or the like) based on the initiating of the BFR procedure. The UE may detect the beam failure based on a determination that a quality of beam pair link(s) of an associated control channel is unsatisfactory (e.g., having an error rate higher than an error rate threshold, a received signal power lower than a received signal power threshold, an expiration of a timer, and/or the like).

The UE may measure a quality of a beam pair link using one or more reference signals (RSs) comprising one or more SS/PBCH blocks, one or more CSI-RS resources, and/or one or more demodulation reference signals (DMRSs). A quality of the beam pair link may be based on one or more of a block error rate (BLER), an RSRP value, a signal to interference plus noise ratio (SINR) value, a reference signal received quality (RSRQ) value, and/or a CSI value measured on RS resources. The base station may indicate that an RS resource is quasi co-located (QCLed) with one or more DM-RSs of a channel (e.g., a control channel, a shared data channel, and/or the like). The RS resource and the one or more DMRSs of the channel may be QCLed when the channel characteristics (e.g., Doppler shift, Doppler spread, average delay, delay spread, spatial Rx parameter, fading, and/or the like) from a transmission via the RS resource to the UE are similar or the same as the channel characteristics from a transmission via the channel to the UE.

A network (e.g., a gNB and/or an ng-eNB of a network) and/or the UE may initiate a random access procedure. A UE in an RRC_IDLE state and/or an RRC_INACTIVE state may initiate the random access procedure to request a connection setup to a network. The UE may initiate the random access procedure from an RRC_CONNECTED state. The UE may initiate the random access procedure to request uplink resources (e.g., for uplink transmission of an SR when there is no PUCCH resource available) and/or

acquire uplink timing (e.g., when uplink synchronization status is non-synchronized). The UE may initiate the random access procedure to request one or more system information blocks (SIBs) (e.g., other system information such as SIB2, SIB3, and/or the like). The UE may initiate the random access procedure for a beam failure recovery request. A network may initiate a random access procedure for a handover and/or for establishing time alignment for an SCell addition.

FIG. 13A illustrates a four-step contention-based random access procedure. Prior to initiation of the procedure, a base station may transmit a configuration message 1310 to the UE. The procedure illustrated in FIG. 13A comprises transmission of four messages: a Msg 1 1311, a Msg 2 1312, a Msg 3 1313, and a Msg 4 1314. The Msg 1 1311 may include and/or be referred to as a preamble (or a random access preamble). The Msg 2 1312 may include and/or be referred to as a random access response (RAR).

The configuration message 1310 may be transmitted, for example, using one or more RRC messages. The one or more RRC messages may indicate one or more random access channel (RACH) parameters to the UE. The one or more RACH parameters may comprise at least one of the following: general parameters for one or more random access procedures (e.g., RACH-configGeneral); cell-specific parameters (e.g., RACH-ConfigCommon); and/or dedicated parameters (e.g., RACH-configDedicated). The base station may broadcast or multicast the one or more RRC messages to one or more UEs. The one or more RRC messages may be UE-specific (e.g., dedicated RRC messages transmitted to a UE in an RRC_CONNECTED state and/or in an RRC_INACTIVE state). The UE may determine, based on the one or more RACH parameters, a time-frequency resource and/or an uplink transmit power for transmission of the Msg 1 1311 and/or the Msg 3 1313. Based on the one or more RACH parameters, the UE may determine a reception timing and a downlink channel for receiving the Msg 2 1312 and the Msg 4 1314.

The one or more RACH parameters provided in the configuration message 1310 may indicate one or more Physical RACH (PRACH) occasions available for transmission of the Msg 1 1311. The one or more PRACH occasions may be predefined. The one or more RACH parameters may indicate one or more available sets of one or more PRACH occasions (e.g., prach-ConfigIndex). The one or more RACH parameters may indicate an association between (a) one or more PRACH occasions and (b) one or more reference signals. The one or more RACH parameters may indicate an association between (a) one or more preambles and (b) one or more reference signals. The one or more reference signals may be SS/PBCH blocks and/or CSI-RSs. For example, the one or more RACH parameters may indicate a number of SS/PBCH blocks mapped to a PRACH occasion and/or a number of preambles mapped to a SS/PBCH blocks.

The one or more RACH parameters provided in the configuration message 1310 may be used to determine an uplink transmit power of Msg 1 1311 and/or Msg 3 1313. For example, the one or more RACH parameters may indicate a reference power for a preamble transmission (e.g., a received target power and/or an initial power of the preamble transmission). There may be one or more power offsets indicated by the one or more RACH parameters. For example, the one or more RACH parameters may indicate: a power ramping step; a power offset between SSB and CSI-RS; a power offset between transmissions of the Msg 1 1311 and the Msg 3 1313; and/or a power offset value

between preamble groups. The one or more RACH parameters may indicate one or more thresholds based on which the UE may determine at least one reference signal (e.g., an SSB and/or CSI-RS) and/or an uplink carrier (e.g., a normal uplink (NUL) carrier and/or a supplemental uplink (SUL) carrier).

The Msg 1 1311 may include one or more preamble transmissions (e.g., a preamble transmission and one or more preamble retransmissions). An RRC message may be used to configure one or more preamble groups (e.g., group A and/or group B). A preamble group may comprise one or more preambles. The UE may determine the preamble group based on a pathloss measurement and/or a size of the Msg 3 1313. The UE may measure an RSRP of one or more reference signals (e.g., SSBs and/or CSI-RSs) and determine at least one reference signal having an RSRP above an RSRP threshold (e.g., rsrp-ThresholdSSB and/or $\text{rsrp-ThresholdCSI-RS}$). The UE may select at least one preamble associated with the one or more reference signals and/or a selected preamble group, for example, if the association between the one or more preambles and the at least one reference signal is configured by an RRC message.

The UE may determine the preamble based on the one or more RACH parameters provided in the configuration message 1310. For example, the UE may determine the preamble based on a pathloss measurement, an RSRP measurement, and/or a size of the Msg 3 1313. As another example, the one or more RACH parameters may indicate: a preamble format; a maximum number of preamble transmissions; and/or one or more thresholds for determining one or more preamble groups (e.g., group A and group B). A base station may use the one or more RACH parameters to configure the UE with an association between one or more preambles and one or more reference signals (e.g., SSBs and/or CSI-RSs). If the association is configured, the UE may determine the preamble to include in Msg 1 1311 based on the association. The Msg 1 1311 may be transmitted to the base station via one or more PRACH occasions. The UE may use one or more reference signals (e.g., SSBs and/or CSI-RSs) for selection of the preamble and for determining of the PRACH occasion. One or more RACH parameters (e.g., $\text{ra-ssb-OccasionMskIndex}$ and/or ra-OccasionList) may indicate an association between the PRACH occasions and the one or more reference signals.

The UE may perform a preamble retransmission if no response is received following a preamble transmission. The UE may increase an uplink transmit power for the preamble retransmission. The UE may select an initial preamble transmit power based on a pathloss measurement and/or a target received preamble power configured by the network. The UE may determine to retransmit a preamble and may ramp up the uplink transmit power. The UE may receive one or more RACH parameters (e.g., $\text{PREAMBLE_POWER_RAMPING_STEP}$) indicating a ramping step for the preamble retransmission. The ramping step may be an amount of incremental increase in uplink transmit power for a retransmission. The UE may ramp up the uplink transmit power if the UE determines a reference signal (e.g., SSB and/or CSI-RS) that is the same as a previous preamble transmission. The UE may count a number of preamble transmissions and/or retransmissions (e.g., $\text{PREAMBLE_TRANSMISSION_COUNTER}$). The UE may determine that a random access procedure completed unsuccessfully, for example, if the number of preamble transmissions exceeds a threshold configured by the one or more RACH parameters (e.g., preambleTransMax).

The Msg 2 1312 received by the UE may include an RAR. In some scenarios, the Msg 2 1312 may include multiple RARs corresponding to multiple UEs. The Msg 2 1312 may be received after or in response to the transmitting of the Msg 1 1311. The Msg 2 1312 may be scheduled on the DL-SCH and indicated on a PDCCH using a random access RNTI (RA-RNTI). The Msg 2 1312 may indicate that the Msg 1 1311 was received by the base station. The Msg 2 1312 may include a time-alignment command that may be used by the UE to adjust the UE's transmission timing, a scheduling grant for transmission of the Msg 3 1313, and/or a Temporary Cell RNTI (TC-RNTI). After transmitting a preamble, the UE may start a time window (e.g., ra-ResponseWindow) to monitor a PDCCH for the Msg 2 1312. The UE may determine when to start the time window based on a PRACH occasion that the UE uses to transmit the preamble. For example, the UE may start the time window one or more symbols after a last symbol of the preamble (e.g., at a first PDCCH occasion from an end of a preamble transmission). The one or more symbols may be determined based on a numerology. The PDCCH may be in a common search space (e.g., a Type1-PDCCH common search space) configured by an RRC message. The UE may identify the RAR based on a Radio Network Temporary Identifier (RNTI). RNTIs may be used depending on one or more events initiating the random access procedure. The UE may use random access RNTI (RA-RNTI). The RA-RNTI may be associated with PRACH occasions in which the UE transmits a preamble. For example, the UE may determine the RA-RNTI based on: an OFDM symbol index; a slot index; a frequency domain index; and/or a UL carrier indicator of the PRACH occasions. An example of RA-RNTI may be as follows:

$$\text{RA-RNTI} = 1 + s_id + 14 \times t_id + 14 \times 80 \times f_id + 14 \times 80 \times 8 \times ul_carrier_id$$

where s_id may be an index of a first OFDM symbol of the PRACH occasion (e.g., $0 \leq s_id < 14$), t_id may be an index of a first slot of the PRACH occasion in a system frame (e.g., $0 \leq t_id < 80$), f_id may be an index of the PRACH occasion in the frequency domain (e.g., $0 \leq f_id < 8$), and $ul_carrier_id$ may be a UL carrier used for a preamble transmission (e.g., 0 for an NUL carrier, and 1 for an SUL carrier).

The UE may transmit the Msg 3 1313 in response to a successful reception of the Msg 2 1312 (e.g., using resources identified in the Msg 2 1312). The Msg 3 1313 may be used for contention resolution in, for example, the contention-based random access procedure illustrated in FIG. 13A. In some scenarios, a plurality of UEs may transmit a same preamble to a base station and the base station may provide an RAR that corresponds to a UE. Collisions may occur if the plurality of UEs interpret the RAR as corresponding to themselves. Contention resolution (e.g., using the Msg 3 1313 and the Msg 4 1314) may be used to increase the likelihood that the UE does not incorrectly use an identity of another the UE. To perform contention resolution, the UE may include a device identifier in the Msg 3 1313 (e.g., a C-RNTI if assigned, a TC-RNTI included in the Msg 2 1312, and/or any other suitable identifier).

The Msg 4 1314 may be received after or in response to the transmitting of the Msg 3 1313. If a C-RNTI was included in the Msg 3 1313, the base station will address the UE on the PDCCH using the C-RNTI. If the UE's unique C-RNTI is detected on the PDCCH, the random access procedure is determined to be successfully completed. If a TC-RNTI is included in the Msg 3 1313 (e.g., if the UE is

in an RRC_IDLE state or not otherwise connected to the base station), Msg 4 **1314** will be received using a DL-SCH associated with the TC-RNTI. If a MAC PDU is successfully decoded and a MAC PDU comprises the UE contention resolution identity MAC CE that matches or otherwise corresponds with the CCCH SDU sent (e.g., transmitted) in Msg 3 **1313**, the UE may determine that the contention resolution is successful and/or the UE may determine that the random access procedure is successfully completed.

The UE may be configured with a supplementary uplink (SUL) carrier and a normal uplink (NUL) carrier. An initial access (e.g., random access procedure) may be supported in an uplink carrier. For example, a base station may configure the UE with two separate RACH configurations: one for an SUL carrier and the other for an NUL carrier. For random access in a cell configured with an SUL carrier, the network may indicate which carrier to use (NUL or SUL). The UE may determine the SUL carrier, for example, if a measured quality of one or more reference signals is lower than a broadcast threshold. Uplink transmissions of the random access procedure (e.g., the Msg 1 **1311** and/or the Msg 3 **1313**) may remain on the selected carrier. The UE may switch an uplink carrier during the random access procedure (e.g., between the Msg 1 **1311** and the Msg 3 **1313**) in one or more cases. For example, the UE may determine and/or switch an uplink carrier for the Msg 1 **1311** and/or the Msg 3 **1313** based on a channel clear assessment (e.g., a listen-before-talk).

FIG. **13B** illustrates a two-step contention-free random access procedure. Similar to the four-step contention-based random access procedure illustrated in FIG. **13A**, a base station may, prior to initiation of the procedure, transmit a configuration message **1320** to the UE. The configuration message **1320** may be analogous in some respects to the configuration message **1310**. The procedure illustrated in FIG. **13B** comprises transmission of two messages: a Msg 1 **1321** and a Msg 2 **1322**. The Msg 1 **1321** and the Msg 2 **1322** may be analogous in some respects to the Msg 1 **1311** and a Msg 2 **1312** illustrated in FIG. **13A**, respectively. As will be understood from FIGS. **13A** and **13B**, the contention-free random access procedure may not include messages analogous to the Msg 3 **1313** and/or the Msg 4 **1314**.

The contention-free random access procedure illustrated in FIG. **13B** may be initiated for a beam failure recovery, other SI request, SCell addition, and/or handover. For example, a base station may indicate or assign to the UE the preamble to be used for the Msg 1 **1321**. The UE may receive, from the base station via PDCCH and/or RRC, an indication of a preamble (e.g., ra-PreambleIndex).

After transmitting a preamble, the UE may start a time window (e.g., ra-ResponseWindow) to monitor a PDCCH for the RAR. In the event of a beam failure recovery request, the base station may configure the UE with a separate time window and/or a separate PDCCH in a search space indicated by an RRC message (e.g., recoverySearchSpaceId). The UE may monitor for a PDCCH transmission addressed to a Cell RNTI (C-RNTI) on the search space. In the contention-free random access procedure illustrated in FIG. **13B**, the UE may determine that a random access procedure successfully completes after or in response to transmission of Msg 1 **1321** and reception of a corresponding Msg 2 **1322**. The UE may determine that a random access procedure successfully completes, for example, if a PDCCH transmission is addressed to a C-RNTI. The UE may determine that a random access procedure successfully completes, for example, if the UE receives an RAR comprising a preamble identifier corresponding to a preamble transmitted by the UE

and/or the RAR comprises a MAC sub-PDU with the preamble identifier. The UE may determine the response as an indication of an acknowledgement for an SI request.

FIG. **13C** illustrates another two-step random access procedure. Similar to the random access procedures illustrated in FIGS. **13A** and **13B**, a base station may, prior to initiation of the procedure, transmit a configuration message **1330** to the UE. The configuration message **1330** may be analogous in some respects to the configuration message **1310** and/or the configuration message **1320**. The procedure illustrated in FIG. **13C** comprises transmission of two messages: a Msg A **1331** and a Msg B **1332**.

Msg A **1331** may be transmitted in an uplink transmission by the UE. Msg A **1331** may comprise one or more transmissions of a preamble **1341** and/or one or more transmissions of a transport block **1342**. The transport block **1342** may comprise contents that are similar and/or equivalent to the contents of the Msg 3 **1313** illustrated in FIG. **13A**. The transport block **1342** may comprise UCI (e.g., an SR, a HARQ ACK/NACK, and/or the like). The UE may receive the Msg B **1332** after or in response to transmitting the Msg A **1331**. The Msg B **1332** may comprise contents that are similar and/or equivalent to the contents of the Msg 2 **1312** (e.g., an RAR) illustrated in FIGS. **13A** and **13B** and/or the Msg 4 **1314** illustrated in FIG. **13A**.

The UE may initiate the two-step random access procedure in FIG. **13C** for licensed spectrum and/or unlicensed spectrum. The UE may determine, based on one or more factors, whether to initiate the two-step random access procedure. The one or more factors may be: a radio access technology in use (e.g., LTE, NR, and/or the like); whether the UE has valid TA or not; a cell size; the UE's RRC state; a type of spectrum (e.g., licensed vs. unlicensed); and/or any other suitable factors.

The UE may determine, based on two-step RACH parameters included in the configuration message **1330**, a radio resource and/or an uplink transmit power for the preamble **1341** and/or the transport block **1342** included in the Msg A **1331**. The RACH parameters may indicate a modulation and coding schemes (MCS), a time-frequency resource, and/or a power control for the preamble **1341** and/or the transport block **1342**. A time-frequency resource for transmission of the preamble **1341** (e.g., a PRACH) and a time-frequency resource for transmission of the transport block **1342** (e.g., a PUSCH) may be multiplexed using FDM, TDM, and/or CDM. The RACH parameters may enable the UE to determine a reception timing and a downlink channel for monitoring for and/or receiving Msg B **1332**.

The transport block **1342** may comprise data (e.g., delay-sensitive data), an identifier of the UE, security information, and/or device information (e.g., an International Mobile Subscriber Identity (IMSI)). The base station may transmit the Msg B **1332** as a response to the Msg A **1331**. The Msg B **1332** may comprise at least one of following: a preamble identifier; a timing advance command; a power control command; an uplink grant (e.g., a radio resource assignment and/or an MCS); a UE identifier for contention resolution; and/or an RNTI (e.g., a C-RNTI or a TC-RNTI). The UE may determine that the two-step random access procedure is successfully completed if: a preamble identifier in the Msg B **1332** is matched to a preamble transmitted by the UE; and/or the identifier of the UE in Msg B **1332** is matched to the identifier of the UE in the Msg A **1331** (e.g., the transport block **1342**).

A UE and a base station may exchange control signaling. The control signaling may be referred to as L1/L2 control signaling and may originate from the PHY layer (e.g., layer

1) and/or the MAC layer (e.g., layer 2). The control signaling may comprise downlink control signaling transmitted from the base station to the UE and/or uplink control signaling transmitted from the UE to the base station.

The downlink control signaling may comprise: a downlink scheduling assignment; an uplink scheduling grant indicating uplink radio resources and/or a transport format; a slot format information; a preemption indication; a power control command; and/or any other suitable signaling. The UE may receive the downlink control signaling in a payload transmitted by the base station on a physical downlink control channel (PDCCH). The payload transmitted on the PDCCH may be referred to as downlink control information (DCI). In some scenarios, the PDCCH may be a group common PDCCH (GC-PDCCH) that is common to a group of UEs.

A base station may attach one or more cyclic redundancy check (CRC) parity bits to a DCI in order to facilitate detection of transmission errors. When the DCI is intended for a UE (or a group of the UEs), the base station may scramble the CRC parity bits with an identifier of the UE (or an identifier of the group of the UEs). Scrambling the CRC parity bits with the identifier may comprise Modulo-2 addition (or an exclusive OR operation) of the identifier value and the CRC parity bits. The identifier may comprise a 16-bit value of a radio network temporary identifier (RNTI).

DCIs may be used for different purposes. A purpose may be indicated by the type of RNTI used to scramble the CRC parity bits. For example, a DCI having CRC parity bits scrambled with a paging RNTI (P-RNTI) may indicate paging information and/or a system information change notification. The P-RNTI may be predefined as "FFFF" in hexadecimal. A DCI having CRC parity bits scrambled with a system information RNTI (SI-RNTI) may indicate a broadcast transmission of the system information. The SI-RNTI may be predefined as "FFFF" in hexadecimal. A DCI having CRC parity bits scrambled with a random access RNTI (RA-RNTI) may indicate a random access response (RAR). A DCI having CRC parity bits scrambled with a cell RNTI (C-RNTI) may indicate a dynamically scheduled unicast transmission and/or a triggering of PDCCH-ordered random access. A DCI having CRC parity bits scrambled with a temporary cell RNTI (TC-RNTI) may indicate a contention resolution (e.g., a Msg 3 analogous to the Msg 3 1313 illustrated in FIG. 13A). Other RNTIs configured to the UE by a base station may comprise a Configured Scheduling RNTI (CS-RNTI), a Transmit Power Control-PUCCH RNTI (TPC-PUCCH-RNTI), a Transmit Power Control-PUSCH RNTI (TPC-PUSCH-RNTI), a Transmit Power Control-SRS RNTI (TPC-SRS-RNTI), an Interruption RNTI (INT-RNTI), a Slot Format Indication RNTI (SFI-RNTI), a Semi-Persistent CSI RNTI (SP-CSI-RNTI), a Modulation and Coding Scheme Cell RNTI (MCS-C-RNTI), and/or the like.

Depending on the purpose and/or content of a DCI, the base station may transmit the DCIs with one or more DCI formats. For example, DCI format 0_0 may be used for scheduling of PUSCH in a cell. DCI format 0_0 may be a fallback DCI format (e.g., with compact DCI payloads). DCI format 0_1 may be used for scheduling of PUSCH in a cell (e.g., with more DCI payloads than DCI format 0_0). DCI format 1_0 may be used for scheduling of PDSCH in a cell. DCI format 1_0 may be a fallback DCI format (e.g., with compact DCI payloads). DCI format 1_1 may be used for scheduling of PDSCH in a cell (e.g., with more DCI payloads than DCI format 1_0). DCI format 2_0 may be used for providing a slot format indication to a group of UEs.

DCI format 2_1 may be used for notifying a group of UEs of a physical resource block and/or OFDM symbol where the UE may assume no transmission is intended to the UE. DCI format 2_2 may be used for transmission of a transmit power control (TPC) command for PUCCH or PUSCH. DCI format 2_3 may be used for transmission of a group of TPC commands for SRS transmissions by one or more UEs. DCI format(s) for new functions may be defined in future releases. DCI formats may have different DCI sizes, or may share the same DCI size.

After scrambling a DCI with a RNTI, the base station may process the DCI with channel coding (e.g., polar coding), rate matching, scrambling and/or QPSK modulation. A base station may map the coded and modulated DCI on resource elements used and/or configured for a PDCCH. Based on a payload size of the DCI and/or a coverage of the base station, the base station may transmit the DCI via a PDCCH occupying a number of contiguous control channel elements (CCEs). The number of the contiguous CCEs (referred to as aggregation level) may be 1, 2, 4, 8, 16, and/or any other suitable number. A CCE may comprise a number (e.g., 6) of resource-element groups (REGs). A REG may comprise a resource block in an OFDM symbol. The mapping of the coded and modulated DCI on the resource elements may be based on mapping of CCEs and REGs (e.g., CCE-to-REG mapping).

FIG. 14A illustrates an example of CORESET configurations for a bandwidth part. The base station may transmit a DCI via a PDCCH on one or more control resource sets (CORESETs). A CORESET may comprise a time-frequency resource in which the UE tries to decode a DCI using one or more search spaces. The base station may configure a CORESET in the time-frequency domain. In the example of FIG. 14A, a first CORESET 1401 and a second CORESET 1402 occur at the first symbol in a slot. The first CORESET 1401 overlaps with the second CORESET 1402 in the frequency domain. A third CORESET 1403 occurs at a third symbol in the slot. A fourth CORESET 1404 occurs at the seventh symbol in the slot. CORESETs may have a different number of resource blocks in frequency domain.

FIG. 14B illustrates an example of a CCE-to-REG mapping for DCI transmission on a CORESET and PDCCH processing. The CCE-to-REG mapping may be an interleaved mapping (e.g., for the purpose of providing frequency diversity) or a non-interleaved mapping (e.g., for the purposes of facilitating interference coordination and/or frequency-selective transmission of control channels). The base station may perform different or same CCE-to-REG mapping on different CORESETs. A CORESET may be associated with a CCE-to-REG mapping by RRC configuration. A CORESET may be configured with an antenna port quasi co-location (QCL) parameter. The antenna port QCL parameter may indicate QCL information of a demodulation reference signal (DMRS) for PDCCH reception in the CORESET.

The base station may transmit, to the UE, RRC messages comprising configuration parameters of one or more CORESETs and one or more search space sets. The configuration parameters may indicate an association between a search space set and a CORESET. A search space set may comprise a set of PDCCH candidates formed by CCEs at a given aggregation level. The configuration parameters may indicate: a number of PDCCH candidates to be monitored per aggregation level; a PDCCH monitoring periodicity and a PDCCH monitoring pattern; one or more DCI formats to be monitored by the UE; and/or whether a search space set is a common search space set or a UE-specific search space set.

A set of CCEs in the common search space set may be predefined and known to the UE. A set of CCEs in the UE-specific search space set may be configured based on the UE's identity (e.g., C-RNTI).

As shown in FIG. 14B, the UE may determine a time-frequency resource for a CORESET based on RRC messages. The UE may determine a CCE-to-REG mapping (e.g., interleaved or non-interleaved, and/or mapping parameters) for the CORESET based on configuration parameters of the CORESET. The UE may determine a number (e.g., at most 10) of search space sets configured on the CORESET based on the RRC messages. The UE may monitor a set of PDCCH candidates according to configuration parameters of a search space set. The UE may monitor a set of PDCCH candidates in one or more CORESETs for detecting one or more DCIs. Monitoring may comprise decoding one or more PDCCH candidates of the set of the PDCCH candidates according to the monitored DCI formats. Monitoring may comprise decoding a DCI content of one or more PDCCH candidates with possible (or configured) PDCCH locations, possible (or configured) PDCCH formats (e.g., number of CCEs, number of PDCCH candidates in common search spaces, and/or number of PDCCH candidates in the UE-specific search spaces) and possible (or configured) DCI formats. The decoding may be referred to as blind decoding. The UE may determine a DCI as valid for the UE, in response to CRC checking (e.g., scrambled bits for CRC parity bits of the DCI matching a RNTI value). The UE may process information contained in the DCI (e.g., a scheduling assignment, an uplink grant, power control, a slot format indication, a downlink preemption, and/or the like).

The UE may transmit uplink control signaling (e.g., uplink control information (UCI)) to a base station. The uplink control signaling may comprise hybrid automatic repeat request (HARQ) acknowledgements for received DL-SCH transport blocks. The UE may transmit the HARQ acknowledgements after receiving a DL-SCH transport block. Uplink control signaling may comprise channel state information (CSI) indicating channel quality of a physical downlink channel. The UE may transmit the CSI to the base station. The base station, based on the received CSI, may determine transmission format parameters (e.g., comprising multi-antenna and beamforming schemes) for a downlink transmission. Uplink control signaling may comprise scheduling requests (SR). The UE may transmit an SR indicating that uplink data is available for transmission to the base station. The UE may transmit a UCI (e.g., HARQ acknowledgements (HARQ-ACK), CSI report, SR, and the like) via a physical uplink control channel (PUCCH) or a physical uplink shared channel (PUSCH). The UE may transmit the uplink control signaling via a PUCCH using one of several PUCCH formats.

There may be five PUCCH formats and the UE may determine a PUCCH format based on a size of the UCI (e.g., a number of uplink symbols of UCI transmission and a number of UCI bits). PUCCH format 0 may have a length of one or two OFDM symbols and may include two or fewer bits. The UE may transmit UCI in a PUCCH resource using PUCCH format 0 if the transmission is over one or two symbols and the number of HARQ-ACK information bits with positive or negative SR (HARQ-ACK/SR bits) is one or two. PUCCH format 1 may occupy a number between four and fourteen OFDM symbols and may include two or fewer bits. The UE may use PUCCH format 1 if the transmission is four or more symbols and the number of HARQ-ACK/SR bits is one or two. PUCCH format 2 may occupy one or two OFDM symbols and may include more

than two bits. The UE may use PUCCH format 2 if the transmission is over one or two symbols and the number of UCI bits is two or more. PUCCH format 3 may occupy a number between four and fourteen OFDM symbols and may include more than two bits. The UE may use PUCCH format 3 if the transmission is four or more symbols, the number of UCI bits is two or more and PUCCH resource does not include an orthogonal cover code. PUCCH format 4 may occupy a number between four and fourteen OFDM symbols and may include more than two bits. The UE may use PUCCH format 4 if the transmission is four or more symbols, the number of UCI bits is two or more and the PUCCH resource includes an orthogonal cover code.

The base station may transmit configuration parameters to the UE for a plurality of PUCCH resource sets using, for example, an RRC message. The plurality of PUCCH resource sets (e.g., up to four sets) may be configured on an uplink BWP of a cell. A PUCCH resource set may be configured with a PUCCH resource set index, a plurality of PUCCH resources with a PUCCH resource being identified by a PUCCH resource identifier (e.g., pucch-Resourceid), and/or a number (e.g. a maximum number) of UCI information bits the UE may transmit using one of the plurality of PUCCH resources in the PUCCH resource set. When configured with a plurality of PUCCH resource sets, the UE may select one of the plurality of PUCCH resource sets based on a total bit length of the UCI information bits (e.g., HARQ-ACK, SR, and/or CSI). If the total bit length of UCI information bits is two or fewer, the UE may select a first PUCCH resource set having a PUCCH resource set index equal to "0". If the total bit length of UCI information bits is greater than two and less than or equal to a first configured value, the UE may select a second PUCCH resource set having a PUCCH resource set index equal to "1". If the total bit length of UCI information bits is greater than the first configured value and less than or equal to a second configured value, the UE may select a third PUCCH resource set having a PUCCH resource set index equal to "2". If the total bit length of UCI information bits is greater than the second configured value and less than or equal to a third value (e.g., 1406), the UE may select a fourth PUCCH resource set having a PUCCH resource set index equal to After determining a PUCCH resource set from a plurality of PUCCH resource sets, the UE may determine a PUCCH resource from the PUCCH resource set for UCI (HARQ-ACK, CSI, and/or SR) transmission. The UE may determine the PUCCH resource based on a PUCCH resource indicator in a DCI (e.g., with a DCI format 1_0 or DCI for 1_1) received on a PDCCH. A three-bit PUCCH resource indicator in the DCI may indicate one of eight PUCCH resources in the PUCCH resource set. Based on the PUCCH resource indicator, the UE may transmit the UCI (HARQ-ACK, CSI and/or SR) using a PUCCH resource indicated by the PUCCH resource indicator in the DCI.

FIG. 15 illustrates an example of a wireless device 1502 in communication with a base station 1504 in accordance with embodiments of the present disclosure. The wireless device 1502 and base station 1504 may be part of a mobile communication network, such as the mobile communication network 100 illustrated in FIG. 1A, the mobile communication network 150 illustrated in FIG. 1B, or any other communication network. Only one wireless device 1502 and one base station 1504 are illustrated in FIG. 15, but it will be understood that a mobile communication network may include more than one UE and/or more than one base station, with the same or similar configuration as those shown in FIG. 15.

The base station **1504** may connect the wireless device **1502** to a core network (not shown) through radio communications over the air interface (or radio interface) **1506**. The communication direction from the base station **1504** to the wireless device **1502** over the air interface **1506** is known as the downlink, and the communication direction from the wireless device **1502** to the base station **1504** over the air interface is known as the uplink. Downlink transmissions may be separated from uplink transmissions using FDD, TDD, and/or some combination of the two duplexing techniques.

In the downlink, data to be sent to the wireless device **1502** from the base station **1504** may be provided to the processing system **1508** of the base station **1504**. The data may be provided to the processing system **1508** by, for example, a core network. In the uplink, data to be sent to the base station **1504** from the wireless device **1502** may be provided to the processing system **1518** of the wireless device **1502**. The processing system **1508** and the processing system **1518** may implement layer 3 and layer 2 OSI functionality to process the data for transmission. Layer 2 may include an SDAP layer, a PDCP layer, an RLC layer, and a MAC layer, for example, with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. Layer 3 may include an RRC layer as with respect to FIG. 2B.

After being processed by processing system **1508**, the data to be sent to the wireless device **1502** may be provided to a transmission processing system **1510** of base station **1504**. Similarly, after being processed by the processing system **1518**, the data to be sent to base station **1504** may be provided to a transmission processing system **1520** of the wireless device **1502**. The transmission processing system **1510** and the transmission processing system **1520** may implement layer 1 OSI functionality. Layer 1 may include a PHY layer with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. For transmit processing, the PHY layer may perform, for example, forward error correction coding of transport channels, interleaving, rate matching, mapping of transport channels to physical channels, modulation of physical channel, multiple-input multiple-output (MIMO) or multi-antenna processing, and/or the like.

At the base station **1504**, a reception processing system **1512** may receive the uplink transmission from the wireless device **1502**. At the wireless device **1502**, a reception processing system **1522** may receive the downlink transmission from base station **1504**. The reception processing system **1512** and the reception processing system **1522** may implement layer 1 OSI functionality. Layer 1 may include a PHY layer with respect to FIG. 2A, FIG. 2B, FIG. 3, and FIG. 4A. For receive processing, the PHY layer may perform, for example, error detection, forward error correction decoding, deinterleaving, demapping of transport channels to physical channels, demodulation of physical channels, MIMO or multi-antenna processing, and/or the like.

As shown in FIG. 15, a wireless device **1502** and the base station **1504** may include multiple antennas. The multiple antennas may be used to perform one or more MIMO or multi-antenna techniques, such as spatial multiplexing (e.g., single-user MIMO or multi-user MIMO), transmit/receive diversity, and/or beamforming. In other examples, the wireless device **1502** and/or the base station **1504** may have a single antenna.

The processing system **1508** and the processing system **1518** may be associated with a memory **1514** and a memory **1524**, respectively. Memory **1514** and memory **1524** (e.g., one or more non-transitory computer readable mediums) may store computer program instructions or code that may

be executed by the processing system **1508** and/or the processing system **1518** to carry out one or more of the functionalities discussed in the present application. Although not shown in FIG. 15, the transmission processing system **1510**, the transmission processing system **1520**, the reception processing system **1512**, and/or the reception processing system **1522** may be coupled to a memory (e.g., one or more non-transitory computer readable mediums) storing computer program instructions or code that may be executed to carry out one or more of their respective functionalities.

The processing system **1508** and/or the processing system **1518** may comprise one or more controllers and/or one or more processors. The one or more controllers and/or one or more processors may comprise, for example, a general-purpose processor, a digital signal processor (DSP), a micro-controller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) and/or other programmable logic device, discrete gate and/or transistor logic, discrete hardware components, an on-board unit, or any combination thereof. The processing system **1508** and/or the processing system **1518** may perform at least one of signal coding/processing, data processing, power control, input/output processing, and/or any other functionality that may enable the wireless device **1502** and the base station **1504** to operate in a wireless environment.

The processing system **1508** and/or the processing system **1518** may be connected to one or more peripherals **1516** and one or more peripherals **1526**, respectively. The one or more peripherals **1516** and the one or more peripherals **1526** may include software and/or hardware that provide features and/or functionalities, for example, a speaker, a microphone, a keypad, a display, a touchpad, a power source, a satellite transceiver, a universal serial bus (USB) port, a hands-free headset, a frequency modulated (FM) radio unit, a media player, an Internet browser, an electronic control unit (e.g., for a motor vehicle), and/or one or more sensors (e.g., an accelerometer, a gyroscope, a temperature sensor, a radar sensor, a lidar sensor, an ultrasonic sensor, a light sensor, a camera, and/or the like). The processing system **1508** and/or the processing system **1518** may receive user input data from and/or provide user output data to the one or more peripherals **1516** and/or the one or more peripherals **1526**. The processing system **1518** in the wireless device **1502** may receive power from a power source and/or may be configured to distribute the power to the other components in the wireless device **1502**. The power source may comprise one or more sources of power, for example, a battery, a solar cell, a fuel cell, or any combination thereof. The processing system **1508** and/or the processing system **1518** may be connected to a GPS chipset **1517** and a GPS chipset **1527**, respectively. The GPS chipset **1517** and the GPS chipset **1527** may be configured to provide geographic location information of the wireless device **1502** and the base station **1504**, respectively.

FIG. 16A illustrates an example structure for uplink transmission. A baseband signal representing a physical uplink shared channel may perform one or more functions. The one or more functions may comprise at least one of: scrambling; modulation of scrambled bits to generate complex-valued symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; transform precoding to generate complex-valued symbols; precoding of the complex-valued symbols; mapping of precoded complex-valued symbols to resource elements; generation of complex-valued time-domain Single Carrier-Frequency Division Multiple Access (SC-FDMA) or CP-

OFDM signal for an antenna port; and/or the like. In an example, when transform precoding is enabled, a SC-FDMA signal for uplink transmission may be generated. In an example, when transform precoding is not enabled, an CP-OFDM signal for uplink transmission may be generated by FIG. 16A. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

FIG. 16B illustrates an example structure for modulation and up-conversion of a baseband signal to a carrier frequency. The baseband signal may be a complex-valued SC-FDMA or CP-OFDM baseband signal for an antenna port and/or a complex-valued Physical Random Access Channel (PRACH) baseband signal. Filtering may be employed prior to transmission.

FIG. 16C illustrates an example structure for downlink transmissions. A baseband signal representing a physical downlink channel may perform one or more functions. The one or more functions may comprise: scrambling of coded bits in a codeword to be transmitted on a physical channel; modulation of scrambled bits to generate complex-valued modulation symbols; mapping of the complex-valued modulation symbols onto one or several transmission layers; precoding of the complex-valued modulation symbols on a layer for transmission on the antenna ports; mapping of complex-valued modulation symbols for an antenna port to resource elements; generation of complex-valued time-domain OFDM signal for an antenna port; and/or the like. These functions are illustrated as examples and it is anticipated that other mechanisms may be implemented in various embodiments.

FIG. 16D illustrates another example structure for modulation and up-conversion of a baseband signal to a carrier frequency. The baseband signal may be a complex-valued OFDM baseband signal for an antenna port. Filtering may be employed prior to transmission.

A wireless device may receive from a base station one or more messages (e.g. RRC messages) comprising configuration parameters of a plurality of cells (e.g. primary cell, secondary cell). The wireless device may communicate with at least one base station (e.g. two or more base stations in dual-connectivity) via the plurality of cells. The one or more messages (e.g. as a part of the configuration parameters) may comprise parameters of physical, MAC, RLC, PCDP, SDAP, RRC layers for configuring the wireless device. For example, the configuration parameters may comprise parameters for configuring physical and MAC layer channels, bearers, etc. For example, the configuration parameters may comprise parameters indicating values of timers for physical, MAC, RLC, PCDP, SDAP, RRC layers, and/or communication channels.

A timer may begin running once it is started and continue running until it is stopped or until it expires. A timer may be started if it is not running or restarted if it is running. A timer may be associated with a value (e.g. the timer may be started or restarted from a value or may be started from zero and expire once it reaches the value). The duration of a timer may not be updated until the timer is stopped or expires (e.g., due to BWP switching). A timer may be used to measure a time period/window for a process. When the specification refers to an implementation and procedure related to one or more timers, it will be understood that there are multiple ways to implement the one or more timers. For example, it will be understood that one or more of the multiple ways to implement a timer may be used to measure a time period/window for the procedure. For example, a random access response window timer may be used for measuring a win-

dow of time for receiving a random access response. In an example, instead of starting and expiry of a random access response window timer, the time difference between two time stamps may be used. When a timer is restarted, a process for measurement of time window may be restarted. Other example implementations may be provided to restart a measurement of a time window.

FIG. 17 illustrates joint TCI states and separate TCI states as per an aspect of an example embodiment of the present disclosure. A transmission configuration indicator (TCI) state pool (e.g., a joint TCI state pool, a separate TCI state pool comprising UL TCI states and/or DL TCI states) may be a TCI state group, a TCI state set, a TCI state list, or a TCI state sequence. The TCI state pool may be common TCI state pool, for example, each TCI state (e.g., common TCI state or common beam) of the TCI state pool may be used by wireless device (e.g., may be configured by the base station via RRC messages) for receptions of one or more downlink channels or signals (e.g., PDCCH, PDSCH, and/or CSI-RS) and/or transmissions of one or more uplink channels or signals (e.g., PUCCH, PUSCH, and/or SRS). The TCI state pool may comprise one or more TCI states. The one or more TCI states may comprise: one or more joint TCI states, one or more downlink TCI states, and/or one or more uplink TCI states. In an example, a joint TCI state pool may comprise the one or more joint TCI states. Each of the one or more joint TCI states may be used, by a wireless device, to receive downlink signals (e.g., PDCCH, PDSCH, and/or CSI-RS), for example, the wireless device may determine a downlink spatial domain filter of a reception of the downlink signals. Each of the one or more joint TCI states may be used, by the wireless device, to transmit uplink signals (e.g., PUCCH, PUSCH, and/or SRS), for example, the wireless device may determine an uplink spatial domain filter of a transmission of the uplink signals. A separate TCI state pool may comprise the UL TCI state pool, and/or the DL TCI state pool. For example, a UL TCI state pool may comprise the one or more uplink TCI states. Each of the one or more uplink TCI states may be used, by the wireless device, (or may be configured by the base station via RRC messages) to transmit uplink signals (e.g., PUCCH, PUSCH, and/or SRS), for example, the wireless device may determine an uplink spatial domain filter of a transmission of the uplink signals.

In an example, the separate TCI state pool may comprise the DL TCI state pool. For example, the DL TCI state pool may comprise the one or more downlink TCI states. Each of the one or more downlink TCI states may be used, by the wireless device, to receive downlink signals (e.g., PDCCH, PDSCH, and/or CSI-RS), for example, the wireless device may determine a downlink spatial domain filter of a reception for the downlink signals. Each TCI state of the TCI state pool (e.g., joint TCI state pool, UL TCI state pool, or DL TCI state pool) may be referred to as a common beam (or a unified TCI state) used for a reception and/or a transmission by the wireless device. For example, a joint TCI state may be used for the receptions, by the wireless device, via PDCCH, PDSCH, and/or CSI-RS, and/or the transmissions via PUCCH, PUSCH, PRACH, and/or sounding channel (e.g., SRS transmission). For example, a downlink TCI state may be used by the wireless device for the receptions of downlink signals via PDCCH, PDSCH, and/or CSI-RS. For example, an uplink TCI state may be used by the wireless device for the transmissions of uplink signals via PUCCH, PUSCH, PRACH, and/or sounding channel (e.g., SRS transmission).

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In an example, an uplink TCI state may be associated with (or may comprise) a quasi-colocation (QCL) information RS, a spatial relation information RS, a pathloss reference RS (or a pathloss RS), and/or a power control parameter setting. For example, an information element of the uplink TCI state may be configured, via RRC messages by the base station, with (or may comprise) the QCL information RS, the spatial relation information RS, the pathloss reference RS, and/or the power control parameter setting. The power control parameter setting may comprise one or more of: p_0 for PUSCH power control (e.g., a target received power), Alpha for PUSCH power control (e.g., a pathloss compensation factor), or close loop index for PUSCH power control. In an example, the uplink TCI state may be configured with (or may comprise) the pathloss reference RS (or the pathloss RS), and/or the power control parameter setting via RRC messages. In an example, the information element of the uplink TCI state may be configured, via RRC messages, with an association with the QCL information RS, the spatial relation information RS, the pathloss reference RS (or a pathloss RS), and/or the power control parameter setting. The association may comprise that the pathloss reference RS (or the pathloss RS), and/or the power control parameter setting is linked with an uplink TCI state identity (e.g., in RRC configuration parameters and as illustrated in FIG. 18A). The wireless device may determine the pathloss reference RS, and/or the power control parameter setting based on the uplink TCI state identity. In an example, the wireless device may determine the QCL information RS, the spatial relation information RS, the associated pathloss reference RS, and/or the power control parameter setting based on the association and/or the information element of the uplink TCI state.

In an example, a joint TCI state may be associated with (or may comprise) a QCL information RS, a spatial relation information RS, a pathloss reference RS (or a pathloss RS), and/or a power control parameter setting. For example, an information element of the joint TCI state may be configured, via RRC messages by the base station, with (or may comprise) the QCL information RS, the spatial relation information RS, the pathloss reference RS, and/or the power control parameter setting. The power control parameter setting may comprise one or more of: p_0 for PUSCH power control, Alpha for PUSCH power control, or close loop index for PUSCH power control. In an example, the joint TCI state may be configured with (or may comprise) the pathloss reference RS (or the pathloss RS), and/or the power control parameter setting via RRC messages. In an example, the information element of the joint TCI state may be configured, via RRC messages, with an association with the QCL information RS, the spatial relation information RS, the pathloss reference RS (or a pathloss RS), and/or the power control parameter setting. The association may comprise that the pathloss reference RS (or the pathloss RS), and/or the power control parameter setting is linked with a joint TCI state identity (e.g., in RRC configuration parameters and as illustrated in FIG. 18B). The wireless device may determine the pathloss reference RS, and/or the power control parameter setting based on the joint TCI state identity. In an example, the wireless device may determine the QCL-info RS, the spatial relation information RS, the associated pathloss reference RS, and/or the power control parameter setting based on the association and/or the information element of the joint TCI state.

In an example, a downlink TCI state may be associated with (or may comprise) one or more QCL information RSs. For example, an information element of the downlink TCI

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state may be configured, via RRC messages, with (or may comprise) the one or more QCL information RSs. The wireless device may determine a spatial domain filter for reception of multiple downlink channels or signals (e.g., PDCCH, PDSCH, and/or CSI-RS) based on a joint TCI state or a downlink TCI state. For example, the wireless device may determine the spatial domain filter same as a spatial domain filter used for reception of quasi collocated RS (e.g., QCL information RS) of the joint TCI state or the downlink TCI state.

FIG. 18A illustrates association between separate TCI states and power control parameters as per an aspect of an example embodiment of the present disclosure. The separate TCI states may be uplink TCI states. In an example, the association may indicate that an uplink TCI state comprises a pathloss reference RS. The association may indicate that the uplink TCI state may comprise a power control parameter setting (e.g., as illustrated in FIG. 17). In an example, the association may indicate that the pathloss reference RS is identified by the uplink TCI state. The association may indicate that the power control parameter setting is identified by the uplink TCI state. FIG. 18B illustrates association between joint TCI states and power control parameters as per an aspect of an example embodiment of the present disclosure. In an example, the association may indicate that a joint TCI state comprises a pathloss reference RS. The association may indicate that the joint TCI state may comprise a power control parameter setting (e.g., as illustrated in FIG. 17). In an example, the association may indicate that the pathloss reference RS is identified by the joint TCI state. The association may indicate that the power control parameter setting is identified by the joint TCI state.

FIG. 19 illustrates indications of joint TCI states and separate TCI states as per an aspect of an example embodiment of the present disclosure. A wireless device may receive (e.g., at time T1) a downlink control information (DCI) from a base station. The DCI may indicate one or more joint TCI states and/or one or more separate TCI states. In an example, the DCI may schedule a transport block via a PDSCH. The wireless device may receive (e.g., at time T2) the transport block via the PDSCH from the base station. In an example, the DCI may not schedule the transport block via the PDSCH. The wireless device may transmit (e.g., at time T3) an acknowledgement (ACK/NACK) of the transport block (e.g., the transport block being scheduled) and/or the DCI (e.g., the transport block not being scheduled) to the base station. The wireless device may apply (e.g., from time T4) the one or more joint TCI states and/or the one or more separate TCI states after a time offset from transmission (by the wireless device) of the acknowledgement (ACK/NACK). The application of the one or more joint TCI states and/or the one or more separate TCI states may comprise transmissions and/or receptions (by the wireless device) of signals and/or channels for uplink and/or downlink with the one or more joint TCI states and/or the one or more separate TCI states. For example, the wireless device may determine a spatial domain filter based on the one or more joint TCI states and/or the one or more separate TCI states. In an example, the wireless device may determine the spatial domain filter based on quasi colocation reference signal(s) (or spatial relation information reference signal(s)) of the one or more joint TCI states and/or the one or more separate TCI states. The wireless device may transmit signals and/or channels for uplink with the spatial domain filter. The wireless device may receive signals and/or channels for downlink with the spatial domain filter. The time offset may comprise: one or more million seconds; or one or more

symbols (e.g., orthogonal frequency division multiplexing (OFDM) symbols). The signals and/or channels may comprise one or more of: PDCCH, PDSCH, PBCH, SSB(s), CSI-RS(s), SRS(s), DMRS(s), PUCCH, PUSCH, or PRACH.

FIG. 20 illustrates a configuration of power control parameters with sounding reference signal resource indicator (SRI) as per an aspect of an example embodiment of the present disclosure. A wireless device may receive one or more RRC messages from a base station. The one or more RRC messages may comprise configuration parameters indicating a plurality of power control parameter sets (e.g., "sri-PUSCH-MappingToAddModList"). Each of the plurality of power control parameter sets may comprise a pathloss reference RS (e.g., "PUSCH-PathlossReferenceRS-Id"), p0-PUSCH-Alpha Set (e.g., "p0-PUSCH-AlphaSetId"), and/or PUSCH-ClosedLoopindex. The wireless device may receive, from the base station, a DCI comprising an uplink grant. The uplink grant may schedule a transport block via a physical uplink shared channel (PUSCH). The uplink grant may comprise a sounding reference signal resources indicator (SRI). The SRI may indicate a power control parameter set from the plurality of power control parameter sets. The wireless device may determine a transmit power based on the power control parameter set comprising a pathloss reference RS (e.g., "PUSCH-PathlossReferenceRS-Id"), p0-PUSCH-Alpha Set (e.g., "p0-PUSCH-AlphaSetId"), and/or PUSCH-ClosedLoopindex. The wireless device may transmit the transport block with the transmit power via the PUSCH.

In existing technologies, a wireless device may receive a first downlink control information (DCI) indicating a joint TCI state and/or an uplink TCI state. In an example, the joint TCI state and/or the uplink TCI state may comprise a pathloss reference RS, and/or a power control parameter setting. The power control parameter setting may comprise one or more of: p0 for PUSCH power control, Alpha for PUSCH power control, or close loop index for PUSCH power control. In an example, the joint TCI state and/or the uplink TCI state may be associated with the pathloss reference RS, and/or the power control parameter setting. The wireless device may determine a transmit power based on the pathloss reference RS, and/or the power control parameter setting. The wireless device may transmit a transport block (e.g., via PUSCH) based on the transmit power. In an example, the wireless device may receive a second DCI comprising an uplink grant. The uplink grant may comprise an SRI field indicating power control parameters. The power control parameters may comprise a pathloss reference RS (e.g., "PUSCH-PathlossReferenceRS-Id"), p0-PUSCH-Alpha Set (e.g., "p0-PUSCH-AlphaSetId"), and/or PUSCH-ClosedLoopindex (e.g., as illustrated in FIG. 20). The uplink spatial relation update signaling/configuration mechanism may be reused by the wireless device to update/configure joint TCI states and/or separate TCI states for common beams. The common beams (unified beams or unified TCI states) may comprise the joint TCI states and/or the separate TCI states.

By implementing the existing technologies of beam indication and/or power control parameter indication, the uplink beam and/or power control parameter determination procedure may comprise a set of disparate frameworks for uplink beam and/or power control parameter determinations. For example, these disparate frameworks may include a unified TCI framework with joint TCI state and separate TCI state as well as the legacy framework for spatial relation information signaling and/or power control parameter signaling.

Additionally, the wireless device may decrease the uplink transmission performance if the wireless device merely determines uplink beam and/or power control parameters based on a random selection of one of the two indications (e.g., based on TCI states or based on SRI field). There is a need to enhance the operation behaviors of the wireless device for uplink beam and/or power control parameter determination procedures based on the unified TCI states and legacy SRI indication, without increasing signaling overhead and detection complexity for the wireless device and the base station. For example, if indications of the joint TCI state (or separate TCI state) and legacy SRI field are received by the wireless device via DCIs, different behaviors of wireless device need to be enhanced to handle the two indications of the joint TCI state (or separate TCI state) and the legacy SRI field for the uplink beam and/or power control parameters determination procedures, without increasing signaling overhead and detection complexity for the wireless device and the base station.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on a selection between the first power control parameter and the second power control parameter. The selection may be based on comparing the first time with an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on a random selection of the first power control parameter and the second power control parameter, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the first time being later than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the second power control parameter if joint TCI state (or separate TCI state) is applicable for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control

parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the first time being equal to or earlier than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the first power control parameter if joint TCI state (or separate TCI state) is not applicable for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the receiving of the second DCI being later than the receiving of the first DCI. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the first power control parameter if SRI indication is a most recent indication for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the receiving of the second DCI being equal to or earlier than the receiving of the first DCI. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the second power control parameter if joint TCI state (or separate TCI state) is a most recent indication for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding

reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the receiving of the second DCI being later than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the first power control parameter if the second DCI is applicable after the application time for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the receiving of the second DCI being equal to or earlier than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on the second power control parameter if the second DCI is applicable before the application time for the wireless device, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

Example embodiments of the present disclosure may comprise a wireless device receiving configuration parameters or a medium access control control element (MAC CE) from a base station. The configuration parameters or the MAC CE may comprise applicability indication for TCI state and sounding reference signal resource indicator (SRI). The wireless device may receive a first downlink control information (DCI) from the base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive, from the base station, a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power by using the TCI state or the SRI field based on the applicability indication. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block. The wireless device may avoid determining a transmit power based on a random selection of the first power control parameter and the second power control parameter, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

FIG. 21 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive, from a base station, one or more RRC messages comprising configuration parameters indicating one or more of: a first association between the power control parameter and a TCI state (e.g., as illustrated in FIG. 18A and FIG. 18B); a second

association between the power control parameter and an SRI field of downlink control information (DCI) (e.g., as illustrated in FIG. 20). The wireless device may receive (e.g., at time T1) a medium access control control element (MAC CE) from a base station. In an example, the MAC CE may update (or indicate) a power control parameter of the first association between the power control parameter and a TCI state. In an example, the MAC CE may update (or indicate) a power control parameter of the second association between the power control parameter and an SRI field of downlink control information (DCI). The power control parameter may comprise one or more of: a pathloss reference signal; a p0-PUSCH-AlphaSet; and a closed loop index for power control. In an example, wireless device may not receive the MAC CE from the base station. The wireless device may receive (e.g., at time T2) a first DCI indicating one or more TCI states. The one or more TCI states may comprise one or more joint TCI states, and/or one or more separate TCI states (e.g., one or more uplink TCI states). The one or more TCI states may be associated with one or more power control parameters. The one or more power control parameters may comprise: a pathloss reference RS, and/or a power control parameter setting (e.g., as illustrated in FIG. 17).

In an example, the first DCI may schedule a transport block via a PDSCH. The wireless device may receive (e.g., at time T3) the transport block via the PDSCH from the base station. In an example, the DCI may not schedule the transport block via the PDSCH. The wireless device may receive (e.g., at time T4) a second DCI comprising an uplink grant. The uplink grant may comprise an SRI field. The SRI field may indicate one or more second power control parameters. The one or more second power control parameters may comprise: a pathloss reference RS, and/or a power control parameter setting (e.g., as illustrated in FIG. 17). The uplink grant may schedule an uplink transport block via PUSCH. The wireless device may transmit (e.g., at time T5) an acknowledgement (ACK/NACK) of the transport block (e.g., the transport block being scheduled) and/or the DCI (e.g., the transport block not being scheduled) to the base station. The wireless device may apply (e.g., from time T6) the one or more TCI states after a time offset from transmission (by the wireless device) of the acknowledgement (ACK/NACK). The wireless device may determine power control parameters based on the one or more TCI states, or the SRI field with one or more criteria. The one or more criteria may comprise the wireless device determine the power control parameters (or determine a transmit power) based on one or more of: a time of the first DCI, a time of the second DCI, the application time, a time of transmission of the transport block via PUSCH, or an applicability indication. The wireless device may determine the transmit power based on the determined power control parameters. The wireless device may transmit (e.g., at time T7) the uplink transport block based on the transmit power and via PUSCH. The wireless device may avoid determining a transmit power based on a random selection between first power control parameter associated with the one or more TCI states and second power control parameter indicated by the SRI, which reduces system unreliability of power control procedure and increases power control performance of uplink transmission for the wireless device.

FIG. 22 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. In an example, a wireless device may receive (e.g., at time T1), from a base station, a second DCI comprising an uplink grant (e.g., an SRI). The uplink grant may comprise an SRI field. The SRI field may indicate one

or more second power control parameters. The one or more second power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18). The uplink grant may schedule an uplink transport block via PUSCH at a time T7. The wireless device may receive (e.g., at time T2) a first DCI indicating one or more TCI states. The one or more TCI states may comprise one or more joint TCI states, and/or one or more separate TCI states (e.g., one or more uplink TCI states). The one or more TCI states may be associated with one or more first power control parameters. For example, each TCI may be associated with respective first power control parameters. The one or more first power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18).

In an example, as shown in the dotted lines, the first DCI may schedule a transport block via a PDSCH at time T3. The wireless device may receive (e.g., at time T3) the scheduled transport block via the PDSCH from the base station. In another example, the DCI does not schedule the transport block via the PDSCH. In an example, instead of the second DCI being received at a time T1 that is before receiving the first DCI at time T2, the wireless device may receive the second DCI from the base station at a time T4 that is after receiving the first DCI at time T2. The wireless device may transmit (e.g., at time T5) an acknowledgement (ACK/NACK) of the transport block (e.g., if the transport block is scheduled by the first DCI) and/or the first DCI (e.g., if the transport block is not scheduled by the first DCI) to the base station. The wireless device may apply the one or more TCI states at an application time (e.g., at time T6) that is after a time offset from transmission (by the wireless device) of the acknowledgement (ACK/NACK). In an example, the wireless device may determine a transmit power based on the SRI (or based on the one or more second power control parameters indicated by the SRI) in response to the receiving of the second DCI being later than the receiving of the first DCI. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the receiving of the second DCI being equal to or earlier than the receiving of the first DCI. The wireless device may transmit (e.g., at time T7) the uplink transport block based on the transmit power and via PUSCH.

In an example, the wireless device may determine a spatial domain filter based on the SRI (or based on one or more SRS resources indicated by the SRI) in response to the receiving of the second DCI being later than the receiving of the first DCI. For example, the wireless device may determine the spatial domain filter same as used for transmission of the one or more SRS resources (e.g., in a most recent transmission of the one or more SRS resources). In an example, the wireless device may determine the spatial domain filter based on the one or more TCI states (or based on one or more quasi colocation RSs associated with the one or more TCI states) in response to the receiving of the second DCI being equal to or earlier than the receiving of the first DCI. For example, the wireless device may determine the spatial domain filter same as used for transmission or reception of the one or more quasi colocation RSs associated with the one or more TCI states. The wireless device may transmit (e.g., at time T7), via PUSCH, the uplink transport block based on the transmit power and the spatial domain filter determined by the wireless device.

FIG. 23 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive a first DCI indicating one or more TCI states. The one or more TCI states may be associated with one or more first power control parameters (e.g., as illustrated in FIG. 17 and FIG. 18). For example, each TCI state may be associated with respective power control parameters. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise an SRI field indicating one or more second power control parameters (e.g., as illustrated in FIG. 20 and FIG. 22). Not all of the steps of the flow diagram in FIG. 23 need to be in the order shown. For example, the second DCI may be received before the first DCI.

In an example, the wireless device may determine a transmit power based on the SRI field (or based on the one or more second power control parameters indicated by the SRI field) in response to the receiving of the second DCI being later than the receiving of the first DCI. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the receiving of the second DCI being equal to or earlier than the receiving of the first DCI. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block to the base station.

FIG. 24 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive (e.g., at time T1), from a base station, a first DCI indicating one or more TCI states. The one or more TCI states may comprise one or more joint TCI states, and/or one or more separate TCI states (e.g., one or more uplink TCI states). The one or more TCI states may be associated with one or more first power control parameters. For example, each TCI state (of the one or more TCI states) may be associated with respective first power control parameters. The one or more first power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18). In an example, the first DCI may schedule a transport block via a PDSCH at time T2. The wireless device may receive (e.g., at time T2) the scheduled transport block via the PDSCH from the base station. In another example, the DCI does not schedule the transport block via the PDSCH. In an example, the wireless device may receive (e.g., at time T3) a second DCI comprising an uplink grant. The uplink grant may comprise an SRI field. The SRI field may indicate one or more second power control parameters. The one or more second power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18). The uplink grant may schedule an uplink transport block via PUSCH at, e.g., time T5 or T7. The wireless device may transmit (e.g., at time T4) an acknowledgement (ACK/NACK) of the transport block (e.g., if the transport block is scheduled by the first DCI) and/or the DCI (e.g., if the transport block is not scheduled by the first DCI) to the base station. In an example, the wireless device may apply the one or more TCI states at a time (e.g., at or after time T6) that is after a time offset from the transmission (by the wireless device) of the acknowledgement (ACK/NACK).

In an example, the wireless device may transmit the uplink transport block via PUSCH at a time (e.g., time T5)

scheduled by the second DCI and that is before the application time at T6. In another example, the wireless device may transmit the uplink transport block via PUSCH at a time (e.g., time T7) scheduled by the second DCI and that is after the application time at T6. In an example, the wireless device may determine a transmit power based on the SRI (or based on the one or more second power control parameters indicated by the SRI) in response to the transmission of the uplink transport block being equal to or earlier than the application time of the one or more TCI states. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the transmission of the uplink transport block being later than the application time of the one or more TCI states. The wireless device may transmit (e.g., at time T7) the uplink transport block based on the transmit power and via the PUSCH indicated by the second DCI.

In an example, the wireless device may determine a spatial domain filter based on the SRI (or based on one or more SRS resources indicated by the SRI) in response to the transmission of the uplink transport block being equal to or earlier than the application time of the one or more TCI states. For example, the wireless device may determine the spatial domain filter same as used for transmission of the one or more SRS resources (e.g., in a most recent transmission of the one or more SRS resources). In an example, the wireless device may determine the spatial domain filter based on the one or more TCI states (or based on one or more quasi colocation RSs associated with the one or more TCI states) in response to the transmission of the uplink transport block being later than the application time of the one or more TCI states. For example, the wireless device may determine the spatial domain filter same as used for transmission or reception of the one or more quasi colocation RSs associated with (or comprised by) the one or more TCI states. The wireless device may transmit (e.g., at time T7), via PUSCH, the uplink transport block based on the transmit power and the spatial domain filter determined by the wireless device.

FIG. 25 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive a first DCI indicating one or more TCI states. The one or more TCI states may be associated with one or more first power control parameters (e.g., as illustrated in FIG. 17 and FIG. 18). For example, each TCI state may be associated with respective power control parameters. The one or more TCI states may be applicable by the wireless device after an application time. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise an SRI field indicating one or more second power control parameters (e.g., as illustrated in FIG. 20 and FIG. 22). Not all of the steps of the flow diagram in FIG. 25 need to be in the order shown. For example, the second DCI may be received before the first DCI.

In an example, the wireless device may determine a transmit power based on the SRI field (or based on the one or more second power control parameters indicated by the SRI field) in response to the first time (i.e., the time of the scheduled transport block) being equal to or earlier than the application time of the one or more TCI states. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the

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one or more first power control parameters associated with the one or more TCI states) in response to the first time being later than the application time. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block to the base station. The first time may comprise one or more of: a slot; an orthogonal frequency division multiplexing symbol; a mini-slot; a subframe; or a frame. The application time may comprise one or more of: a slot; an orthogonal frequency division multiplexing symbol; a mini-slot; a subframe; or a frame.

FIG. 26 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive (e.g., at time T1), from a base station, a first DCI indicating one or more TCI states. The one or more TCI states may comprise one or more joint TCI states and/or one or more separate TCI states (e.g., one or more uplink TCI states). The one or more TCI states may be associated with one or more first power control parameters. The one or more first power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18). In an example, the first DCI may schedule a transport block via a PDSCH. If the first DCI schedules the transport block, the wireless device may receive (e.g., at time T2) the transport block via the PDSCH from the base station. In another example, the DCI does not schedule the transport block via the PDSCH. In an example, the wireless device may transmit (e.g., at time T3) an acknowledgement (ACK/NACK) of the transport block (e.g., when the transport block is scheduled by the first DCI) and/or the DCI (e.g., when the transport block is not scheduled by the first DCI) to the base station.

In an example, as shown in the dotted lines, the wireless device may receive (e.g., at time T4) a second DCI comprising an uplink grant. The uplink grant may comprise an SRI field. The SRI field may indicate one or more second power control parameters. The one or more second power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17 and FIG. 18). The uplink grant may schedule an uplink transport block via PUSCH (e.g., at time T7). The wireless device may apply (e.g., from time T5) the one or more TCI states at an application time, which is a time after a time offset from the transmission (by the wireless device) of the acknowledgement (ACK/NACK). In an example, instead of the second DCI being received at, for example, time T4, the second DCI may be received by the wireless device after the application time T5 (e.g., the second DCI may be received at time T6). The wireless device may transmit (e.g., at time T7) the uplink transport block via PUSCH. In an example, the wireless device may determine a transmit power based on the SRI (or based on the one or more second power control parameters indicated by the SRI) in response to the reception of the second DCI being later than the application time of the one or more TCI states. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the reception of the second DCI being equal to or earlier than the application time of the one or more TCI states. The wireless device may transmit (e.g., at time T7) the uplink transport block based on the transmit power and via PUSCH.

In an example, the wireless device may determine a spatial domain filter based on the SRI (or based on one or more SRS resources indicated by the SRI) in response to the

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receiving of the second DCI being later than the application time of the one or more TCI states. For example, the wireless device may determine the spatial domain filter same as used for transmission of the one or more SRS resources (e.g., in a most recent transmission of the one or more SRS resources). In an example, the wireless device may determine the spatial domain filter based on the one or more TCI states (or based on one or more quasi colocation RSs associated with the one or more TCI states) in response to the receiving of the second DCI being equal to or earlier than the application time of the one or more TCI states. For example, the wireless device may determine the spatial domain filter same as used for transmission or reception of the one or more quasi colocation RSs associated with (or comprised by) the one or more TCI states. The wireless device may transmit (e.g., at time T7), via PUSCH, the uplink transport block based on the transmit power and the spatial domain filter determined by the wireless device.

FIG. 27 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive a first DCI indicating one or more TCI states. The one or more TCI states may be associated with one or more first power control parameters (e.g., as illustrated in FIG. 17 and FIG. 18). For example, each TCI state may be associated with respective power control parameters. The one or more TCI states may be applicable by the wireless device after an application time. The wireless device may receive, at a first time, a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise an SRI field indicating one or more second power control parameters (e.g., as illustrated in FIG. 20 and FIG. 22). Not all of the steps of the flow diagram in FIG. 27 need to be in the order shown. For example, the second DCI may be received before the first DCI.

In an example, the wireless device may determine a transmit power based on the SRI field (or based on the one or more second power control parameters indicated by the SRI field) in response to the first time (i.e., the time of the scheduled transport block) being later than the application time of the one or more TCI states. In an example, the wireless device may determine the transmit power based on the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the first time being equal to or earlier than the application time. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block to the base station. The first time may comprise one or more of: a slot; an orthogonal frequency division multiplexing symbol; a mini-slot; a subframe; or a frame. The application time may comprise one or more of: a slot; an orthogonal frequency division multiplexing symbol; a mini-slot; a subframe; or a frame.

FIG. 28 illustrates a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive (e.g., at time T1) an RRC message or a medium access control control element (MAC CE) from a base station. In some embodiments, a message (such as the RRC message or the MAC CE) may comprise an applicability indication for TCI state and SRI. In an example, the applicability indication may indicate which one between the TCI state and the SRI is used by the wireless device to determine transmit power for uplink transmissions. In an example, the applicability indication may separately or jointly indicate whether the

TCI(s) or the SRI is applicable for determining the transmit power for uplink transmissions. In an example, the applicability indication may indicate which one between the TCI states and the SRI is used by the wireless device to determine spatial domain filter for uplink transmissions. In an example, the applicability indication may separately or jointly indicate whether the TCI(s) or the SRI is applicable for determining the spatial domain filter for uplink transmissions. In an example, the applicability indication may indicate which channels (e.g., PUSCH, PUCCH, PRACH) or signals (e.g., SRS, DMRS) for uplink transmission are used by the wireless device for uplink transmission based on the TCI state.

In an example, the wireless device may receive (e.g., at time T2) a first DCI indicating one or more TCI states. The one or more TCI states may comprise one or more joint TCI states and/or one or more separate TCI states (e.g., one or more uplink TCI states). The one or more TCI states may be associated with one or more first power control parameters. The one or more first power control parameters may comprise: a pathloss reference RS, and/or a power control parameter setting (e.g., as illustrated in FIG. 17). In an example, the first DCI may schedule a transport block via a PDSCH. The wireless device may receive (e.g., at time T3) the transport block via the PDSCH from the base station. In another example, the first DCI (received at time T2) does not schedule the transport block via the PDSCH.

The wireless device may receive (e.g., at time T4) a second DCI comprising an uplink grant. The uplink grant may comprise an SRI field. The SRI field may indicate one or more second power control parameters. The one or more second power control parameters may comprise: a pathloss reference RS and/or a power control parameter setting (e.g., as illustrated in FIG. 17). The uplink grant may schedule an uplink transport block via PUSCH. The wireless device may transmit (e.g., at time T5) an acknowledgement (ACK/NACK) of the transport block (e.g., when the transport block is scheduled by the first DCI) and/or the first DCI (e.g., when the transport block is not scheduled by the first DCI) to the base station. The wireless device may apply (e.g., from time T6) the one or more TCI states after a time offset from the transmission (by the wireless device) of the acknowledgement (ACK/NACK). As shown in FIG. 28, this time is the application time of the one or more TCI states.

In an example, the wireless device may determine a transmit power by using the one or more TCI states or the SRI based on the applicability indication. In an example, the wireless device may determine the transmit power by using the SRI (or based on the one or more second power control parameters) in response to the applicability indication indicating the one or more TCI states are not applicable to PUSCH. In an example, the wireless device may determine the transmit power by using the SRI (or based on the one or more second power control parameters) in response to the applicability indication indicating the SRI is used by the wireless device. In an example, the wireless device may determine the transmit power by using the one or more TCI states (or based on the one or more first power control parameters associated with the one or more TCI states) in response to the applicability indication indicating the one or more TCI states are used by the wireless device. The wireless device may transmit (e.g., at time T7) the uplink transport block based on the transmit power and via PUSCH.

In an example, the wireless device may determine a spatial domain filter by using the one or more TCI states or the SRI based on the applicability indication. In an example, the wireless device may determine a spatial domain filter

based on the SRI (or based on one or more SRS resources indicated by the SRI) in response to the applicability indication indicating the one or more TCI states are not applicable to PUSCH. For example, the wireless device may determine the spatial domain filter same as used for transmission of the one or more SRS resources (e.g., in a most recent transmission of the one or more SRS resources). In an example, the wireless device may determine the spatial domain filter based on the SRI (or based on one or more SRS resources indicated by the SRI) in response to the applicability indication indicating the SRI is used by the wireless device. In an example, the wireless device may determine the spatial domain filter based on the one or more TCI states (or based on one or more quasi colocation RSs associated with the one or more TCI states) in response to the applicability indication indicating the one or more TCI states are used by the wireless device. For example, the wireless device may determine the spatial domain filter same as used for transmission or reception of the one or more quasi colocation RSs associated with (or comprised by) the one or more TCI states. The wireless device may transmit (e.g., at time T7), via PUSCH, the uplink transport block based on the transmit power and the spatial domain filter determined by the wireless device.

FIG. 29 illustrates a flow diagram of a power control procedure with common beams as per an aspect of an example embodiment of the present disclosure. A wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block (TB) at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on a selection between the first power control parameter and the second power control parameter. The selection may be based on comparing the first time with an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block (TB).

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on a selection between the first power control parameter and the second power control parameter. The selection may be based on comparing the first time with an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, the comparing the first time with the application time of the TCI state may comprise one of: the first time is later than the application time; and the first time is equal to or earlier than the application time. The wireless device may determine the transmit power based on the selection of the first power control parameter in response to the first time being later than the application time. The wireless device may determine the transmit power based on

the selection of the second power control parameter in response to the first time being equal to or earlier than the application time. The first power control parameter may comprise one or more of: a pathloss reference signal; a p0-PUSCH-AlphaSet; and a closed loop index for power control. The second power control parameter may comprise one or more of: a pathloss reference signal; a p0-PUSCH-AlphaSet; and a closed loop index for power control. The TCI state may be a joint TCI state, a separate TCI state, or an uplink TCI state. The association between the TCI state and the first power control parameter may comprise at least one of: the TCI state comprises the first power control parameter; and the first power control parameter is linked with (or identified by) an index of the TCI state. The first DCI may comprise a downlink assignment. The downlink assignment may schedule a second transport block via a physical downlink shared channel.

In an example, the first time may comprise one or more of: a slot; an orthogonal frequency division multiplexing symbol; a mini-slot; a subframe; and a frame. The SRI may further indicate a sounding reference signal resource. The first power control parameter may be different from the second power control parameter. The application time of the TCI state may be used by the wireless device to apply the TCI state for an uplink transmission or a downlink reception after the application time. The application of the TCI state may comprise that the wireless device determines a spatial domain filter for the uplink transmission, or the downlink reception based on the TCI state.

In an example, the determination of the spatial domain filter may be further based on a quasi-colocation reference signal of the TCI state. The wireless device may determine an uplink spatial domain filter based on a selection between the TCI state and SRI. The selection may be based on second comparing the first time with the application time of the TCI state. The second comparing the first time with the application time of the TCI state may comprise one of: the first time is later than the application time; and the first time is equal to or earlier than the application time. The wireless device may determine the uplink spatial domain filter based on the selection of the TCI state in response to the first time being later than the application time. The wireless device may determine the uplink spatial domain filter based on the selection of the SRI in response to the first time being equal to or earlier than the application time. The transmitting the transport block may be further based on the uplink spatial domain filter.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the first time being later than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a

second DCI comprising an uplink grant. The uplink grant may schedule a transport block at a first time. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the first time being equal to or earlier than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the receiving of the second DCI being later than the receiving of the first DCI. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the receiving of the second DCI being equal to or earlier than the receiving of the first DCI. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the second power control parameter in response to the receiving of the second DCI being later than an application time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive a first downlink control information (DCI) from a base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power based on the first power control parameter in response to the receiving of the second DCI being equal to or earlier than an appli-

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cation time of the TCI state. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

In an example, a wireless device may receive configuration parameters or a medium access control control element (MAC CE) from a base station. The configuration parameters or the MAC CE may comprise applicability indication for TCI state and sounding reference signal resource indicator (SRI). The wireless device may receive a first downlink control information (DCI) from the base station. The first DCI may indicate a transmission configuration indicator (TCI) state. The TCI state may be associated with a first power control parameter. The wireless device may receive, from the base station, a second DCI comprising an uplink grant. The uplink grant may schedule a transport block. The uplink grant may comprise a sounding reference signal resource indicator (SRI) indicating a second power control parameter. The wireless device may determine a transmit power by using the TCI state or the SRI field based on the applicability indication. The wireless device may transmit, based on the transmit power and via a physical uplink shared channel, the transport block.

What is claimed is:

1. A method comprising:
 - receiving, by a wireless device, a first downlink control information (DCI):
 - indicating a first transmission configuration indicator (TCI) state of TCI states, wherein the first TCI state is associated with a first power control parameter and is applicable after a first time instance for uplink channel transmissions;
 - receiving a second DCI comprising an uplink grant:
 - scheduling a transport block for transmission via a physical uplink shared channel (PUSCH) at a second time instance; and
 - indicating a second power control parameter;
 - selecting, from the first power control parameter and the second power control parameter, a power control parameter associated with an earlier time instance based on a comparison of the first time instance with the second time instance, wherein the first time instance indicates an application time of the first TCI state after a time offset from transmission of an acknowledgement of the transport block;
 - determining, based on the selecting, a transmit power associated with the power control parameter; and
 - transmitting, based on the transmit power and via the PUSCH, the transport block.
2. The method of claim 1, further comprising receiving configuration parameters of the TCI states, wherein each of the TCI states is associated with a respective power control parameter of power control parameters.
3. The method of claim 1, wherein the selecting the power control parameter comprises selecting the first power control parameter as the power control parameter in response to the first time instance being equal to or earlier than the second time instance.
4. The method of claim 1, wherein selecting the power control parameter comprises selecting the second power control parameter as the power control parameter in response to the first time instance being later than the second time instance.
5. The method of claim 1, wherein the first power control parameter or the second power control parameter comprises one or more of:

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a pathloss reference signal;
a p0-PUSCH-AlphaSet; and
a closed loop index for power control.

6. The method of claim 1, wherein the first TCI state is a joint TCI state, a separate TCI state, or an uplink TCI state.
7. The method of claim 1, wherein the first TCI state is associated with the first power control parameter based on at least one of:
 - the first TCI state comprising the first power control parameter; or
 - the first power control parameter being linked with an index of the first TCI state.
8. The method of claim 1, wherein the uplink grant comprises a sounding reference signal resource indicator (SRI) indicating the second power control parameter.
9. The method of claim 8, further comprising:
 - determining an uplink spatial filter based on a selection between the first TCI state and the SRI; and
 - transmitting the transport block further based on the uplink spatial filter.
10. The method of claim 1, wherein the first power control parameter is different from the second power control parameter.
11. A wireless device comprising:
 - one or more processors; and
 - memory storing instructions that, when executed by the one or more processors, cause the wireless device to:
 - receive a first downlink control information (DCI):
 - indicating a first transmission configuration indicator (TCI) state of TCI states, wherein the first TCI state is associated with a first power control parameter and is applicable after a first time instance for uplink channel transmissions;
 - receive a second DCI comprising an uplink grant:
 - scheduling a transport block for transmission via a physical uplink shared channel (PUSCH) at a second time instance; and
 - indicating a second power control parameter;
 - select, from the first power control parameter and the second power control parameter, a power control parameter associated with an earlier time instance based on a comparison of the first time instance with the second time instance, wherein the first time instance indicates an application time of the first TCI state after a time offset from transmission of an acknowledgement of the transport block;
 - determine, based on the power control parameter, a transmit power associated with the power control parameter; and
 - transmit, based on the transmit power and via the PUSCH, the transport block.
12. The wireless device of claim 11, wherein the instructions further cause the wireless device to receive configuration parameters of the TCI states, wherein each of the TCI states is associated with a respective power control parameter of power control parameters.
13. The wireless device of claim 11, wherein the instructions further cause the wireless device to select the first power control parameter as the power control parameter in response to the first time instance being equal to or earlier than the second time instance.
14. The wireless device of claim 11, wherein the instructions further cause the wireless device to select the second power control parameter as the power control parameter in response to the first time instance being later than the second time instance.
15. The wireless device of claim 11, wherein the first power control parameter or the second power control parameter comprises one or more of:

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a pathloss reference signal;
 a p0-PUSCH-AlphaSet; and
 a closed loop index for power control.

16. The wireless device of claim 11, wherein the first TCI state is a joint TCI state, a separate TCI state, or an uplink TCI state. 5

17. The wireless device of claim 11, wherein the first TCI state is associated with the first power control parameter based on at least one of:

the first TCI state comprising the first power control parameter; or
 the first power control parameter being linked with an index of the first TCI state. 10

18. The wireless device of claim 11, wherein the uplink grant comprises a sounding reference signal resource indicator (SRI) indicating the second power control parameter. 15

19. The wireless device of claim 18, wherein the instructions further cause the wireless device to:

determine an uplink spatial filter based on a selection between the first TCI state and the SRI; and
 transmit the transport block further based on the uplink spatial filter. 20

20. A non-transitory computer-readable medium comprising instructions that, when executed by one or more processors of a wireless device, cause the wireless device to:

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receive a first downlink control information (DCI):
 indicating a first transmission configuration indicator (TCI) state of TCI states, wherein the first TCI state is associated with a first power control parameter and is applicable after a first time instance for uplink channel transmissions;

receive a second DCI comprising an uplink grant:
 scheduling a transport block for transmission via a physical uplink shared channel (PUSCH) at a second time instance; and

indicating a second power control parameter;
 select, from the first power control parameter and the second power control parameter, a power control parameter associated with an earlier time instance based on a comparison of the first time instance with the second time instance, wherein the first time instance indicates an application time of the first TCI state after a time offset from transmission of an acknowledgement of the transport block;

determine, based on the power control parameter, a transmit power associated with the power control parameter; and

transmit, based on the transmit power and via the PUSCH, the transport block.

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